

HOW DO RISK MITIGATION MECHANISMS AND STRUCTURAL RESILIENCE ALLOW RISK-ADJUSTMENT PERFORMANCE FOR MACRO HEDGE FUNDS DURING MACROECONOMIC REGIMES?



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Summary:

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Prelude of the thesis:

Before commencing my applied research thesis work, I would like to present briefly the ideas of the work, as well as some general gratitude towards the people who have helped me at completing this thesis.

The main idea of the applied research thesis is to explore the world of hedge funds, a domain that is sophisticated and mystical into the domain of private equity, while also not being totally known by the public. At the same time, the domain of private equity, and the possibilities of being able to handle hedge funds while also exploring the diversification possibilities offered across hedge funds represent one of the fields, I would like to commit myself to in the foreseeable future. Furthermore, the idea of having the chance at working in a domain where the dynamics shift quickly would allow me to integrate myself better in the different perspectives of growth and changes that are shaping our current society, while also having a certain intellectual stimulation at completing most of the tasks related to the domain of hedge funds. The idea of exploring the way that hedge funds are behaving during the different successive macroeconomic cycles is also allowing me at understanding the perspective that investors adapt during certain economic phases, while also comprehending the ways in which investors are managing to diversify their portfolios, as well as understanding the mechanisms that large private equity firms and asset management firms, such as KKR or Blackstone do adopt during economic crises. Lastly, this research is also inspired from the different courses I had across my recent years of my undergraduate studies, and especially with the courses I had taken during my double-degree experience at Sungkyunkwan University in Seoul, as per the example of Current Topics in Finance and Intermediate Investments during which I was able to garner my skills related to the different equity strategies that firms do adopt, the ways of diversifying portfolios, while also better comprehending the impact that macroeconomic announcements do have on different portfolios and investments realized by the large firms.

As a conclusion, I would like to thank my different professors, Dr. HK Choi, Dr. Andy Kim and Professor Shin Young Soo for their teaching and wisdom during the courses I had the chance to participate in, as well as for their support for finding the different resources that helped me at understanding further the topic of hedge funds. Moreover, I would like to thank Mr. Richard Ruble for his expertise and recommendations of papers, as well as his support throughout the whole completion of my applied research thesis, as well as Mrs. Bayet-Robert for her help at understanding the requirements of the applied research thesis.

I – Introduction to the applied research thesis.

1.1 – Introduction:

In a world that is shaped by a large magnitude of different events that are shifting our daily lives, and where the different technological advancements are impacting the way we are regarding our vision to the world, companies working in the finance industry are also working at finding different investment strategies that enable further profit maximization, while also enabling these companies to hedge any further risk related to the different macroeconomic cycles, or announcements that could damper the growth and expansion of these firms across the world. Within this diverse domain that the finance industry is incorporating, the world of private equity, and especially the domain of hedge funds has been able to capture further the intention of specialized professionals, such as investment bankers or equity researchers or specialized institutions. Longtime seen to alternate the external and market premium risks that are related to the sudden shifts in the macroeconomic environment, hedge funds have been an opportunity for investors to enhance their gains even during uncertain periods.

1.1.1 - Definition of hedge funds:

To better comprehend the main theme of the applied research thesis, we must understand further what hedge funds are. Although no clear definition has been given regarding hedge funds (according to the SEC – Securities and Exchange Commission), people have defined nowadays hedge funds as being an “actively managed private investment fund” that uses money regrouped into a pool and that is managed by professional fund managers. The concept came in effect in 1949 when investor Alfred W. Jones created a fund that could hedge assets from other portfolios, and thus neutralizing the effect that market changes had on the respective assets by buying assets that were expected to increase in value while also short-selling assets expected to see their market value decreasing. Hedge funds have often been characterized by specific characteristics that are delimiting them from other funds, especially open-ended funds:

- Flexibility related to the investments: a hedge fund allows an investor or institution to hedge potential risks related to sudden market changes, such as the FED Interest Rate, or the US YoY CPI by investing into long-positioned assets that are due to increase their market value, while also short-selling assets with values expected to decrease in the defined range.
- Diversity of tools: hedge funds use a lot of risky strategies, leverage (practice of borrowing money and investing into funds to generate better future returns), and derivative securities such as futures or forwards.
- Nature of the investment: hedge funds are mostly illiquid, meaning that the assets included within the hedge funds cannot be sold directly to the stock exchange, and with proceeds promptly available in cash. This is mainly due to investors having their money back restricted throughout lock-up agreements, which function as a stated commitment

from the investor at paying an upfront sum for the contribution to the hedge fund's management. At the same time, withdrawals may only happen under certain circumstances, and certain periods (either quarterly or biannually).

- Lack of regulations: unlike mutual funds or ETFs, hedge funds are not directly regulated as their investments are mainly comprising private positions and assets. Moreover, the shares are not required to be redeemable daily based on the fund's net asset value (NAV) or ensuring fairness in the pricing of fund shares.

Across hedge funds, one of the key characteristics that distinguishes this type of fund from the other traditional funds is the inclusion of performance and management fees. The fee structure of a hedge fund, usually known under the rule of "2-20", will represent the structuration of the fees that are earned by the different actors implicated in the management of the hedge fund. Unlike a traditional fund, such as mutual funds, hedge funds' fee structure includes both a performance fee and a management fee. A performance fee represents the reward that hedge funds managers get for generating profits, while the management fee is mandatorily set in the agreements between the hedge fund managers, regardless of the returns generated, and that is more focused on paying the administrative and operational services that the investment manager provides to the fund. Generally, hedge funds' management fees represent 2% of the fund's NAV (net asset value), while the performance fees are roughly 20% of the NAV. This rule is mainly implemented since hedge funds target mostly absolute returns, which are mainly incorporating the returns that do not depend on the market shifts within the macroeconomic environment.

1.2 – History of hedge funds:

The history of hedge funds, unlike the other fund vehicles within the domain of finance, is more recent and more sophisticated, especially given the origins and vague definition of the term itself. One of the first hedge funds' core idea originated in the 1940s, and more precisely in 1949 when Alfred Winslow Jones, a famous Australian investor and the first "accredited" hedge fund manager, is proposing the first hedge fund strategy when he started to work on his thesis to separate two risks involved in stock investments with the aim to create a market-neutral portfolio, a portfolio that is combining assets that are limiting macroeconomic factors. The first risk included was the market risk, which is the potential losses that may be incurred from financial markets fluctuations (and mostly associated to movements into both macroeconomic aggregates like the interest rate, or the stock prices), while the second risk being more associated to the nature and characteristics of the stocks within the portfolio, and thus called the systematic risk.

Within his strategy, he used financial leverage to facilitate his purchase of shares while also yielding further potential financial profit, while also short selling some of the less performative shares to counter market risk. The quantities of both bought and sold shares was the same, so that the market-neutral portfolio could be shaped, while also charging to the 99 investors within the fund a total of 2% of management fees and 20% of performance fees, a structure that wasn't

common for the period (and especially regarding performance fees, which aren't always granted for ETFs and mutual funds) and is now the main reference regarding the fee structure of modern hedge funds.

These concepts came to light in 1952 when the fund's structuration came to light, with the further inclusion of leverage and investors under a limited partnership. The ideas of Jones quickly became prominent and the following decade, the 1960s, became the period where hedge funds had commenced to rise. The sudden interest into hedge funds was triggered by the publishing of an article in the Fortune magazine highlighting the astonishing performances obtained by Jones' fund, described as a method enabling to outperform the best mutual fund over the previous 5 years by "44%", albeit the management-incentive fee structure that had been described as being a risky gamble regarding the possibility to reward investors based on the performance of the fund. More and more hedge funds saw the day, including the infamous Quantum Fund, which was advised by George Soros. Some analysts, such as Caldwell described that the combination of incentive fees and leverage in an environment highly market by bullish movements have motivated managers to participate into hedge fund vehicles, where high margins with low hedging opportunities were applied as the main strategies. This structure, though, lost idea, as the idea backfired during the recessions of 1969-1970 and 1973-1974, which led to the closure of traditional hedge funds, including Jones' fund finding a nadir point during the period.

The resurgence of hedge funds was more visible in the following decade, with the sector continuing to evolve over the 1980s, especially thanks to the changed scaled of major markets. Hedge funds started to combine unconstrained investment styles, such as leverage but also focusing on specific new markets, especially fixed-income markets that have become one of the main focus for hedge funds (under what we know as arbitrage hedge fund strategies). The main characteristics of hedge funds were that most of them were small, but with high returns. The example of Tiger Fund, launched by investor Julian Robertson, highlights the alpha opportunities generated: while being launched with roughly \$8 million (today's equivalent), a \$22 billion-peak had been reached.

The more recent decades, and especially the 2010s, brought some serious changes to the hedge fund industry. In effect, some agreements such as the Dodd-Frank Wall Street Reform and Consumer Protection Act came into effect, with more and more reporting being mandated and obligatory towards the SEC. In Europe, the Alternative Investment Fund Manager Directive and the Vockler rule forced hedge funds to upgrade their compliance and operational framework, and limiting banks to invest into certain investment activities, and thus resulting into an influx of new funds as prop-traders, with traders using the firm's money instead of external money source.

1.3 – Investment ecosystem and comparison: hedge funds vs other types of funds.

In this section, we will focus mostly on explaining how the mechanisms are involved regarding the actions a hedge fund undertakes, and how it differentiates itself from the main classic funds used within the public sphere.

As we have mentioned previously, the goal for a hedge fund will be to seek out money-making opportunities through these main principles:

- **Hedging positions:** this strategy represents the core of how hedge funds do function. In effect, the hedge fund will regroup an amount of assets from diverse or specific sectors, and which can be used in diverse ways. Some of the assets will be invested into securities would move inversely to how the assets and securities within the hedge fund are moving, creating an offset opportunity for the hedge fund against losses in the core holdings.
- **Integration of major events:** some of the strategies used within hedge funds will rely on significant events, which could be both corporate and macroeconomic. As an example, hedge funds can take opportunities into different situations, such as bankruptcies, mergers, policy changes, or shifts into the global market interest rate. In this scenario, the objective of the hedge fund will not be only to gain any further profits, but also secure potential ownership or control over potential prospective operations.
- **Core of the strategies:** the main axes where hedge funds will surge will occur into equity, fixed-income, and event-driven scenarios. Usually, a hedge fund can deploy a simple long/short, which is part of the pairs trading core (using long and short positions into two stocks that are highly correlated between each other), and where stocks are evaluated based on their relative valuation. Alternatively, a fixed-income hedge fund strategy will guarantee investors solid returns with a minimal level of volatility thanks to the nature of fixed-income securities. Finally, an event-driven focus is mainly occurring into different macroeconomic announcements (such as interest rates, or emergent markets) or corporate events (bankrupts, insolvencies, mergers or acquisitions).
- **Exploitation of price gaps:** hedge funds, as stated earlier, will focus on making profits by correcting minor mispriced levels between investments with similar term structures and cash flows. Hedge funds will have their focus on breakaway gaps, which represent price gaps above a support resistance area defining the range of trade, as well as runaway gaps due to the interest investors have in specific stocks.

At the same time, hedge funds have the distinguishable future of being, for most of them, open-ended, meaning that investors can continually redeem or issue new shares at the fund's total Net Asset Value (NAV). At the same time, the shares are not listed on the stock exchange, meaning that investors will have to transact directly from the fund. By being able to redeem shares enables hedge funds to capture spreads and further profits between the securities that are either underlying, or over the fund's NAV to stabilize the risks, while also maximizing risks. These

features are very distinctive from the traditional funds that exist across market finance, which will be defined and compared (in a theoretical perspective) with hedge funds.

1.3.1 - Mutual funds:

A mutual fund is a fund garnering money from many investors to purchase different financial instruments that are regrouped into a highly diversified portfolio of stocks or bonds. Generally, these can be regrouped into either securities funds (bond type, equity type, hybrid equity type, hybrid bond type), or non-securities funds such as short-term finance (composed mostly of short-term instruments such as call loans), real-estate instruments, special assets, or mixed assets (regrouping different assets to enhance diversification). Within the mutual fund, investors pool their money to buy collectively assets by buying shares within the fund, and thus investors will become the fund's owners of the holdings. The performance of the fund will depend mainly on the assets regrouped: in a scenario where the stock or bonds' prices will go up, the fund's NAV will increase, and vice versa. The main mutual funds would include the Vanguard 500 Index Fund (VFIAX) and regrouping the analysis of the S&P 500, Fidelity 500 Index Fund (FXAIX) that has a lower expense ratio than Vanguard's, or T. Rowe's Price Dividend Growth Fund (PRDGX).

Although both the mutual funds and hedge funds can be open-ended, there are a few differences that are enabling the hedge fund to be distinguishable from mutual funds. From the objectives of the fund, in which a mutual fund would approach its returns to a specific benchmark across an index to ensure more stable returns, to risk-adverse strategies employed by mutual fund managers to garner long positions, all while more impacted by a strict regulatory framework where the NAV and yields generated by the fund should be reported at the SEC, mutual funds represent a safer option for investors that seek long-term performance driven by specific benchmarks and rules, which are very different from the high-risk and hedge nature of hedge funds.

1.3.2 - ETFs:

An ETF (Exchange-Traded Fund) is a mutual fund holding multiple underlying assets. It can be "bought and sold" on a stock market. Much more upheld regarding the applied regulations (since the ETF must be registered with the SEC, subject to the Investment Company Act of 1940), an ETF is also more constraining generally to register as it needs first authorized participants that will create the portfolio where the ETF's holdings will be stored, while also ensuring that the creation units are generated to create the shares of the ETF, which will be eventually either hold or sold on the stock exchange to other investors or strategic buyers. ETFs can hold a diverse amount of financial instruments, although diversification is much more rare than on usual mutual funds. Different ETFs can exist, such as passive or actively managed ETFs, bond ETF, commodity, or currency ETF. The main advantages of the ETF lay around the exposure to many stocks across various industries, while the mechanism of creation and redemption of shares will

allow the shares to keep the ETF's NAV close to its market price, enabling further the process of arbitrage. At the same time, higher fees can be occurred (especially if the ETF is actively managed), while the lack of liquidity will constrain the costs, and thus increasing further the exposure to either premiums or discounts.

1.3.3 - Private Equity Funds:

The last type of funds that will be covered are the private equity funds (PEFs). These funds are mainly structured around private equity firms, which are often known as being limited partnership companies, and thus being closed-end funds.

Within a private equity fund, the main actors are as the following:

- **General partner:** also known as the private equity firm, this will represent the main vehicle managing the fund and the investment decisions.
- **Limited partner(s):** known as the investors (which can be fund-of-funds, institutions, or high net worth individuals), they will commit the capital within the fund's pool.
- **Private Equity Fund:** forming the limited partnership, this will be the main pooled investment vehicle.
- **NewCo:** known as the investment vehicles and that will constitute the legal entities that are used to acquire and hold target companies.

PEFs are generally known for employing long-term strategies to yield positive returns, regardless of the macroeconomic context. Leveraged buyouts (LBOs) is the most popular approach, during which the PE vehicle would approach a target company based on borrowed debt and contributions of shareholders to buy the specific target company, all while handling the operational structure before selling the vehicle through an IPO (Initial Public Offering). Another strategy, the capital growth strategy is more conservative, where the vehicle would provide minority equity so that the target firm could restructure its operations or finance further activities expanding the latter's activities. Other major strategies would include mezzanine capital for financing equilibrium of funds, as well as venture capital funding which targets new or promising funds.

Compared to hedge funds, a private equity fund will espouse differences both into the scope of investments, its horizons, and mechanisms employed. Private Equity Funds will privilege a long-term approach of acquiring and merging companies or acquiring the stake of public entities to enhance growth, while hedge funds will have their main focus on seizing short-term squeezes to maximize profits. While both strategies combine risk management by applying diversification to their portfolios, hedge funds will tolerate a higher amount of risks than private equity funds, especially since hedge funds will always use short and long positions to hedge away any risk. Lastly, both hedge funds and private equity funds will require large balances per each investor:

nonetheless, hedge funds may then lock the provided funds for months, while the period will be longer for a private equity fund since the nature of the investment will be less liquid.

1.4 – Limits and difficulties of hedge fund mechanisms:

Although hedge funds have often been touted as being the most performant funds within the domain of quantitative finance, limits do exist, especially related to the ways that performance is actually garnered, as well as the hedge funds' real exposure to volatility and macroeconomic cycles.

1.4.1 - Issues regarding the performance of hedge funds:

One of the most common reasons over the failure of hedge funds is regarding their bad or inconsistent performance, which can usually be linked to the poor operations management surrounding the handling of the fund. In effect, ¹50% of hedge funds fail due to issues into the managerial operations of the hedge fund. A hedge fund is requiring careful attention to different details that must be managed and monitored on a daily basis, such as the costs (labor, software, insurance, taxes, legal), legal and compliance matters (monthly audits, reporting to the SEC), employing staffing, or having an efficient monitoring into middle-backed offices. According to different researches, the investment flows that hedge fund managers bring to the fund are chasing past performance, while the funds are becoming more and more passively managed as their age increases, and thus reducing the likelihood of maximized performances. In most of the cases, benchmarks are not completely beaten since investors will “competitively supply funds to managers and there are decreasing returns for managers deploying their superior ability”. In this regard, although the size of the fund increases (and thus, theoretically the performance of the hedge fund), the past performances might still remain below the benchmarks that investors set on the hedge fund's aims. Furthermore, the younger and more prospective the fund is, the higher performances are: in fact, funds will have higher prospects because of the ease at generating returns, which could be constrained in the future as the fund will reach an optimal size in a quick period (at least 2 years), with another factor explaining the low performance of older funds being the inconsistent and non-persistence of alphas, a factor decreased by the cummulation of performance and management fees over time. ²One study conducted about young funds (less than a year of holding) or middle-sized hedge funds revealed the following results (source: Nicole M; Boyson, “Hedge Fund Performance Persistence: A New Approach”):

¹ Ackermann C. (2002). The Performance of Hedge Funds: Risk, Return, and Incentives. *The Journal of Finance*, 54(3), 833-874

² Boyson M. (2018). Hedge Fund Performance Persistence: A New Approach. *Taylor & Francis Online*, 64(6), 27-44.

Table 9. Performance of Portfolios of Funds Formed on Independent Sorts on Size, Age, and Past Performance
(*t*-statistics in parentheses)

Portfolio	One-Year Holding Period		Two-Year Holding Period	
	Alpha	Difference	Alpha	Difference
<i>A. All Funds</i>				
Small, young, past high <i>t</i> -statistics	0.63%	0.77%***	0.67%	0.68%***
Large, old, past low <i>t</i> -statistics	-0.14	(4.10)	-0.01	(4.03)
<i>B. All but FOF</i>				
Small, young, past good alpha	0.59%	0.84%***	0.86%	0.87%***
Large, old, past bad alpha	-0.25	(3.21)	-0.01	(4.05)
<i>C. FOF Only</i>				
Small, young, past good alpha	0.60%	0.18%	0.26%	0.07%
Large, old, past bad alpha	0.42	(0.78)	0.19	(0.36)
<i>D. Breakout for All but FOF</i>				
	Security Selection	Directional	Relative Value	Multiprocess
Difference for small, young, past good alpha minus large, old, past bad alpha: one-year holding period	0.92%*** (2.31)	Too few obs	Too few obs	Too few obs
Difference for small, young, past good alpha minus large, old, past bad alpha: two-year holding period	1.19%*** (3.85)	Too few obs	Too few obs	Too few obs

Notes: Hedge funds were sorted into quintile portfolios on the basis of independent tercile sorts on lagged 12-month size (in millions of dollars), lagged 12-month age (in months), and past performance (*t*-statistic of 36-month seven-factor alpha).

***Statistically significant at the 1 percent level.

As we can analyze based on this table, we can see that there are different metrics that could impact the way hedge funds perform over time. Funds that are mainly concentrated on the strategy of funds of funds (investment strategy consisting of holding a portfolio of multiple investment funds rather than buying separately stocks) might perform better, especially into into small and young funds that have a one-year holding period.

This part brings us to the issue that is related to the type of strategy the hedge fund is adopting. While the strategy can be chosen by the scopes of the hedge fund vehicle (and managers), the macroeconomic environment, and the products that the hedge fund will invest into, a hedge fund strategy will mostly be influenced by the strategy and the way the chosen strategy is deployed. Generally, hedge funds that are sensible to macro environments (those who are either using a macro or emergent strategy), can react differently to potential macroeconomic shocks: for example, emergent hedge funds could suffer from sudden interest rate hikes or worsening CPI in the emergent country in which they do invest.

One of the most prominent cases of failed operational management of hedge funds could be the infamous example of Bernie Madoff, known for one of the biggest Ponzi schemes in his history: in effect, along with his main society (Wood River Capital Management), he didn't disclose his financials and audits with the SEC regarding a conflict of interest with its investments. 85% of the the funds were invested into a firm closs to Wood River Capital Management, which resulted in the assets being ripped off as the firm eventually declared bankruptcy.

1.4.2 - Downside risk and risk-return trade-offs:

Hedge funds can also be prone to both market and operational risks that could impact the performance of the hedge fund. As a general knowledge, the calculation of an asset or of a fund will be to determine the return standard deviation or developing a factor model related to the variations of the return, which could be, nonetheless, affected by the skewness and excessive kurtosis in the overall performance of the fund. Generally, well-adjusted hedge funds that are combining perfect weights in their analysis do display higher results, especially regarding the portfolios that might have a high alpha and optimal standard return (as per the study of Christos Argyropoulos and Ekaterini Panopoulou entitled “Downside risk and hedge fund returns”), where their top 30 alpha seem to perform in the most optimal ways.

Panel A: AR Hedge Funds 100 Million	AR	Alpha	SR	Sortino	Upside	MDD	SD	1 % VaR	5 %VaR	10 %VaR
Top 30 SR	0.561	0.381	0.401	0.649	1.161	8.717	0.984	3.858	0.864	0.255
Top 30 MSR	0.472	0.297	0.281	0.418	0.91	10.249	1.088	4.519	1.183	0.481
Top 30 Alpha	0.809	0.442	0.196	0.355	0.951	19.102	3.275	7.033	4.326	2.871
Previous “Month” Top 30 Return	1.082	0.315	0.168	0.283	0.82	35.694	5.447	11.988	8.606	5.438
Optimal SR	0.523	0.358	1.061	2.652	4.015	3.151	0.335	0.396	-0.014	-0.14
Optimal MSR	0.535	0.366	0.640	1.753	2.814	4.027	0.576	0.978	0.391	0.161

On the contrary, hedge funds that do perform in optimal regimes can also be the funds that are the most prone at being exposed to volatile risks and drawdowns. In effect, one factor that could influence the value that a fund might lose on a daily or monthly basis will be estimation of either the DEAR or MEAR (daily/monthly earnings at risk): this indicator shows how the value of the fund can be impacted the daily volatility of the hedge fund’s earnings, which can severe further the risks of the hedge fund amid uncertain macroeconomic conditions, especially for different

strategies, such as macro hedge funds (which can see their earnings being impacted by the changes in the exchange rates and daily volatility related to the exchange rates) or fixed-income hedge funds (if the value of bonds is changing, albeit more rarely and less severe, especially in the US). This can result into what we call the Value at Risk (VaR), a factor that can also be impacted by the duration and sensitivity of the hedge fund.

At the same time, liquidity and leverage constraints can also represent a major factor that might constraint the performance and management of hedge funds. Most of the assets constituting hedge funds are illiquid, while redemptions or creation of funds is more frequent due to the real nature of hedge funds (which are open-ended). At the same time, this movement could limit further investors or high-net worth individuals from participating into the hedge fund, and thus resulting into an increased liquidity risk for investors since the potential defaults could grow while the number of investors might remain stable or decreasing. At the same time, while higher leverages help funds at extracting any further returns from their different positioning (especially into long/short positions), it can also increase the margin-call risk (a mechanism that, while amplifying potential returns, may also increase the risk of substantial risks into volatile markets): this can increase the risk of forced liquidation, especially during volatility risks, as a result of amplified price movements resulting into equity losses, and thus surpassing the expected margin quotas, forcing the hedge fund through assets liquidation, and amplifying market stress into the macroeconomic environment. One notorious example includes the investment management firm Archegos Capital Management: a limited partnership managing the assets of Bill Hwang (convicted US trader for fraud allegations), the firm default on margin calls in 2021 from several global investment banks, including Credit Suisse and Nomura Holdings due to the severity of the firm's derivatives contracts and the liquidation of assets from brokers, resulting into sharp prices drop into key stocks and consequential losses into the banks aforementioned.

1.4.3 - Lack of transparency:

Because of the lower regulatory framework that is applied to hedge funds, the industry may also be prone to further biases or unreported data about the performance of hedge funds. Known under the phenomenon of intentional returns smoothing, hedge fund managers tend to “misreport” returns so that investors can be attracted to invest into the hedge funds, while also increasing cash inflows and decreasing outflows, including negative macroeconomic regimes. Because of this lack of transparency preferred by managers, manager will use this practice to increase the performance fees they could earn, which is also helped by the peculiar fee structure of hedge funds. Furthermore, another form of dishonest practices could also be explained by the different biases that exist among hedge fund managers:

- Survivorship bias: this represents a tendency to view the performance of existing stocks or funds as a “representative comprehensive sample” without taking into account the non-performant stocks. This also represents the probability that an investor will misreport the gains within the hedge fund based on the published investment fund return data. This

behaviour is mainly manifested since the investors of the hedge fund will be directly impacted by the announcements of the fund's closing.

- Backfill bias: this bias manifests when a new fund's past and strong returns is included in the latest dataset of the fund. This bias is manifested in the same way and for the same motives as the survivorship bias, since managers could cash out on further performance fees from the hedge fund. This bias tends to "inflate" historical investment returns, misleading investors into aggregated returns, while having lower returns in the real life.

Some researchers do advocate for the introduction of further regulations to limit the potential risks or biases reported within hedge funds, especially regarding the potential liquidity risk that is associated with the lack of transparency. This is mainly advocated by researchers ³Viral V. Acharya, Lasse H. Pedersen, Thomas Philippon, and Matthew Richardson in the article "How to calculate systematic risk surcharges", in which they would implement a regulatory system fighting against systematic risk, a problem that is recurring within the field of hedge funds due to a lack of capitalization or accredited observation of the real returns yielded by hedge funds. Furthermore, the institution should face a "surcharge" that is based on the extent to which the hedge funds contribute to both their own systematic risk, and the market risk in the macroenvironments in which they do operate.

II – Literature review:

2.1 – Major players and hedge fund firms:

2.1.1 - Citadel:

⁴Citadel is the most well-known hedge fund company in the world, with a total AUM evaluated roughly at US \$65.9 billion as per 2025. Founded by the hedge fund manager Kenneth Griffin in 1990, the firm has a different range of activities. One of the most prominent is managing different investment strategies, ranging from equities, commodities, or fixed income. Moreover, the firm provides reinsurance and risk management mechanisms by allocating risk capital, handling liquidity management, or hedge against excessive stress exposure to potential adverse movements in the market. In this sense, Citadel is known for calculating sometimes more than 500 scenarios that could represent the risks a firm may face amid a negative market outlook.

2.1.2 - Bridgewater Associates:

⁵Bridgewater Associates is based in Connecticut, and was created in 1975 by Ray Dalio, a former hedge fund manager, and one of the first prominent managers within the industry. The firm is known for pioneering the early investment strategies into hedge funds through different aspects,

³ Viral V. Acharya, Lasse Heje Pedersen (2005). Asset pricing with liquidity risk. *Science Direct*, 77(2), 375-410.

⁴ Citadel. (2025). Citadel – Who we are. <https://www.citadel.com/who-we-are/>

⁵ Bridgewater Associates. Bridgewater Associates. <https://www.bridgewater.com/>

such as the currency overlay (risk management of currency risks that is separated from the underlying portfolio in order to protect investors and the portfolio from exchange rates changes on global markets), separation of Alpha and Beta strategies while evaluating the overall risk of firms, as well as risk parity. Products proposed by the firm include:

- **Pure Alpha:** called by the firm a “diversified alpha source”, it is a package that enables investment opportunities across different asset classes.
- **All Weather:** launched in 1996, this strategy aims at keeping the focus on global fixed-income securities, and diversifying any risks related to bonds by putting long and short positions. The goal was to create adjusted-risk returns aimed at surpassing the global market returns.

2.1.3 - AQR Capital Management:

Unlike the two previous companies,⁶ AQR Capital Management aims at utilizing quantitative analysis to develop strategies related to equities and alternatives. This comes as quantitative analysis allows traders to use mathematical and statistical concepts to analyze financial data, especially for forecasting potential future market trends, or the movement of stock prices. Just like Bridgewater Associates, AQR focuses on diversification in its portfolios, while also adding strategies involving low correlation to traditional assets. While being mainly focused on alternative investments, the firm has also recently specialized its services into alternative mutual funds, with a combination of both mutual funds’ principals and alternative financial instruments.

2.2 – Type of hedge fund investment strategies:

As the world of hedge funds is regrouping different actors and perspective markets, different strategies do exist that are incorporating different techniques common to hedge fund investors and managers. Although there are many hedge fund strategies that are heavily used within the industry of hedge funds, the ones that will be at the core of our research will be Macro-based Hedge Funds, as well as Event-Driven Hedge Funds to a lesser extent.

2.2.1 - Macro-driven:

This category of hedge funds refers to all hedge funds that aim at mixing the trading strategy with the different macroeconomic indicators that could influence the position of different stocks, while also diversifying the portfolio by investing into stocks from different markets that could be either established or emerging. Just like the equity-based strategy, this sphere will also serve as a main pillar supporting the completion of this work. The two common strategies present will be:

- **Global macro:** often being the most common strategy in the macro-driven ecosystem, the global macro is the possibility to use leveraged bets on anticipated price movements of stock markets, interest rates, FX, or commodities. Global macro is positioned around a particular outcome to either predict a potential macroeconomic outcome, or the final

⁶ AQR Capital Management. (2025). Our Approach. <https://www.aqr.com/What-We-Do/Our-Approach>

aggregate that is imminent to arrive within the market in which the global macro is operating. These funds offer investors exposure to high-level bets that are regrouping assets and other financial instruments, offering a diversification that can help investors at hedging against global financial events, while also comprising higher fees.

- Emerging markets: this strategy, which has a smaller scale of operation, is focused on investing a major share of portfolio in securities of companies or sovereign debt of developing or emerging countries/markets. The investments are primarily long, as investors are expecting long gains in markets that have the potential to have a sizeable influence on the long-term. It has a variety of financial instruments, including stocks or bonds from the emerging markets. These markets seek to capitalize on the return opportunity presented by the attractiveness of the newly incoming markets.

2.2.2 - Event-driven:

In this category, hedge fund managers will take advantage of specific announcements that could occur both on a corporate or macroeconomic level to generate value by taking advantage of temporary stock mispricing at any point in time that is related to a specific macroeconomic or corporate event. The main strategies included within the event-driven sphere would include:

- Distressed securities: in this strategy, managers will aim at investing in companies facing distressed situations, such as bankruptcies, insolvency, or restructuring phases. The firms can put long positions onto firms that are expected to recover from their distressed situation or who might not face a difficult situation, while short positions are put into firms expected to face serious or difficulties at recovering in the long-term.
- Merger arbitrage: this strategy is the most common event-driven, as the main aim for managers will be to generate returns by going long on the target company and short on the acquirer company's stock, expecting a decrease in the value as M&A's long-term effects will be visible. Unlike other strategies, the merger arbitrage approach is heavily influenced by diverse factors, such as the stock price movement of both the target and acquirer company, as well as the type of deal that is concretized between the 2 sides: as an example, if the acquirer firm offers an all-in-cash merger, investors would generally aim at opting for long positions in the target firm. In a stock-to-stock merger, the arbitrageur would buy shares from the target company's stock and shorten the shares into the acquiring firm, the short position is converted, and thus a profit can be yielded.
- Activism: while being more unusual due to its implication, this approach is used by an investor aiming at obtaining representation in a target firm's board of directors to ensure future control over the firm's company policy and strategic direction. The investor would first commence to start accumulating a small stake in the public firm's equity. Once the stake reaches at least 5%, the investor would use the legal filling from the US government to announce their respective position and start a public campaign. The investor would further release a presentation about the firm's problems and the hedge fund's main purpose in fixing those problems, while also contacting other shareholders or

members of the board of directors to propose the strategy so that the investor can win votes within the Board of Directors to have its strategy implemented.

2.3 – Fundamental concepts – portfolio behavior:

Different concepts and statistical variables have enabled macro-hedge funds to assess their performance while also adjusting their risk-adjustment strategies across different macroeconomic environments.

2.3.1 - Beta:

Market beta is another indicator that is crucial for finance investors. It indicates the price volatility of a stock or another financial asset in comparison with the general market and thus displaying the activity of a security return's as it responds to the swings within the market. Defined initially within the ⁷Capital Asset Pricing Model (CAPM) concept by Sharpe (1964), an investor will use the Market Beta to analyze how much risk the investor will add to their portfolio by ensuring that a specific stock will be compared to the right benchmark, as well as having a relevant R-squared value relative to the benchmark. The estimation of the beta is made by following this formula:

$$\beta = \frac{\text{Covariance}(R_e, R_m)}{\text{Variance}(R_m)}$$

With:

- Covariance (R_e, R_m) = how changes in the returns of one stock (R_e) are related to the changes of the market returns (R_m).
- Variance (R_e, R_m) = how far the market's data is spreading out from their average value.

As a main interpretation of the beta, a beta whose score is equal to 1 means that the price of the stock perfectly correlated with the market returns, and thus showing a stability in the returns that the market portfolio would generate. Meanwhile, a beta whose score is superior to 1 will bear a higher amount of market risk given the higher volatility to which the stock will be exposed while also increasing the potential of excessive returns. Lastly, a market beta with a score less than 1 will show that the individual stock will be less volatile than the market returns and thus diminish the potential for excessive returns while also increasing the likelihood of risk-aversion for the market portfolio.

At the same time, both the systematic risk and unsystematic risk will be used for the evaluation: the first one represents the risk of the entire market declining while the other one is associated with the individual stock or industry of the stock, which can be mitigated through diversification.

Nonetheless, the interpretation of the beta regarding hedge funds could be more different. In fact, given the sole inclusion of systematic and unsystematic risks, betas will fluctuate more often for

⁷ William F. Sharpe (1966). Mutual fund performance. *The Journal of Business*. 39(1), 119-138.

hedge funds since these funds will generally include dynamic trading, continuous change of dynamics and asset allocation, as well as higher active management, showing that betas may fail at capturing the whole essence of the market volatility to which hedge funds are exposed to.

2.3.2 – Alpha – CAPM:

The Alpha metric is one of the most important metrics used by hedge fund managers regarding the assessment of the hedge fund's performance. Described more commonly as the ⁸Jensen's alpha (α) in 1968, the alpha is used to measure the "risk-adjusted" return of a security from the CAPM (Capital Asset Pricing Model) concept which has been defined by ⁹Treynor (1961), and ¹⁰Sharpe (1964). It also calculates the excess return generated by the portfolio over the expected return, which is obtained from the CAPM, which is used for most of the assets, including stocks and bonds. The general calculation formula for the Alpha would be:

$$\text{Alpha} = R(i) - [R(f) + \beta * (R(m) - R(f))]$$

Where:

- $R(i)$ = realized return of the portfolio or investment.
- $R(m)$ = the realized return of the estimate market index.
- $R(f)$ = the risk-free rate of return in the period of evaluation.
- β = beta of the investment portfolio relative to the chosen market index.

In a more general way, the alpha can also be interpreted as a way for financial security to outperform or bear the market's overall performance, which is crucial for hedge funds who are seeking excessive returns. For non-efficient markets, managers and investors will usually spend time finding the assets with the highest and most stable number of positive returns. Within the statistical framework, alpha has been used as the regression intercept estimate with ordinary least squares (OLS) as an estimate to see the opportunities of returns for hedge funds, albeit not always concrete given the returns being not normally distributed, with risks of skewness (the asymmetry of the statistical model compared to the tail directions of the distribution) or kurtosis (how outliers are present within the distribution model) being often present, and thus leading to potential biases regarding the interpretation, or the accuracy of the alphas for hedge funds.

The concept around the traditional CAPM might be limited, however, in the context of hedge funds. Most of the factors integrated by the CAPM, especially regarding the implied risk-free rate return and risk-free rate are too static, and base themselves on the stability of the existing macroeconomic environment, a concept that is not compatible with the core of modern hedge funds that rely mostly on the spikes and movements that potential variables might experience for driving their strategy. At the same time, the model doesn't integrate multiple factors that might

⁸ Jensen. MC (1968). The performance of mutual funds in the period 1945-1964. *The Journal of Finance*. 23(2), 389-416.

⁹ L. Treynor (1961). Market value, Time, and Risk. *SSRN*,

¹⁰ William F. Sharpe (1966). Mutual fund performance. *The Journal of Business*. 39(1), 119-138.

explain the truth about the behavior that hedge funds do espouse, as both the market premium and risk-free are only capturing the surface-level macroeconomic level of an environment, and thus not explicitly mentioning or searching for the underlying factors which may influence hedge fund returns.

2.3.3 - Multi-factor alpha model (Fama-French Model):

While alpha and beta have been usually used as the main reference for investors when assessing their profit opportunities and risk-adjustment, these 2 factors might not always be suited for the functioning or the assessment of the way that hedge funds function, often characterized as an insufficient tool to incorporate and assess all the variables affecting the performance of hedge funds, with the alpha being described as being a “static” tool that can only function with a precise amount of tools that are directly to the market risk. In this perspective, the work of ¹¹Fama & French (1993) through the multi-factor model, which is often described as a reviewed CAPM.

A general model as per defined within the work of Fama & French would show the multi-factor model as a linear regression defined as such:

$$R_i - R_f = \alpha + \beta_1(R_m - R_f) + \beta_2 \text{ SMB} + \beta_3 \text{ HML} + \varepsilon.$$

Where:

- R_i : total return of a stock or portfolio.
- R_f : risk free return of the stock or portfolio.
- α : alpha of the whole aggregates.
- R_m : total market portfolio return at the aggregated time.
- SMB: size premium of the market.
- HML: value premium of high and low stocks.
- ε : margin of error.

Although the Fama & French model tried to explain the way some other macroeconomic factors could impact the whole return performance of market portfolios, most of the constructs and variables used are hypothetical and fictional, based mostly on the observations that have been made within the targeted industry. By integrating the size premium of the market (SMB) and the value premium of high and low stocks (HML) across an industry, analysts have managed to capture some underlying factors that could explain the behavior that equity funds adapt during specific macroeconomic conditions.

¹¹ Fama, E.F., & French, K.R. (1993) – Common risk factors in the returns of stocks and bonds. *Journal of Financial Economics*, 33(1), 3-56.

According to traditional research around the Fama & French model, the idea of including the amount of size premium and value premium for the observation of returns in a specific industry.

However, the Fama-French model presents its simple limitations in its usage for evaluating the financial performance of hedge funds. In fact, although the Fama French adds substantial underlying factors explaining performance, it is mostly retained around the idea of equity portfolios which do not integrate specific variables such as volatility, as the core of the model assumes that higher market beta will bring in higher returns, which hasn't been proven as a factor that correlates highly with the way hedge fund behave, as most of hedge funds will bet on adverse movements compared to the market beta, and thus making the correlation relationship at -1 between the two factors. Furthermore, just like the CAPM, Fama French represents a statistical model which will regroup statistical figures from the reported data, a principle that is incompatible with the core of hedge funds, which is about capturing every variable part of the active management structure such as short selling, derivatives, or long/short positions structured. Lastly, the Fama French is mostly adapted to cover the behavior and performance across specific industries or market sections, something that is not always the case for hedge funds, which regroup specific asset classes that have each a specific weight allocated within the hedge fund and that may represent characteristics different from traditional equity portfolios.

2.4 – Hedge-Fund Multi-Factor Models:

Based on the previous sections, the main research highlights the actual limitations that the Fama-French model implies for hedge funds, which can be demonstrated through the lack of observed relationship between the model's macroeconomic variables and actual underlying economic factors that might explain the behavior of hedge funds. Other works, on the other hand, have demonstrated that actual non-linear variables that shape our macroeconomic environment could drive some explanations about how hedge funds might react to potential macroeconomic shocks.

2.4.1 - Fung and Hsieh (2001):

One of the main works that have explored the underlying causes explaining the behavior of hedge funds is William Fung and David A. Hsieh's work (2001) that explores the amount of risk that underlies around hedge fund strategies. By trying to build a multi-factor that follows hedge fund return dynamics, the two authors have integrated specific straddle options, a strategy that bets on the idea that an underlying asset or position would experience a significant price movement, especially by relying on high movements in the position's volatility, which is expected to increase significantly.

Thanks to this article, Fung and Hsieh demonstrate that trend-following strategies based on macroeconomic underlying factors will yield returns higher than traditional equity portfolios, with the lookback straddle being the main driver. The latter represents an options strategy paying off based on the maximum price movement of the asset across a period, showing a non-linear and state-dependent condition for the fund itself. Another crucial aspect of the article is the idea

of having underlying macroeconomic factors that translate into direct hedge fund performance, with their research demonstrating the main limit of the Fama French model due to $^{12}R^2$ values being close to 0 during regressions due to the different nature of the variables included within the Fama French Model, which shows the distinctive difference between equity portfolios and specific macro hedge funds. The idea of adding underlying macroeconomic factor is justifying the usage of specific macroeconomic factors for this research, as well as the conduction of OLS models across structured regimes.

The idea of Fung & Hsieh (2001) shows the non-linear and static nature of macro hedge funds, which is driven by unexplained yet indirect underlying macroeconomic factors that explain the performance and dynamics of hedge funds.

2.4.2 - Fung and Hsieh (2004):

Another research from Fung and Hsieh (2004) explores the evaluation of traditional hedge funds benchmarks by taking a risk-based approach. For this article, the two authors are seeking deeper from the asset-based factors that had been included in the first paper research, with the inclusion of specific arbitrage pricing theory and dynamic risk-beta coefficients that could translate into a further understanding of volatility's impact on hedge funds, as well as the risk mitigation mechanisms that hedge fund managers may utilize, all while trying to counter potential hedge fund limits, ¹³including the lack of transparency, presence of "survivorship bias" in the hedge fund data, as well as the choice of index weights, given that hedge fund managers would generally skew the weight attribution among top-performing positions in the hedge fund. By including different other factors, including bond market risk, credit-spread risk, or the observation of trend factors in the macroeconomic environment, the two authors realized that specific macroeconomic factors and risks might be the actual factors that do potentially explain the behavior of hedge funds, unlike the size of the market or the returns of funds in specific industries, as previous research around the Fama & French model have demonstrated.

By developing specific structures around variables reflecting macroeconomic conditions, Fung and Hsieh enhance their previous research by including ¹⁴specific factors such as bond market risk, credit risk spread between high yields and high investment grade bonds and FX markets, with the model signing a different framework to the purpose of Fama and French model, with the developed model searching for the implicit factors explaining the exposure degree and level of performance that these exposure degrees will result in the performance of hedge funds. Their

¹² Fung, W., & Hsieh, D. A. (2001). The risk in hedge fund strategies: Theory and evidence from trend followers. *Review of Financial Studies*, 14(2), 313–341.

¹³ Fung, W., & Hsieh, D. A. (2004). Hedge fund benchmarks: A risk-based approach. *Financial Analysts Journal*, 60(5), 65–80.

¹⁴ Fung, W., & Hsieh, D. A. (2004). Hedge fund benchmarks: A risk-based approach. *Financial Analysts Journal*, 60(5), 65–80.

idea is also to develop trend-following factors that are based on the most direct underlying macroeconomic factors that will influence hedge funds' exposure to market risks, as well as standard equity factor which corresponds to the article's search around market drivers during tight monetary policies or macroeconomic uncertainty.

Their results demonstrate that underlying factors can explain a large majority of the hedge fund returns, up to ¹⁵50 to 80% of fund's index proxies, explaining that the power of underlying factors on hedge fund behavior may justify the attribution of specific weights and classification of macroeconomic variables for this article.

2.4.3 - Ang and Timmermann (2012):

The last significant research shaping multi-factor models and their application towards hedge funds is the article of Andrew Ang and Allan Timmermann, an article that mentions for the first time the idea of regime-switching models that can capture the way markets are swinging based on macroeconomic variables, stating clearly that a macroeconomic environment could remain for a certain period given the consistency or the observation of specific macroeconomic variables. Asset return dynamics are directly linked to the way that underlying macroeconomic factors vary across prolonged periods, and not heterogeneous periods that might extend for a long time without capturing the essence of factors driving the macroeconomic environment. Thanks to the evaluation of specific regimes by evaluating returns, volatility, and cross-regime analysis shows that the changes are not sudden but gradually occurring as underlying factors are changing over time, all while finding that a macroeconomic regime might persist across time before transitioning to a new context.

By demonstrating asset return distributions thanks to the usage of returns, volatility and variance evaluations across different regimes, this article adds another dimension to the evaluation of hedge funds, which is about the dynamic nature and changes that directly impact the hedge fund's performance in time. It represents a structural change for other asset pricing models since usual models would yield generic weighted averages and coefficients that might not capture the changing dynamics of regimes or underlying macroeconomic factors which will enable to capture means, variances and cross-regime analysis will structure the OLS methodology that will be applied and presented shortly in this paper research.

The dynamics around transition regimes also represent another major indication, with the two researchers finding that transitions are not sudden or spontaneous, with different macroeconomic regimes being observed for a significant period, while periods of transition are needed before changing the existing regime. The main signals are shown by the changes in cross-asset correlations, coefficients reported for hedge funds, as well as indicators around volatility or

¹⁵ Fung, W., & Hsieh, D. A. (2004). Hedge fund benchmarks: A risk-based approach. *Financial Analysts Journal*, 60(5), 65–80.

inflation, which are also crucial to the structuration of this current research. The point to prove is that classification methods or specific interpretation of macroeconomic factors is crucial at understanding the main factors that could cause shifts in the performance of hedge funds, all while understanding whether the changes in those underlying factors are also implying prolonged effects to the existing regimes that are applied to the funds, instead of relying on a single or multiple static indicators as it has been shown across the CAPM or the Fama French Models. Regime transitions and their probabilities of switch/stay momentum are also important from the researcher's perspectives as every movement is not static or spontaneous, but rather a result from the existing and changing macroeconomic conditions, which is crucial for the optimization of portfolio weights in hedge funds as it will be portrayed later in the data analysis of the conducted research of this article.

The dynamic around cross-asset correlations during specific regimes are also another major point that is proving as a catalyst to the research of Ang and Timmermann. In this context, correlations or specific dynamics might see disruptions when macroeconomic conditions worsen, which is contradictory to the previous beliefs among researchers when investors or analysts would assume a linear and perfect relationship between assets during normal conditions. This justifies further an outlook around specific regimes during this research, all while proving the calculation of specific regime-dynamic probabilities that will seek to evaluate the way a regime's dynamics can also affect the correlation between assets and hedge fund strategies. This is crucial since the final argument for Ang and Timmermann is that an investment outlook for hedge funds should be structured rationally through the knowledge of the existing macroeconomic regime, which would eliminate in theory existing biases such as the survivorship bias. In this sense, structured portfolios would have a natural functioning, all while recalibrating its performance in the horizon as conditions are changing.

2.5 – Mechanisms of return generation and risk mitigation:

Based on the three major models that have been defined through the presentation of the three major paper articles above, we can continue further the nature of the hedge funds that explains some of the mechanisms that will be mentioned further.

In fact, one of the key characteristics aforementioned of hedge funds (and especially domestic hedge funds) is that they are largely unregulated. This is mainly explained by the fact that most hedge fund firms are mostly limited partnerships with fewer than 100 investors accounted, exempting most of the hedge funds from the Company Act of 1940, and allowing thus hedge funds to keep their investment strategies flexible. Hedge funds are using different simple mechanisms which allow them to reduce systematic risk, including:

- ¹⁶**Short selling:** short selling (known commonly as short position) is a trading strategy that is used by investors to speculate on the stock's positioning and future movement on the market exchange. The investors will bet on a decrease of the stock's price, who have to open a margin account with a broker and pay interest on the value of the borrowed shares while the position is open. This concept is mainly integrated into the primary objective of hedging, which is the protection of speculation, since opportunity costs and operational costs could be expensive for the hedge fund.
- **Long buying:** on the other hand, the long buying (also known simply as long position) represents the situation when an investor expects that a stock (or portfolio composed of individual stocks) will see an increase in its market price, and betting on further future growths. Investors can either have a long position by holding the investment, or simply trying to lock-in futures contracts to hedge against price movements from short-positioned stocks. This can limit the potential systematic losses and locking the price, while also being more sensitive to any sudden hike or nadir point in the stock prices of the portfolio.
- **Financial leverage:** this practice represents borrowing money, investing into funds, and planning for future returns to be greater than the debt funding which is incurred within the potential costs. Regarding hedge funds, several forms of leverage are used; this would include credit lines (trading on margin but by borrowing directly from a third-party lender and not a broker), derivative trading (especially with the inclusion of option to capture further profits), or buying on margin (with a margin account being borrowing money from a broker to invest into specific securities).

Another core feature that is important regarding the mechanisms employed by hedge funds would include the usage of high-frequency dynamic exposures, which is measuring the speed at which hedge funds alter their investments given the existing macroeconomic environment. Throughout different performance evaluations, hedge funds may have variations that are occurring within a monthly basis, with the performance being measured on a daily basis for certain hedge funds. The norms of having long and short positions can also vary very specifically for certain short-term instruments in which hedge funds are investing, and thus exposing furthermore the idea of having a potential macroeconomic influence over certain hedge funds. Other models may also support the idea of a certain exposure and influence of the periods, especially regarding the daily observations that could occur within specific months or days: in effect, the daily exposure could seek higher values for certain periods (and especially the beginning of the month, followed by a calm and then a potential decrease in the returns given the occurring of different economic announcements, such as the FED's Interest Rate decisions), while certain models being more or less affected by the phenomenon and the correlation to daily and monthly events, including equity hedge and directional hedge funds, and thus revealing a

¹⁶ H. Jacobs, S. Muller (2020). Anomalies across the globe: Once public, no longer existent. *Journal of Financial Economics*. 135, 213-230.

consistency in the potential operational constraints or strategic reporting incentives given to hedge fund managers.

Lastly, there is also a conception that hedge fund managers can display a certain amount of threshold behavior, with de-risk strategies occurring when volatility or illiquidity spikes, or in response to significant market events. One of the most observed trends is that hedge funds will cut their exposure into smaller stocks, with different documents such as ¹⁷Brunnermeier and Nagel (2004) are providing, and who document that hedge funds are sensible to certain industries that could be impacted by major shocks (such as the early technology years, and especially during the bubble crash in 1998). At the same time, when the S&P 500 volatility is increasing significantly, hedge funds cut their positions into risky assets.

One other major point that defines the way hedge funds are seeking their gains will be through the employment of liquidity management strategies. The main core concept surrounding hedge funds is that liquid funds can still be liquidity-risk takers, with the liquidity being highly exposed by monthly redemptions, with fund managers agreeing that gates (aka the characteristics of the hedge fund that they're taking part to) protect investors as they permit funds to liquidate in an orderly fashion and avoid selling assets at fire sale prices, or when the fund has an undercapitalization process, with the focus being to pursue the business operations rather than offering a protection for investors, with one of the main risks being the asset-liquidity mismatch risk, a risk that could accelerate the demise of hedge funds that are experiencing difficulties and make them susceptible to further losses. This mechanism can also be explained through different theories, with the losing spiral / funding liquidity logic showing that the initial losses within a hedge fund would encourage managers to seek for investor redemptions (that is enabled, given the close-end fund nature of hedge funds) that could lead to forced sales, which could yet train to more losses and more withdrawals.

The idea of gates regarding liquidity management reveals different interesting concepts:

- Flow shocks: the sudden shocks that could occur into the macroeconomic environment reveal the potential performance, and the spreading of the so-called fire-sale effect, in which investors are obliged to redeem their shares or positions within the hedge fund, and thus creating potential discount or premium risks for the hedge fund. Within most of liquid hedge funds, there is a certain linear relationship between the spread of the systematic risk and the fund's performance, with potential similar movements occurring in the hedge fund's performance regarding how both market and systematic risk manifest themselves.
- The effect of flow seems to be stronger into certain segments and parameters. Generally, high inflow hedge funds might perform better than low inflow hedge funds, with other factors such as the fees incurred being also responsible for the performance of the hedge

¹⁷ Brunnermeier, M.K. & Nagel, S. (2004). Hedge funds and the technology bubble. *Journal of Finance*, 59(5), 2013-2040.

funds. The effect has also been noticed when the market liquidity (the ease with which assets can be traded), which is dependent and relied to the funding liquidity (the ease with which traders can earn their financing). Furthermore, hedge fund flows are more impactful when economy-wide funding liquidity is tight, with different parameters such as the TED (Treasury-Eurodollar) spread is low.

- Another factor could also be the financial leverage used by investors or hedge fund managers. In effect, the more financing is required for ensuring the right flow of funds within the fund, the further asset liquidation risk could be increased. Although leverage can be useful for normal macroeconomic situations, as it helps the fund to strengthen further its returns, it can also transform itself into a pain when investors are forced to sell their assets.

Furthermore, more and more research have demonstrated that hedge funds may tend to display potential liquidity mismatches, which may be similar to the asset-liability mismatch concept described earlier. These mismatches can be found among larger funds, funds with lower leverage, funds with greater personal stakes employed by investors, or unused cash borrowings. The mismatch can be described under three main concepts:

- Asset illiquidity: this represents the level of illiquid assets, which are securities or properties that aren't sold easily, or who cannot be exchanged for cash without incurring any significant losses. It can also represent the time it takes to liquidate the fund's portfolio.
- Financing illiquidity and investor illiquidity: this represents the way that the illiquid assets will be acquired and managed within a specific liquid hedge fund, as well as the amount that creditors and investors have committed their loan and equity facilities to the hedge fund.

The effect of liquidity mismatch can worsen for those illiquid hedge funds when the hedge fund is confronted with financial stress, or when the interest rates of a macroeconomic environment worsen. At the same time, volatility factors such as a surge into the VIX (Volatility Index) or dip into the DXY (Dollar Index) can also accentuate those effects. To counter those effects, hedge fund managers will often employ two major tools:

- Unencumbered cash: they allow the fund to fill investor redemption orders without having to liquidate their non-cash assets
- Unused borrowing capacity, which is coming essentially from undrawn credits or margin capacities.

The last part we will talk about will concern the capital allocation, and the flexibility capacities that hedge funds have at their own disposal. Unlike most of funds, hedge funds are not fixing their capital allocation upon the fund's launch, but will rather adjust the capital and asset allocation based on the different metrics existing into the macroeconomic environment, such as the decisions intended by the FED regarding the interest rates, the movement of the VIX (which

should remain under a total of 20 points for a stable and non-volatile environment), or the DXY (with a Dollar Index above 100 indicating a strong economic momentum). At the same time, the potential systematic risks, funding conditions, focus of investment, or the participation of investors are also crucial factors determining how hedge funds will allocate their capital.

For example, as we have seen in the previous sub-section, macro hedge funds are crucially impacted by the potential movements of any macroeconomic aggregate. A hedge fund could change the allocation of the assets, or the national bonds in which the funds invests actively given any potential negative movement in any crucial macroeconomic indicator.

Furthermore, the main focus for hedge funds when allocating their capital will not be to follow raw returns, but rather recent risk-adjusted performance. In effect, investors will learn about the managerial ability and will direct further capital into performances that have been steady and performative across their allocated period. The risk-adjusted performance can be heavily influenced by specific macroeconomic aggregates or financial instruments, such as the excess returns of the S&P 500, the evolution of the FX, or bonds as an example. The allocation of weighted alphas, or weighted assets can also influence the allocation, making the risk exposure changing over the period, and thus the capital allocation at the same time, with alpha-only funds (funds measuring how much a mutual fund has outperformed or underperformed its benchmark index after adjusting its market risk) benefiting from a greater protection against potential macroeconomic exposure, unlike beta-only funds (funds that are betting on how volatile a market can be).

Capital allocation can also be strained by different conditions, especially regarding financial leverage and volatility present across the macroeconomic cycle. Because of the different risks occurring, a financial leverage used by hedge funds won't be constant, but will also move based on factors similar to the way capital is enforced within the hedge fund. Leverage is cut during highly-volatile cycles, as a high risk leverage could result into further incurred losses, while the capital is shifted away from relative-value trades towards further liquid exposures, as the arbitrage opportunities allow the funds to have a greater flexibility over the capital's deployment. Furthermore, these macro and leverage constraints could be reinforced by existing balance sheet constraints, especially in situations where assets could be distressed or illiquid, and thus making hedge funds more sensible regarding their capital allocation during intensive regimes. This can lead investors to adopt a potential capital allocation angle in which capital contraction is enforced so that the fund is less exposed and allocated to potential volatile assets. This can lead a better protection for certain hedge funds, especially the ones impacted by macroeconomic events or corporate events such as macro hedge funds or M&A hedge funds.

2.6 – Hedge funds and impact of macroeconomic cycles:

As we have mentioned throughout the previous sections, hedge funds are mainly affected by how the macroeconomic environment is evolving, with a macroeconomic regime being composed of the amount and evolution of volatility, the available liquidity within stock exchanges, the access to different fund means, and how leverage can be used by hedge funds to enhance the further long-term returns. Different macroeconomic regimes can exist, and hedge funds have behaved in different ways depending on how each parameter of the existing macroeconomic regime.

2.6.1 - Expansionary or low-volatility regimes:

An expansionary regime is the situation where a market is characterized by the ease of restrictive economic policies or measures, such as the presence of austerity measures or high interest rates. This is mainly coming mostly during periods in which the economic growth is steadily high, as well as when different indicators are showing signs of recovery on the long-term, such as the decrease of the inflation rate, the increase of the Dollar Index, or the ease of market interest rates. Based on the research of Ang and Timmermann (2012), those regimes face a low amount of volatility, all while showing normal returns, with betas being stable, and the correlation between assets are stable and low. Any form of adverse movement is cancelled out given the stability of the macroeconomic factors, while mispricings between different asset classes is low, giving a low opportunity for funds to generate alpha while putting different hedge positions.

This is also resulting into stock exchanges soaring their returns, which also boosts the investments of different investors and funds. In this context, the volatility is lower as the market becomes more stable, while the level of uncertainty is decreasing significantly. One of the main indicators evaluating the level of volatility represents the Volatility Index, which represents a real-time market index measuring the market sentiment and perceived risk, with the evaluation being concluded over the volatility regime of the S&P 500 over the last 30 days. The more dramatic the price swings within the S&P 500 occur, the higher the Volatility Index will appear. In periods where lower interest rates get announced (such as potential cuts of the FED), or higher-than-expected increases into the GDP, the Volatility Index can decrease, meaning that the confidence of investors is higher over the general market. On the other hand, political events could also trigger a positive interpretation of the Volatility Index, such as the election of Donald Trump as the next US President: in effect, the Volatility Index's value saw consecutive decreases, which were attributed to the hopes of investors for a better macroeconomic environment, as it can be seen in the following picture, which traces the evolution of the Volatility Index following the election of Donald Trump:

ov 14, 2024	14.31	14.17	14.32	13.59	+2.07%
ov 13, 2024	14.02	15.09	15.26	13.77	-4.69%
ov 12, 2024	14.71	15.09	15.37	14.69	-1.74%
ov 11, 2024	14.97	15.33	15.56	14.89	+0.20%
ov 08, 2024	14.94	15.13	15.33	14.66	-1.71%
ov 07, 2024	15.20	15.86	15.86	15.13	-6.58%
ov 06, 2024	16.27	16.06	16.82	15.44	-20.60%

In this case, the main objective of hedge funds won't be to protect investors against potential macroeconomic risks, but rather to collect a higher amount of alpha. Defined as the active return on an investment compared to a benchmark or market index, the goal for the hedge fund will be to spur a further return that is above the average market return during the regime, or compared to other traditional hedge funds during similar periods. Furthermore, in this period hedge funds will decrease their capital allocation into risky assets due to the unnecessary and costly exposure to the long positions that are usually hold over risky assets within the portfolio. And yet, the general findings indicate that the strategies of alpha are lower when capital flows into the strategy is high, evidenced by the findings of ¹⁸Naik (2007 – “Capacity constraints and Hedge Fund Strategy Returns”), which in return will decrease future returns, especially under heavy arbitrage regimes, while also potentially boosting the percentage of incentive and management fees within the hedge fund. Nonetheless, some specific strategies, such as fixed incomes would see significant losses on their value, given their risk-adverse approach that is constrained by the continuous erosion of the strategy-level alpha of hedge funds. During a low-volatility regime, hedge funds can use different leverages that are cyclical: in fact, certain leverages, such as long-only leverages (putting long positions per share of the fund's NAV), while others could benefit, as it is the case of net leverages which will examine the difference between long and short positions per share, mainly due to the low risk carried by short hedges. Given the different adjustments of investors, leverage will be highly used by investors, strategy that will enhance further the returns of the hedge funds, albeit the possibility of having a fragility of the returns and NAV costs on the long-term run.

Lastly, the main focus of hedge fund investors will surround mostly hedge funds that could be positively impacted by a positive signal coming from the macroeconomic and business environment. In effect, one of the main hedge fund strategies that will be privileged will be

¹⁸ Y. Naik. T. Ramadorai (2007). Capacity Constraints and Hedge Fund Strategy Returns. *European Financial Management*, 13(2), 239-256.

relative value: because of the tighter spread between the positions undertaken by the hedge fund, the returns could be enhanced since pricing discrepancies can be exploited further. However, the main risk is that multiple hedge funds could adopt a similar approach within the market environment, and thus being exposed to the risk magnitude of adverse movements coming from other investments or economic aggregates. Event-driven and macro strategies can also benefit during a low-volatile regime: although the returns could be lower in most of the cases due to the stable nature of the macroeconomic aggregates, the ability of having a successive amount of deals can increase the liquidity flow of the funds.

2.6.2 - Tightening or rising-rate regimes:

The second defined regime that has been explained across the recent years is the tight regime, which is the situation when the macroeconomic shows signals of an eventual decline which could affect the growth of the economy, and the tightening of stock exchanges. In this context, central banks and authorities will play a key role, as they could decide on the raise of policy and interest rates, and thus restricting the ease of funding or diversification of portfolios.

Furthermore, the uncertainty within the market could increase further, with the Volatility Index generally starting to rise slightly above the general yearly average, while the Dollar Index might see a decrease, as the Dollar could weaken its positions against other currencies amid fears of a surge within the inflation rate.

The most interesting idea is that transition regimes are not coming suddenly: according to the Ang & Timmermann article (2012), transitions between the different macroeconomic regimes is not synonymous to the rate of change in the different underlying macroeconomic factors, but rather a process that takes time. As we can link to the VIX movements, their findings have shown that volatility experiences some spikes but which are not prolonged over time, correlations between assets starts to erode but not gradually, while the risk premium on equity positions might increase, which might translate into a lower power of currency, which may coincide with the potential small decreases that could be observed on DXY's level.

In this context, a hedge fund won't use its traditional tools or strategies to benefit from potential market discrepancies, but rather try to emphasis on a dynamic exposure management, and thus requiring managers to have a more active risk management approach. In some cases, hedge fund managers could pursue low-beta strategies to have a lower exposure given subsequent reporting dates of their performance, or release date of their fund's performance. This is mainly explained due to the intention of reducing the exposure to the rising volatility or downside risks within the market, as well as the further reporting costs that could be incurred within the balance sheet. Given the exposure and the level of strategies, the adjustments within the beta exposure of hedge funds is done gradually, and not directly from a single attempt, and thus highlighting the potential of dynamic budgeting ¹⁹(J.Patton and Ramadorai - 2013).

¹⁹ J. Patton. Ramadorai (2013). On the High-Frequency Dynamics of Hedge Fund Risk Exposures, *The Journal of Finance*. 68 (2), 597-635.

On the other hand, hedge funds can also reverse their strategy by using their monthly returns into their portfolio by the inverse of the previous month's variance, while also decreasing exposure to risk if the variance is too high ²⁰(Moreira and Muir – 2017). Given precise volatility timing, managers could enhance their returns by earning large alphas given the parameters of their asset pricing factors, with the importance being emphasized on the deep-in analysis of the financial statements of the hedge fund (such as the market value, the return on equity, or currency carry trade), while investors will usually take stock market risk during high market volatility, or when the regime of the major stock exchanges starts to dip slightly. This strategy allows managers to realize higher average returns during highly-volatile conditions, as well as when the probability of market crash or economic recession is significantly high. This mechanism can be deployed for specific hedge fund strategies, such as macro funds as they will try to take advantage of the market swings by putting positions into a diverse amount of financial products, such as currencies, futures, bonds, or commodities (and especially physical metals which can offer a greater protection, such as gold or silver). Relative value hedge funds will see their leverage decrease substantially, with a mixture of long positions being preferred to bear the long-term risk that could be present, albeit accentuating the potential risk of crisis as the hedge fund would still fund for its new positions. Lastly, event-driven hedge funds will be severely impacted, as the likelihood of deals or mergers announced could reach potential nadir points, and thus resulting in a potential financing uncertainty.

2.6.3 - Crisis and stress regimes:

This scenario represents the most alarming possibility for hedge funds, as most of the advantages surrounding the ability of hedge funds at capturing macroeconomic discrepancies is challenged. During stress regimes, a general macroeconomic recession is occurring, with sharp volatility spikes (often indicated by the Volatility Index, which reaches a level above 20-points basis) and where liquidity is unavailable for the general investors. This level is generally risky for most of the banks, as their lack of access to supply money reserves and funds can pose the risk of a potential collapse. In this horizon, central banks and authorities will play the most crucial role, as they will seek the direct supply of money (especially M2 type money) across banks, while also requiring banks to reach a certain level of ratios concerning their funding sources (and especially their debt-to-equity ratio). From the different research conducted by Ang & Timmermann (2012), cross-asset correlations become more negative, with adverse movements being observed between the different asset classes in the economic environment, which cancels out the possibility of diversification and efficient allocation of weights for risky assets, which may represent alpha opportunities for portfolios. This idea is also supported by their observations of consistent and high volatility levels during a prolonged period, which might also translate into alpha opportunities: given the non-linear nature of hedge funds, as per the idea of Fung & Hsieh (2001), positive convexity can be yielded by hedge funds that are exposed to high-risk

²⁰ Moreira, Muir. (2017). Volatility-Managed Portfolios. *The Journal of Finance*. 72(4), 1611-1644.

macroeconomic variables, especially when safe-haven futures, including gold, might be included as a hedge against any further macroeconomic downturn.

This could explain further constraints for hedge funds, as predictability of the potential outcomes is challenged, given the lack of liquidity and financial leverage poses serious constraints to the global performance of the fund. In fact, hedge funds will often be forced to sell their risky assets, especially given the fact that market liquidity and funding liquidity will reinforce each other, as the lack of liquidity will require managers to seek other potential sources. The lack of leverage is concrete, as the deleveraging of the hedge fund is accentuated, and leading to potential liquidity spirals, and thus training further risks for the fund ²¹(Brunnermeier & Pedersen – 2009). At the same time, during specific crises, such as the 2008 Great Financial Crisis, hedge fund investors using redemption gates (a gate provision allowing fund managers to limit or halt the withdrawal of investors during redemption periods) saw a concern into their liquidity levels, as the investors started to notice a discrepancy between the liquidity that hedge funds are technically due to provide and the actually liquidity of the underlying assets owned by each investor. This has resulted into a higher drawdowns and drops into the value of the specific funds: as a consequence, the liquidity risk implied from the strategy can amplify the effects of negative capital flows on fund returns, and thus performing lower than hedge fund flows which are highly exposed to the market risk, as the protection mechanisms grant a better redemption for the participating investors. Lastly, these issue can also lead to the quick-sale of risky assets, or other highly-exposed assets within redemption gate funds, and thus constraining the performance of the funds ²²(Teo – 2011). The impact of the funding liquidity shock on asset liquidity is strong enough, the decrease in the asset liquidity forces the hedge fund to tighten funding that will deteriorate both the funds' and funding's liquidity levels, and accentuating a phenomenon entitled "liquidity spirals", which have been noticeably observed across recent financial crises, such as the financial crisis occurring in 2008, albeit the effect started as early as 2007, and which is often the result of speculators who have to finance their trades through collateralized borrowing from financiers, and thus resulting into higher margin costs, and increasing the risk of contagion into hedge funds, leading to the gradual decline of the diversification possibilities of hedge fund managers ²³(Boyson, Stahel & Stulz – 2010).

As a global consequence, multiple hedge fund strategies could perform badly across the regime, including equity long/short positions, which will have a defensive approach due to the reduced beta, as well as the constrained alpha opportunities. Meanwhile, event-driven funds can be vulnerable to the shocks that could be resulting from the failure of the announced deals, changes within the managerial organization of the concerned firms, or the illiquidity provided by potential IPOs (although IPOs are extremely rare during crisis regimes). Nonetheless, macro

²¹ K. Brunnermeier. Pedersen (2009). Market Liquidity and Funding Liquidity. *The Review of Financial Studies*. 22 (6), 2201-2238.

²² M.Teo (2011). The liquidity risk of liquid hedge funds. *Journal of Financial Economics*. 100(1), 24-44.

²³ Boyson. W. Stahel. M.Stulz (2010). Hedge Fund Contagion and Liquidity Shocks. *The Journal of Finance*, 65(5), 1789-1816.

funds could outperform the traditional benchmarks: in effect, by emphasizing their trades and liquidity opportunities across specific products, such as futures contracts or an increased trade of commodities, liquidity could still be provided and convexity ensured.

2.6.4 - Recovery and post-crisis regime:

This macroeconomic environment is more interesting than the previous ones, as it represents the start of a new economic cycle that could affect hedge funds. In this scenario, the volatility levels are decreasing at a certain rate, to a point where it will reach eventually the pre-crisis levels. At the same time, the economic policy imposed by the central bank will start to ease and support further the recovery of stock exchanges, and also allowing for a better access to liquidity.

For hedge funds, this represents an opportunity to increase capital allocation and liquidity flexibility further, while also gradually assessing new methods to evaluate systematic and market risk. In most of the scenarios, funds that are surviving financial crisis would usually attract a new amount of capital flow, and thus increasing the opportunity of having new liquidity channels, while the performance will also increase slightly, but constantly (Fung & Hsieh – 2008).

Furthermore, general funds that are holding illiquid assets will perform better, as they have longer commitment periods for the investment, as well as resulting in better cash holdings and unused borrowings, allowing for a general balance for funds that might also include liquid assets in their portfolio. Additionally, new conservative methods are used by hedge funds: in effect, once the investor and financing liquidity terms are set, the fund will use cash and unused borrowings to dynamically manage the newer liquidity sources of the portfolio, showing a better evaluation of the market and liquidity risks that may still rise. Lastly, the inclusion of other financial instruments, such as lines of credit (and especially cash holdings) are means against liquidity risk, showing that unused credits could also be beneficial for the new perspectives undertaken by hedge fund managers (Aragon, Ergun, Getmansky & Girardi – 2017).

These observations do also provide a context to a potential correlation of this new regime change and asset location. Asset allocation, as seen previously, will depend on the regime in which the hedge fund operates. With the perspectives of having a better adjusted risk evaluation, hedge funds can spur their returns ahead of a sustained recovery of the economy (Ang & Timmerman – 2012). In this new scenario, most strategies can take advantage from the re-access to liquidity opportunities, such as long/short positions which will take advantage further from the discrepancies regarding the prices of stocks, while a macro-regime hedge fund could see its trades normalized as the uncertainty level of the macroeconomic context improves gradually.

2.6.5 - Brief conclusion about macroeconomic regimes:

As a brief conclusion for this crucial subsection, the emphasis at correlating the behavior of hedge funds during specific macroeconomic regimes has always been crucial for managers in order to take the best investment and allocation decisions. Hedge funds should not ignore the importance of the macroeconomic environment and its general aggregates, as any sudden shift

could result into devastating drawdowns that could be caused by restrictions to liquidity means, while also forcing hedge funds to sale their riskier assets, while also putting investors into jeopardy. The effectiveness of hedge funds at mitigating risks is challenged by the diverse changes into the macroeconomic spectrums, which constraints the way hedge fund managers are allocating their resources and capital into the fund.

2.7 – Limits of the current literature review:

While recent literature research about the functioning of hedge funds and their mechanisms during certain periods have shown the behavior that hedge funds are espousing during certain macroeconomic regimes, as well as explaining the way the different strategies are deployed and optimized by managers, there are some current limits that might block the way hedge funds could actually be measured.

2.7.1 - No real-approach and access to hedge funds data:

One of the first issues and challenges that financial and academical researchers face while conducting any research about hedge funds is the lack of concrete data coming from hedge funds or companies specialized into handling hedge funds. In fact, hedge funds are typically not open to the public, with the main disclosures being mainly reserved only for the fund vehicle and investors that are part of the limited partnership. Furthermore, although reports regarding the hedge fund's performance could be reported, albeit not obligatory, no actual report can be disclosed towards the public due to the diverse amount of agreements that are made between the investors and managers of the fund. Moreover, although hedge funds could go publicly by conducting an Initial Public Offering (IPO), few hedge funds do in reality achieve this step, mainly due to the volatility exposure most of the hedge funds face, making high valuation prospects unlikely, while competitive and taxonomy also posing challenges for most of hedge funds. Because of the lack of public data about hedge funds, proxies or estimations of hedge fund data are often used for paper researches (including the current one), such as the Hedge Fund Research (which regroups the main aggregates of hedge fund under indexes which are classified by the strategies deployed by the largest hedge funds), TASS (Trading Advisor Selection Systems, a database regrouping the monthly data for hedge funds), or BarclayHedge. Although these data can be used for data analysis and for interpretations regarding the functioning of specific hedge fund strategies or hedge funds as a whole, it poses a limitation through different perspectives: in effect, there is no position-level data, meaning that ²⁴true exposures and fluctuations of hedge fund returns (including betas and alphas) cannot be directly observed, meaning that the real impact of potential macroeconomic regimes might not be noticed directly. Furthermore, because of concrete lack of data for specific financial items within the hedge funds (including balance sheets), leverages are mainly estimated based on the regime or strategy index

²⁴ 24 Viral V.Acharya, Lasse Heje Pedersen (2005). Asset pricing with liquidity risk. Science Direct, 77(2), 375-410.

in which the observations are carried out. Finally, other issues, such as the accuracy of alphas and betas, or the observation of liquidity is also limited, especially given that most of the alphas and betas of hedge funds are estimated throughout the usage of linear and statistical calculations, such as linear regression.

2.7.2 - Missing political and macroeconomic factors:

Other issues might also be the lack of concrete other variables that might impact the way hedge funds are behaving, or the simple strategies employed by investors. In effect, during recent decades, geopolitical events have become a major focus that have shifted the way stock exchanges have moved, with particular recent events such as the War in Ukraine, the Israel-Palestinian conflict, or the election of Donald Trump and the economic aftermath impacting the responses of major stock exchanges across the world. While real hedge funds could see their strategies shifted by these potential geopolitical risks, literature review hasn't always perceived this sphere as a quantifiable area in the researches conducted about the functioning of hedge funds, mainly due to the lack of concrete quantitative structuration of the different factors within this concrete sphere. Most of literature review structures macroeconomic regimes and the behavior of hedge funds either through the inclusion of business and economic cycles, the monetary policy and decisions made by Central Banks or financial indicators depicting the stress level of investors across major markets, such as the Volatility and Dollar indexes. Researches are often ignoring the amplitude of geopolitical risks, which could affect the way capital flows, foreign exchange rates, or liquidity is available on markets, and thus creating a non-linear pattern affecting hedge funds. Furthermore, other financial instruments, including commodities are rarely included into specific paper researches. As it is often known, commodities like gold and silver futures are representing a buffer protection for investors against potential economic downturns, and hedge funds are usually using commodities to hedge against market risk, like it is the case with macro-based hedge funds. The main issue with paper researches is the lack of concrete evidence of commodities being interpreted, being often associated with large macro factors, which is often not the case, given that commodities futures are lowly correlated with economic aggregates (such as the Volatility Index or the Dollar Index per instance). Unlike equities or bonds, commodities will usually move into an opposite direction from the market's trend, and thus representing a large opportunity for hedge funds to use commodities as a hedging mechanism.

2.7.3 - Lack of concrete aggregation of quantitative methods and variables.

On the quantitative side, hedge fund papers and the calculation methods that are usually employed are mostly approximations based on the data available on the indexes and databases available. For example, liquidity is usually determined based on the estimated level of spreads, alphas, and betas that are calculated after approximations done by using linear regression that comes from the observed data across specific hedge fund indexes. Moreover, the aggregation of specific indicators might be indirect, since the actual weight of impact of financial indicators (such as the

Volatility Index) is not always known, and it could differ between the different hedge fund strategies. In this matter, fund-specific liquidity is missing, with the main assumption being that each regime or specific hedge fund strategy has liquidity reserves stable over time, and thus making alphas and betas estimates always linear in a time interval. This could explain further the mismatch that is often existing between the liquidity constraints and actual alpha availability: in this regard, most of the studies will assume that alpha availability is independent from the reserves and actual economic context. Though, alpha is rising during economic stress which is difficult to obtain in fact, since different constraints, such as lower leverage usage or the forced sales of assets, make the alpha options difficult to obtain. Furthermore, the reliance on simple calculation methods can also pose challenges: one of the challenges is mainly shaped around the timing of results and exposure of hedge funds, which is one of the key factors that currently shapes how the returns of hedge funds evolve. At the current moment, timing is hard to classify given the broad sense of the term, as well as a real lack of concrete data showing how timings are used by hedge fund managers for capitalizing on returns. Paper researches are also emphasizing their results into classic results, such as estimates of alphas and betas, average returns, or sharpe ratios to assess the performance of hedge funds: while these parameters are crucial to understand the trends of hedge funds, the lack of measurements regarding the probability of survival of hedge funds or the maximum drawdowns, as well as real capital and cash holdings is missing, and thus not addressing the way capital allocation is concretely conducted, as well as evaluating the time it takes for hedge funds to return to a normal state.

Lastly, other factors, which are mainly linked to the role of institutions or behavioral finance are also left out of most paper researches. For instance, institutional restrictions and constraints during severe economic regimes are often overlooked by research papers: although findings do often detail some of the consequences that are related to institutions, the direct action of investors or regulators, such as the constraints over balance sheets, the lack of concrete visualization of the limited actions of brokers, or the real margin policy adjustments are not included, which is crucial mostly for the way leverage, capital allocation, or access to liquidity is ensured for hedge funds during periods of crisis.

2.8 – Research issue and hypotheses:

2.8.1 - Research issue:

While the existing literature review allows for a synthetic and global overview of the way hedge funds may interact with various macroeconomic factors, the concrete lack of observation evaluating the way macro and event-driven hedge fund strategies interact with macroeconomic factors, as well as a lack of potential structure of the actual macroeconomic indicators that could impact the performance of highly-exposed hedge funds. At the same time, as it has been highlighted through the limits of the research conducted by Fung & Hsieh (2004) and Ang & Timmermann (2012), the performance of hedge funds, and the ways that these hedge funds adjust their positions and risk mitigation under specific macroeconomic regimes hasn't been

evaluated, especially when it comes to the usage of a stress-score classification methodology based on underlying macroeconomic factors and indicators. Therefore, the intent of this research thesis is to explore the way macro and event-driven hedge funds can see their sensitivity and risk-adjustment mitigation being impacted by macroeconomic regimes that are defined by the presence of volatility, inflation, currency, bond yields, and signals around safe-haven commodities that would shape and structure different macroeconomic regimes. By using the HFRI Macro Index and HFRI Event-Driven Index and macroeconomic data from 2018 to 2025, the aim of this research will be to evaluate how the macroeconomic factors could affect the sensitivity that macro and event-driven hedge funds are exposed to under specific conditions.

2.8.2 - Hypotheses:

Before defining the actual methodology used for responding to the research's issue, different hypotheses have been thought up and formulated based on the different macroeconomic indicators existing, the previous articles written, as well as the dynamics observed while evaluating the returns of the HFRI Macro and HFRI Event-Driven Indexes:

- Hypothesis 1 – Given the research and findings of Fung and Hsieh (2001, 2004), macro hedge funds are prone to exhibit different behaviors based on the different level of macroeconomic variables under different regimes, with DXY, VIX intensifying the most during transitional and crisis periods due to the instability of the macroeconomic environment.
- Hypothesis 2 – VIX is expected to be the macroeconomic factor that will drive excess return variation for macro hedge funds across all the regimes defined within the research, given the significance that volatility and momentum have on hedge funds. Additionally, DXY changes, 5-year treasury yields movements, or gold returns can also provide significant ground to the movements observed by macro hedge funds.
- Hypothesis 3 – Asymmetric performance of macro and event-driven hedge funds: macro hedge funds generate higher risk-adjustment returns and performance during tight macroeconomic regimes by exploiting the potential discrepancies existing by hedging on short positions related to safer positions, including fixed-income and commodities. Conversely, event-driven hedge funds tend to thrive more during positive macroeconomic conditions given the low tightness of monetary policies which would encourage positive economic growth or corporate events, such as the completion of M&A deals or IPOs, which would give direct returns for event-driven hedge funds.
- Hypothesis 4 – Consistency of regimes and signal around regime-switching periods: as the research of Ang & Timmermann (2012) signaled, macroeconomic regimes tend to stay for a certain period, with regime-switch signals occurring only when the different macroeconomic variables would represent a high signal for funds. In the context of macro and event-driven hedge funds, observing regime switches during the period 2018-2025 would allow the research to observe the consistency of the different macroeconomic variables tested. Given the evaluated period including some calmer periods, especially

post-2022, the assumed probabilities of staying in the same macroeconomic regime in the mid-term range (about 6 months or more) should be higher than regime switches because of the prolonged state of different crises or events triggering uncertainty within the market.

- Hypothesis 5 - Optimizing weight allocation of the two strategies: because of the different observed periods that combine both risk-driven and stable periods, a Markov chain mechanism can be used to optimize a portfolio that combines the two hedge fund strategies, all while finding that the optimal macro allocation may fall below 50% of the total portfolio weight due to the macro hedge fund's strategy higher exposure and risk towards macroeconomic fluctuations.

Thanks to these hypotheses highlighting the limits of the existing literature review, the structuration of the methodology used by this research can be stated, during which the development of OLS regression analysis, cross-regime switching, and efficient portfolio weight allocation would be presented. Meanwhile, with the results obtained through this research, we would be able to respond to each of those assumptions, as well as addressing the core question of this research topic.

III – Data & methodology:

3.1 – Prelude of the study research:

3.1.1 - General introduction:

For the overall work and feasibility of this study, different hedge fund strategies have been analyzed, as well as different methods for collecting and conducting the main core of this study research. In effect, the main strategies that this study will shape itself around will be macro-driven hedge fund strategies as the main focus, as well as event-driven strategies. The motivations behind these choices was the nature of the strategies: macro hedge funds are mostly known to be exposed to diverse macroeconomic metrics (Dollar Index, Volatility Index, FED Interest Rate, or geopolitical factors), while event-driven strategies may also have a certain exposure to those factors, in addition to the dynamics surrounding the industry in which the hedge fund operates. At the same time, event-driven hedge funds can actually represent a buffer during macroeconomic regimes that are considered risky, as sudden macroeconomic factors could influence significantly the way hedge funds can shift within the market.

Moreover, for the collection of data, the study is mainly collecting secondary data coming from the different databases that are currently existing depicting the main hedge fund indexes, which will be detailed further in the sub-section of this sub-part. From the secondary data collected, primary data has been garnered thanks to the usage of statistical methods that will also be detailed in the sub-section about primary data. The usage of both secondary and primary data is subject to confidentiality and within the current rights existing across the sources used by the

research. Finally, the main purpose of this section is also to demonstrate the way macro and event-driven hedge funds could behave into different macroeconomic regimes, as well as seeing the potential impact of the newly-improved variables which may have been omitted from previous research papers.

3.1.2 - Methods for collecting secondary data:

The main benchmark and data source used for the secondary data is the database of ²⁵HFRI (Hedge Fund Research Index)'s website and Bloomberg Terminal, with the main focus being the performance of the ²⁶Macro and the Event-Driven indexes, which are noted HFRI Macro Index and HFRI Event-Driven index respectively. The usage of this database is mainly reflected from the fact that hedge fund firms are not obliged to disclose their financial performances, and thus not accessible to the general public given the lack of transparency. In general, using indexes around specific strategies allows a better comprehension of the factors that could impact the hedge funds, while it may also mitigate certain potential behavioral risks, as per the case of survivorship bias. Furthermore, the indexes approximations provided on the HFRI platform are often quoted as being reliable, as well as being available for both medium and long-term study projections, giving a specific weight for the time observation that will be employed across this paper research. At the same time, qualitative data and observations coming from specific market outlooks of major Asset Management departments, such as the departments of ²⁷J.P Morgan, ²⁸Goldman Sachs, or ²⁹Blackrock are also used for this research, with the main purpose being to support the specifications and structure of the different macroeconomic regimes that will be configured through this paper research, as well as for the explanation of the weight that some of the research's variables will carry on.

The data collecting process has been carried through researches on the HFRI website, as well as through the historical database of the asset management departments of the aforementioned companies. The quantitative data thrived through the HFRI website has been stored and organized across a Microsoft Excel file, which has served as the main support of the research. Further platforms that supported the research will also be explained across the variables choice and preparation method section.

²⁵ HFR (Hedge Fund Research). HFRI Event-Driven (Total) Index. <https://www.hfr.com/indices/hfri-event-driven-total-index/>

²⁶ HFR (Hedge Fund Research). HFRI 400 (US) Macro Index (M). <https://www.hfr.com/indices/hfri-400-us-macro-index-m/>

²⁷ J.P Morgan Asset Management (2026). Guide to the Markets. *J.P Morgan Chase & Co.* <https://am.jpmorgan.com/us/en/asset-management/adv/insights/market-insights/guide-to-the-markets/>

²⁸ Goldman Sachs Asset Management (2026). Investment Outlook 2026 – Seeking Catalysts Amid Complexity. *Goldman Sachs.*

²⁹ Blackrock. Tom Becker. (2026). A global macro-outlook for 2026: patience. *Blackrock.* <https://www.blackrock.com/institutions/en-us/insights/2026-macro-outlook>

3.1.3 - Preparation method: variables, strategy, and limits of the method.

For this paper research, the main source of data has been the usage of the HFRI Macro Index, serving as the main source of data collection, as well as HFRI Event-Driven index as a supporting benchmark. The period defined for the research was from January 2018, until December 2025, as this period could reflect the different macroeconomic regimes that will be defined below, with different periods being especially key to the thesis, such as the 2020 COVID Period, the 2022 uncertainty and high-inflation environment caused by the different geopolitical crises (such as the War of Ukraine) being captured to test on whether the model could illustrate the impact of non-economic or financial events for hedge funds. As a general notice, the period is not evaluating the first months of 2026, given the multitude of movements across different indicators (especially the gold and silver prices), as well as the recent geopolitical events (such as the bombing in Venezuela during January 2026, or the bombing strikes between USA, Israel, and Iran in the Middle-East that started end of February 2026).

Along with this secondary data, different variables have been used by the paper research that will reflect the statistical method of this research. By using the rationale of ³⁰Lioui and Maio (2014), as well as the basis around the ideas of Fung and Hsieh (2001, 2004), and the core ideas of Ang and Timmermann (2012), mostly through the usage of macroeconomic variables related to the exposure of hedge funds to macroeconomic sensitivity, the overall volatility of the market, as well as changes in the dynamics of other investments can lead to a better understanding of the underlying variables explaining the behavior of macro and event-driven hedge funds. All of the variables have been extracted through ³¹Investing.com, and the data is based on the monthly averages through January 2018 to December 2025, as well as different calculations being done to ensure a general accuracy of the statistical model. The following variables have been defined for the research:

- **FED Interest Rate (in %):** the most crucial factor for the thesis is illustrated through the different announcements and changes that the FED operates regarding the interest rate. In effect, as the main governing body within the United States of America, the FED and its decisions can impact the way that the market is moving, as well as its response to any potential macroeconomic risks that may arise. For some of the observations within the database, the interest rate was the one announced on the previous month given that the FED isn't making interest rate decisions on a monthly basis.
- **VIX Level (CBOE Volatility Index):** the volatility index is also the other crucial factor, which allows us to evaluate the degree of volatility, and thus of risk that could be present across the macroeconomic environment. The level of volatility will allow to evaluate the

³⁰ Lioui, A., & Maio, P. (2014). Interest rate risk and the cross-section of stock returns. *Journal of Financial and Quantitative Analysis*, 49(2), 483-511.

³¹ Investing.com (2025). Historical data: CBOE Volatility Index, FED Interest Rate, Gold Spot Price, US 5-year treasury yield, Federal Funds Rate, S&P 500. *Investing.com*. <https://www.investing.com/>

sentiment of hedge fund managers and investors, as well as representing the main factor driving asset allocation decisions across hedge funds.

- ΔVIX (in %): changes in volatility index can ensure an approach at understanding any potential shocks, or any unexpected movement that could represent a potential change of macroeconomic regime.
- $|\Delta DXY|$ (in %): the absolute change in the Dollar Index will be another indicator that can showcase the potential consequences of any changes within the interest rate or volatility index, while also underlining the actual strength of the US Dollar, a factor that may impact different trading or asset allocation within macro hedge funds (especially if the macro hedge fund could use FX trading positions to hedge against other macroeconomic risks).
- Δy (5-year, in %): the monthly change in the 5-year treasury yield, which is especially important for the confirmation of the market state, as well as the effects of the monetary policy that is implemented by the FED given its decided interest rate.
- Gold Return (in %): the returns in gold prices could indicate potential movements or capital allocations that hedge fund managers may have completed, especially during macroeconomic transitions and crises, and thus underlining the hedge activity of the hedge fund, as well as any signal for potential safe-havens.

Furthermore, the paper research has also used some institutional frameworks and market outlooks coming from the asset management departments of Goldman Sachs, J.P Morgan, and Blackrock, which help at structuring the way macroeconomic regimes will be defined, as they do offer an insight into how macroeconomic variables can be used to validate the interpretation of the regimes, while also avoiding endogeneity and interpositioning between variables. The following regimes have been resulted as a result:

3.1.3.1 - Expansion regime:

The expansion regime is the paper research's most prolific and profitable macroeconomic regime. In this regime, the alpha opportunities for hedge funds are present, and general returns can be supported, while also being mainly defined by low market stress, steady and stable growths into macroeconomic indicators, as well as policies which are aligning and correlating more or less positively between the different variables.

Furthermore, the regime is characterized by different characteristics, such as a low volatility regime and change, a monetary policy that is steady and accommodative for return opportunities, while also containing potential risks, such as geopolitical and inflation risks. In this regime, hedge fund managers will be able to efficiently allocate their asset classes, while also structure the weights that are emphasized within the hedge fund portfolio.

The main parameters regarding the chosen variables, and their respective ranges will be the following:

- FED Interest Rate: any interest rate that is $\leq 2\%$ (below or equal to 2%), showing that liquidity is abundant within the macroeconomic environment, showing potential financing opportunities for hedge funds, as well as easy access to potential financial leverage.
- VIX Level: VIX figures below or equal to 16 basis-points (≤ 16) will ensure a low level of uncertainty within the market, as well as low risk for any potential spikes that could affect negatively the macroeconomic regime.
- ΔVIX : any change that is flat or negative ($\leq 0\%$) shows that volatility is stable or decreasing, confirming the low level of uncertainty or the potential financial attributions for hedge funds.
- ΔDXY : any change that is below or equal to 1.5 basis-points can represent a safe space for FX trading, underlining that the strength of the US Dollar could advantage further the financial leverage opportunities for hedge funds, while also underlining no cross-asset FX transmission.
- Δy : any evolution below or equal to 0.1% shows a stable evolution for bond markets, while also guaranteeing potential long positions for hedge funds over their macro positions.
- Gold Return: with a range between -2% and +3%, this illustrates that the uncertainty risk is low, while also underlining that no need for safe-haven is needed by managers, and thus indicating positive sentiment along the financial markets.

3.1.3.2 - Transition regime:

The next macroeconomic regimes that is occurring for this research will be the transition regime. This regime represents the situation when the economy starts to stall or slightly decrease, with volatility seeing some sudden spikes, and thus increasing the risks of economic uncertainty for investors and hedge fund managers. In this course, other observed phenomenon would be the tightening of financial conditions, with interest rates remaining steady without any cuts forecasted, as well as evolving cross-assets dynamics, with allocation on assets being more proactively managed by hedge funds.

As monetary policy tightens up, and with interest rates on bank or investment activities starting to rise slightly, risk premia and tightened financial conditions merge. Although signs of distress and decline can be appearing, this period is marked rather by the constant volatility in market conditions, the shifting in the correlation between the main indicators (such as the commodities, who might see higher price spikes, while also simultaneously correlating with the potential increase in the Volatility Index) and the decrease of other indicators (and especially the Dollar Index) will drive macro hedge funds into adjusting their capital allocation by decreasing gradually the weight of risky assets, while also ensuring a better volatility management along with liquidity reservation. Lastly, event-driven hedge funds might also see their activities decline slightly: unlike macro hedge funds, asset allocation could be harder to diversify or to change

(especially between long and short positions) due to a lower amount of M&A or corporate events that could occur during the period.

The main metrics chosen, which are the same as for the first macroeconomic regime, will have the following criteria range:

- FED Interest Rate: a range between 2% and 4% across the observations will indicate a general normal level of normalising cycles across macroeconomic markets, with funding limitations being implemented gradually ahead of any potential risk.
- VIX Level: VIX figures between 16 and 20 will show a sudden increase in the volatility and potential uncertainty that may raise for hedge funds, albeit the consistency and actual risk still being unknown due to hikes that may not be elevated within the period.
- Δ VIX: modest movements between 0% and 3% may confirm potential hikes that could represent temporary shocks across financial markets.
- Δ DX: movements between 1.5 and 3 basis points may indicate any potential risk across FX markets, and thus underlining potential spikes or currency strength dips (on a moderate level) of the US Dollar, and thus representing further macro exposure to potential FX shocks within macro hedge funds.
- Δ y: evolutions ranging from 0.1% to 0.25% may indicate gradual yield and costs for bonds, with the signal confirming potential macro and monetary tightening coming from the FED.
- Gold Return: movements ranging between -5% and +5% could signal potential capital allocation strategies or reduction of risky assets by hedge fund managers, with the safe-haven demand increasing slightly over the period.

3.1.3.3 - Crisis regime:

The last macroeconomic regime in this paper research will be the crisis regime. This regime is characterized by a high number of volatility, with price spikes and swings being more and more common, asset price dynamics being more unpredictable, and with a high uncertain macroeconomic climate, with overall economy decreasing. Tight funding conditions, uncertain asset allocation methods, as well as a high number of sensitivity to abrupt changes expose hedge funds to a potential high risk of alpha decrease.

During this period, most of the regime will be asymmetric and driven by sudden adjustments into the risk allocation, expectations regarding the macroeconomic and monetary policies, as well as liquidity availability. Monetary policies will be characterized by a large emphasis on tight and hawkish interest rates, while financial support for institutions coming from the Central Bank being treated as a feasible option. These regimes won't only just see drops into asset prices, but also market adjustments that might not be coherent and stable, as well as constrained balance sheet activities. Financial leverage will be sustainably limited, while asset allocation into risky assets will cease or barely be active, as those assets are the most prone to liquidity downturns.

Most of the macro hedge funds will be affected by these sudden changes and conditions, as their sensitivity and exposure to constraints can reduce significantly their performance.

The following macro-economic ranges will be used for evaluating crisis regime:

- **FED Interest Rate:** an increase of interest rates above or equal to 4% ($\geq 4\%$) is signaling that the FED will tighten its monetary policy approach, with most of the historical observations depicting that any economic downturn is followed by hikes in the interest rate to tackle any negative effects of the worsening macroeconomic environment. While this represents the usual standard classification used by major asset management and macro evaluators, this also comes with direct limitations, with this indicator mainly serving as a structural signal that can be supported by the other variables and their classification under this regime.
- **VIX Level:** VIX figures above or equal to 20% ($\geq 20\%$) or in the top 25% percentile of the database's observations indicate the potential high stress and uncertainty that is existing across the markets.
- **Δ VIX:** movements that are equal or above 3% ($\geq 3\%$) indicate further volatility spikes and other risks for hedge funds, including a limited access to financial leverage, as well as a forced sale of risky assets across the hedge fund's portfolio.
- **Δ DXY:** increases equal or above 3 point-basis, or in the top 10% of observations, show that the FX trading is disordered across markets, and thus exposing further hedge funds to the macro risks.
- **Δ y:** a surge above or equal to 0.25% ($\geq 0.25\%$) will indicate further the risk for hedge funds.
- **Gold Return:** with ranges above 5% and below -5%, this indicates that the demand for safe-haven is surging strongly, while also showing that hedge funds may be forced to illiquidate their unsafe or risky assets, and thus leading to major capital restructuring processes.

Furthermore, a composite stress scoring system has also been built for each macroeconomic regime that was described above. For each month of the observed period, a stress score will be assigned to ensure the actual classification of each monthly observation. A month will be classified as "Expansion" if the score is between 0 and 2, "Transition" if the score is between 3 and 5, while "Crisis" will be attributed to observations with a score equal or above 6 points. The following scheme has been built based on each indicator that was used for the characterization of the macroeconomic regimes:

Stress Scoring System				
Variable	Role within the model	Expansion regime classification	Transition regime classification	Crisis regime classification
FED Rate	Primary driver	$\leq 1\% \rightarrow 0$ points	1.0%-4.0% $\rightarrow 2$ points	$\geq 4\% \rightarrow 4$ points

VIX (Volatility Index)	Primary	$\leq 16 \rightarrow 0$ points	16 to 20 $\rightarrow 1$ point	$\geq 20 \rightarrow 2$ points
ΔVIX	Secondary (capture any volatility momentum)	$\leq 0 \rightarrow 0$ points	0% to 3% $\rightarrow 0$ points	$\geq 3\% \rightarrow 1$ point
$ \Delta DXY $	Primary – FX risk	$\leq 1.5 \rightarrow 0$ points	1.5 to 3 $\rightarrow 1$ point	$\geq 3 \rightarrow 2$ points
$\Delta y(5\text{-year})$	Secondary – effects of interest rate decisions	$\leq 0.10\% \rightarrow 0$ points	0.10% to 0.25% $\rightarrow 0$ points	$\geq 0.25\% \rightarrow 1$ point
Gold Return	Confirmatory – check on whether asset allocation or liquidation occurs	-2% to 3% range $\rightarrow 0$ points	-5% to 5% range $\rightarrow 0$ points	$\leq -5\%$ or $\geq 5\% \rightarrow 1$ point
Classification		0 to 2 score attributed	3 to 5 score attributed	Score above or equal to 6 attributed

3.1.3.4 - Statistical methodology:

For this research, the main objective will be to evaluate the actual performance of macro and event-driven hedge funds across the defined macroeconomic regimes to see on whether the sensitivity of these strategies to external factors is highly present, whether hedge fund managers employing these strategies can efficiently allocate their capital and asset strategy, while also showing on whether alpha opportunities are more present than on other popularly known strategies. Performance of hedge funds across the different macroeconomic regimes will be first assessed by the usage of regression analysis. For this regression analysis, a few set of factors can be defined, with most of the inspiration coming from the ³²Fung, Hsieh, Naik, Ramadorai (2008) model, which was comprising most of the factors for evaluating hedge fund: even so, as their model is mostly concentrated at using alternative non-linear factor risks, this study emphasis the prompt usage of macro-economic variables that serve as shock proxies for the different defined macroeconomic regimes. Moreover, fundamentals around ³³Harold O.Fried and C.A Knox Lovell's Econometric Analysis (2003) and the way of quantifying economic factors and variables impacting economic output have also served as the main catalyst of defining the OLS equation defined below.

For the macro-based hedge fund strategy, and based on the chosen macroeconomic metrics chosen, we can define for each $k \in \{\text{Expansion, Transition, Crisis}\}$:

$$R_t^M - R_t^F = \alpha_k^m + \beta_{m,k}^M MKT_t + \beta_{v,k}^M \Delta VIX_t + \beta_{r,k}^M \Delta y(5y) + \beta_{f(x),k}^M \Delta DXY_t + \beta_{c,kR}^M Gold_t + \varepsilon_t^M.$$

With:

³² As per the previous citations.

³³ Harold O.Fried & C.A Knox Loyell (2003)

- R_t^M = return of HFRI Macro Hedge Fund index at month t.
- R_t^F = risk-free rate (monthly reported, based on the US risk-free rate evolution, which is the US 1-month treasury yield rate).
- $R_t^M - R_t^F$ = excess return of the macro hedge fund index based on the month's risk-free rate.
- $\beta_{m,k}^M$ = exposure to the equity market risk based on the excess return.
- MKT_t = the market risk exposure which is calculated by doing $MKT_t = (R_B^t - R_t^F)$, the difference between the return from the S&P 500 proxy and the risk-free rate.
- $\beta_{v,k}^M$ = exposure to the changes in volatility for k macroeconomic regime.
- ΔVIX_t = monthly changes occurring in the Volatility Index during period t.
- $\beta_{r,k}^M$ = exposure to the changes in the 5-year US Treasury Bills for k macroeconomic regime.
- $\Delta y_t(5y)$ = monthly changes occurring in the 5-year treasury bills during period t.
- $\beta_{f(x),k}^M$ = exposure to the changes in the Dollar Index for k macroeconomic regime.
- ΔDXY_t = monthly occurring in the Dollar Index during period t.
- $\beta_{c,kR}^{MGold}_t$ = exposure to the changes in the Gold Price for k macroeconomic regime.
- ε_t^M = residual error for period t.

At the same time, we will define the formula used to estimate the excess return that is occurring across macro hedge funds. For each $k \in \{\text{Expansion, Transition, Crisis}\}$:

$$MKT^M = \alpha_k + \beta^M MKT_t + \beta^M_v \Delta VIX_t + \beta^M_r \Delta y_t(5y) + \beta^M_{f(x)} \Delta DXY_t + \beta^{MGold}_t + \varepsilon_t^M.$$

Furthermore, for event-driven fund strategy, we can define for each $k \in \{\text{Expansion, Transition, Crisis}\}$:

$$R_t^E - R_t^F = \alpha_k^s + \beta_{m,k}^E MKT_t + \beta_{v,k}^E \Delta VIX_t + \beta_{r,k}^E \Delta y_t(5y) + \beta_{f(x),k}^E \Delta DXY_t + \beta_{c,kR}^{EGold}_t + \varepsilon_t^E.$$

With:

- R_t^E = return of HFRI Event-Driven Hedge Fund index at time t.
- R_t^F = risk-free rate (monthly reported, based on the US risk-free rate evolution).
- α_k^s = the intercept of the performance for s strategy, with the inclusion of the risk-adjusted performance during k regime.
- $\beta_{m,k}^E$ = exposure to equity market risk (market beta as per defined) within regime k.
- MKT_t = the market risk exposure which is calculated by doing $MKT_t = (R_B^t - R_t^F)$, the difference between the return from the S&P 500 proxy and the risk-free rate.
- $\beta_{v,k}^E$ = exposure to the changes in volatility for k macroeconomic regime.
- ΔVIX_t = monthly changes occurring in the Volatility Index during period t.
- $\beta_{r,k}^E$ = exposure to the changes in the 5-year US Treasury Bills for k macroeconomic regime.
- $\Delta y_t(5y)$ = monthly changes occurring in the 5-year treasury bills during period t.

- $\beta_{f(x),k}^E$ = exposure to the changes in the Dollar Index for k macroeconomic regime.
- ΔDXY_t = monthly changes occurring in the Dollar Index during period t.
- $\beta_{e,kR}^{E,Gold}_t$ = exposure to the changes in the Gold Price for k macroeconomic regime.
- ε_t^E = residual error for period t.

Thanks to the defined equations for both the Macro and Event-Driven hedge funds, we can evaluate whether hedge fund strategies adjust their exposure to the different macroeconomic metrics during specific macroeconomic regimes. For each regime k, the HFRI Macro Hedge Fund Index and HFRI Event-Driven Hedge Fund Index indices will be regressed over a number of defined factors, as per defined above. By comparing the α_k and $\beta_{*,k}$ factors across different regimes and strategies will provide the resilience and possibilities of hedge funds during different environments.

The regression models that are configured across the research are driven by the usage of the Ordinary Least Squares (OLS) method for each defined macroeconomic regime.

3.1.3.5 - Performance and evaluation ratios:

For hedge fund managers, the most important factor when evaluating and choosing a specific hedge fund strategy is the usage of specific ratios to evaluate the volatility of the existing macroeconomic environment, as well as determining the level of risk at which investing strategies should be implemented.

One of the main ratios that is traditionally would incur the presence of the Sharpe Ratio. Developed in 1966 through the analysis of ³⁴Mutual Fund Performance by William F. Sharpe (1966), this ratio compares an investment's return to its risk which has resulted in differences in the performance that funds can yield, all while establishing the main relationship between asset allocation and capital risk across different funds. The main formula of the Sharpe Ratio is the following one:

$$\text{Sharpe Ratio} = \frac{R_p - R_f}{\sigma_p}$$

With:

- R_p = return of the portfolio.
- R_f = risk-free rate.
- σ_p = standard deviation of the portfolio's excess return.

As a general idea, the principle of the Sharpe Ratio is to compare the fund's historical or projected returns to the investment benchmark within the industry, while the risk-free rate will represent the fund's investment premium over safe assets, such as Treasury Bills. It also tells investors whether an investment's excess returns are based on the decided strategies, or the level

³⁴ William F. Sharpe (1966). Mutual fund performance. *The Journal of Business*. 39(1), 119-138.

of risk undertaken. Nonetheless, Sharpe Ratio may also represent its downsides, especially since the formula assumes normal returns for the fund's performance, especially regarding the high tails or risk of kurtosis, while non-normal returns are not captured, which are often caused by deviations from the changing macroeconomic conditions to which macro hedge funds are exposed.

The Sortino Ratio, on the other hand, represents a variation from the Sharpe Ratio in which it uses the asset's standard deviation of negative portfolio returns or downside deviation inside of the portfolio's aggregate standard deviation, mainly defined relative to a minimum acceptable return.

The formula is like the Sharpe Ratio, with the main difference coming from the variables used:

$$\text{Sortino Ratio} = \frac{R_p - r_f}{\sigma_d}$$

Where:

- R_p = actual or expected portfolio return.
- R_f = risk-free rate.
- σ_d = standard deviation of the downside returns.

Just like the Sharpe Ratio, the Sortino Ratio indicates the investment's return for a given level of negative risk, while ignoring upside volatility, a factor that can be beneficial for investors given the potential return gains that investors can garner on. This is more conceptual for hedge fund managers, since it allows them to incorporate return asymmetries, as well as tail risks or peaks that could occur within the performance of the fund.

Another ratio which is used by hedge fund managers would include the Omega ratio. This ratio helps at ³⁵assisting investors to rank the funds in which they do invest by comparing the price of the fund with the risk-adjusted returns, while also describing the high peaks and lows a hedge fund can reach by taking into account both the skewness and kurtosis of the hedge fund, with the key factor being the measurement of the probability that the fund will achieve a target return against the probability of failing to meet the target. The Omega ratio is often defined by scholars through the following formula:

$$\Omega(t) = \frac{\int_t^{\infty} (1 - f(x)) dx}{\int_{-\infty}^t f(x) dx}$$

Where:

- $\Omega(t)$ = the Omega level for a given threshold t .

³⁵ Rambo (2017). An Omega Ratio Analysis of Global Hedge Fund Returns. *The Journal of Applied Business Research*, 33 (3).

- $F(x)$ = cumulative probability distribution of returns.
- $\int_t^{\infty} (1 - f(x))dx$ = probability for a return above a given threshold t.
- $\int_{-\infty}^t f(x)dx$ = probability for a return below the given threshold t.

The last metric that is useful for hedge fund managers is the Value at Risk (VaR). This represents an essential tool for investment and commercial banks, while also used by managers (including hedge fund managers) to understand the probabilities and extents of potential losses within the portfolio or fund, while also assessing the risk exposure and optimizing the capital allocation across the fund. Unlike other metrics, Value at Risk can be calculated either by using the Variance-Covariance method (assuming that gains and losses are normally distributed) or by using the Monte Carlo simulation. However, the historical method return is preferred given the usage of a security or index as a benchmark for simulating potential future returns, as well as having more stable and known parameters regarding distribution of returns.

Even so, the usage of Value at Risk can also pose some important limitations. In fact, no standard protocol or real method is used regarding the assessment of asset or portfolio risk, and thus limiting the way that high-volatile moments could impact the performance of the fund. At the same time, loss risk is also underestimated, as the general parameters would ignore noise or unlikely risks that could occur.

3.1.3.6 - Maximum drawdown:

Another metric that should be defined for understanding the way hedge funds will behave would be the inclusion of the maximum drawdown. For an investment, the maximum drawdown is the lowest point at which an investment can take from a high to a low price. It indicates the downside risk (the potential that an asset or an investment will lose value due to experiencing diverse market conditions). Unlike volatility, maximum drawdown captures the magnitude of capital impairment experience by an investment. With the first representation being portrayed by ³⁶Frank A. Sortino and Robert van der Meer article (1991) in relationship to the Sortino ratio, the Maximum Drawdown can be calculated by using the following formula:

$$MDD = \frac{Trough\ value - Peak\ Value}{Peak\ Value}$$

Where:

- Peak Value: the highest point that the fund had reached.
- Trough Value: the lowest point that the fund had reached.

While the maximum drawdown helps investors and fund managers at comparing the risk of stock screening or capital asset allocation (especially by focusing on preserving capital), it usually measures only the highest losses incurred by the fund, and not the frequency at which the fund losses its value. Nonetheless, it is usually common for investors and fund managers to use

³⁶ A. Sortino. Van Der Meer (1991). Downside risk. *Journal of Portfolio Management*. 17 (4), 27.

maximum drawdown to assess risk evaluation to understand how a fund behaves under certain macroeconomic conditions, as well as enhancing the usage of performance ratios. At the same time, it also allows hedge fund managers to visualize the actual capital risk a hedge fund is incurring.

3.1.4 - Steps undertaken – guideline of the work conducted:

To conduct the actual research, different steps have been taken across the Excel worksheet that has served as the basis for building and observing macro and event-driven hedge funds in the defined macroeconomic environments. The reason over choosing the macro and hedge fund indexes is mostly related to different factors that are related to the purpose of the research, and the previous description of macro and event-driven hedge fund strategies:

- The Index scores provided by the Hedge Funds Research Institute (HFRI) of Macro and Event-Driven hedge funds are the most realistic scores and yields publicly available, as hedge funds rarely disclose publicly their results and returns, given the less regulatory environment of hedge funds.
- The two indexes reflect the most Macro and Event-Driven, which are the two crucial strategies chosen for this research, as these two strategies are the most prone strategies impacted by potential shocks into the economy or macroeconomic indicators, given their nature of investment on long/short positions triggered by economic evolutions or by major announcements of financial and economic indicators.

3.1.4.1 - Macro Index (M):

The first worksheet, entitled “Macro Index (M)” regroups the monthly returns of the Macro-Index Hedge Funds from the Hedge Funds Research Institute (HFRI), which regroups the top 50 biggest macro hedge funds, as per the total Assets Under Management included (AUM). Monthly returns were grouped between 2018 and 2025, period in which the research has been conducted, as well as the major statistical metrics and ratios to evaluate the performance and relationship existing across the macro hedge fund index.

3.1.4.2 - Event-Driven Index (M):

The second worksheet follows the same logic as per the first sub-section, in which the monthly returns of Event-Driven Index Hedge Funds from 2018 to 2025 were collected from the Hedge Funds Research Institute (HFRI), while also including the same major statistical metrics and ratios to evaluate the performance of the index.

3.1.4.3 - Macroeconomic factors:

In order to build the actual macroeconomic regimes, the first step was to regroup and collect the macroeconomic data for each macroeconomic factor that has been included in the linear regression formulas of the research. To do so, monthly data from January 2018 until December 2025 was collected from Investing.com, with the data collected being the FED Interest Rates, the MoM US CPI (which impacts more the regime, given the monthly relationship that can be

defined between the reported CPI data and the other macroeconomic factors), the Volatility Index (CBOE), US Dollar Index (DXY), Gold Prices, as well as the US 5-year treasury bill yield (which relies on the US 1-month treasury bill). However, these raw data cannot be included in the actual research, as the data cannot capture directly the changes that occurred, and thus the potential macroeconomic sensibility that can be indicated for each month of the observed period. To adjust this issue, monthly changes, noted as $\Delta\%$ of change have been calculated for each factor, with the formulas being noted below:

For the FED Interest Rate, we can note ΔFED as the FED Interest Rate change on a monthly period, which is calculated as following:

$$\Delta\text{FED} = \text{FED}_n - \text{FED}_{n-1}$$

With:

- FED_n = the FED interest rate of the current n period.
- FED_{n-1} = the FED interest rate of the previous n period.

Regarding the MoM CPI, we can write ΔCPI as the monthly changes in the CPI, which is calculated as it following:

$$\Delta\text{CPI} = (\text{CPI}_n - \text{CPI}_{n-1}) * 100.$$

With:

- CPI_n = the CPI score of the current n observed period.
- CPI_{n-1} = the CPI score of the previous n observed period.

Same logic can be applied for the Volatility Index, Dollar Index, as well as the US treasury bill, with ΔVIX , ΔDXY , and $\Delta\text{t-bill}$ being the noted variables, which can be calculated as following:

$$\Delta\text{VIX} = \text{VIX}_n - \text{VIX}_{n-1}$$

$$\Delta\text{DXY} = \text{DXY}_n - \text{DXY}_{n-1}$$

$$\Delta\text{t-bill} = \text{t-bill}_n - \text{t-bill}_{n-1}$$

With:

- VIX_n = the VIX score of the current n observed period.
- VIX_{n-1} = the VIX score of the previous n observed period.
- DXY_n = the DXY score of the current n observed period.
- DXY_{n-1} = the DXY score of the previous n observed period.
- T-bill_n = the treasury bill yield of the current n observed period.
- T-bill_{n-1} = the treasury bill yield of the previous n observed period.

For Gold Prices, the monthly gold price return formula has been noted, noted as R_g , which is calculated as following:

$$R_{g} = \left(\frac{\text{Gold price } n}{\text{Gold price } n-1} \right) - 1$$

With :

- Gold_n = the price of gold ounces at current n observed period.
- Gold_{n-1} = the price of gold ounces at previous n observed period.

Note: for some macroeconomic indicators, such as the MoM CPI or the FED Interest Rate, some of the monthly rates used have been a continuity of the latest announced data for the period of observation, as the two indicators are not always posted on a monthly basis.

3.1.4.4 - Reviewes sheet – Macroeconomic Factors:

The next process is one of the most crucial steps for the conducted research, which was the regrouping of all of the macroeconomic factors, along the observed monthly returns in macro and event-driven hedge fund indexes, all while including the monthly S&P 500 returns from January 2018 and December 2025.

To regroup all of the factors, different columns have been created, with the columns being the ΔFED, Dollar Index, Volatility Index, ΔDXY, ΔVIX, R_g (gold price return), Δt-bill, Macro Returns, Event-Driven return, and S&P 500 returns being created. Once these columns have been created, the risk-free rate column was created, which included the data from the US 1-month treasury bill yield that has been collected in the previous step. To simplify the data and apply it for a monthly scale, a monthly rate was calculated by using the following formula:

$$R_{f\text{-monthly}} = \frac{RF_{mn}}{12 \text{ (one-year months)}}$$

With RF_{m n} representing the risk-free rate at observed period n.

However, the first most crucial part of this process will include the calculation of excess macro return, excess event-driven return, as well as market factor excess return. We can define the following terms as per following:

- Excess Macro Return (EMR) = the excessive return the macro hedge fund index is realising each month compared to the monthly risk-free rate to which it is associated.
- Excess Event-Driven Return (EEDR) = the excessive return the event-driven hedge fund index is realising each month compared to the monthly risk-free rate to which it is associated.
- Market Factor Excess Return = noted as MKT, it is the excessive return the S&P 500 is realising each month compared to the monthly risk-free rate to which it is associated.

For calculating each factor, we have defined the following formulas:

$$\text{Excess Macro Return (EMR)} = (R_{m n} - \frac{Rf \text{ monthly}}{100}) * 100.$$

$$\text{Excess Event-Driven Return (EEDR)} = (R_{\text{ev } n} - \frac{R_{f \text{ monthly}}}{100}) * 100.$$

$$\text{MKT} = (\text{S\&P 500 } R_n - \frac{R_{f \text{ monthly}}}{100}) * 100.$$

With:

- $R_{m \ n}$ = macro return for the given period n.
- $R_{\text{ev } n}$ = event-driven return for the given period n.
- S&P 500 R_n = the return in S&P 500 for the given period n.
- $R_{f \text{ monthly } n}$ = the monthly risk-free rate for the given period n.

Once each excess return has been calculated, the most important step was implemented, which is related to evaluating the coefficient scores for each macroeconomic variables included in the formula, which is based on the stress scoring system table that has been written above. To classify a score for each macroeconomic variable, the Excel formula “IF” has been used for each factor, which can be presented mathematically through this form:

$$f(\Delta FEDn) = \begin{cases} 0 & \text{if } \Delta FED * 100 \leq 1 \\ 2 & \text{if } 1 < \Delta FED * 100 \leq 4 \\ 4 & \text{if } \Delta FED * 100 > 4 \end{cases}$$

Note: the classification method for ΔFED is the only one where each monthly observation is multiplied by 100, so that the score can be expressed directly as a percentage.

$$f(VIX) = \begin{cases} 0 & \text{if } VIX \leq 16 \\ 1 & \text{if } 16 < VIX \leq 20 \\ 2 & \text{if } VIX > 20 \end{cases}$$

$$f(|\Delta DXY|) = \begin{cases} 0 & \text{if } |\Delta DXY| < 1.5 \\ 1 & \text{if } 1.5 \leq |\Delta DXY| \leq 3 \\ 2 & \text{if } |\Delta DXY| > 3 \end{cases}$$

Note: the DXY score classification method is the only one using the ABS function on Excel, so that each monthly observation can be written as an absolute value.

$$f(\Delta VIX) = \begin{cases} 1 & \text{if } \Delta VIX > 3 \\ 0 & \text{otherwise} \end{cases}$$

$$f(\Delta t\text{-bill}) = \begin{cases} 1 & \text{if } \Delta t\text{-bill} > 0.25 \\ 0 & \text{otherwise} \end{cases}$$

$$f(R_g) = \begin{cases} 1 & \text{if } R_g > 5 \text{ or } R_g < -5 \\ 0 & \text{otherwise} \end{cases}$$

Note: the gold-return classification method is the only one using the OR function on Excel, given the classification method used for the monthly gold returns, which is a range that is either below -5% or 5%.

Once all of the scores have been attributed, a total score was calculated, which is given by the following formula:

Total Macroeconomic Regime Score (x) = $\sum \Delta \text{FED score; VIX score; DXY score; } \Delta \text{VIX score; } \Delta \text{t-bill score; } R_g \text{ score.}$

With the total score obtained for each monthly observation, the step of classifying each month into a macroeconomic regime was conducted, as it is shown by the formula below:

Regime Classification $\rightarrow \text{IF}(\text{Total Macroeconomic Regime Score} \leq 2; \text{"Expansion"}; \text{IF}(\text{Total Macroeconomic Regime Score} \leq 5; \text{"Transitional"}; \text{"Crisis"}))$

$f(x, \text{Expansion, Transitional, Crisis}) = \text{Expansion if } x \leq 2; \text{Transitional if } x \leq 5; \text{otherwise Crisis.}$

Note: the ranges stated in the regime classification logic formula are based on the total attributed scores that follow the rules established in the stress score regime classification table.

Finally, a next month regime column was added, which is based on the next month's classified regime, with the purpose being to predict the potential upcoming macroeconomic regime.

3.1.4.5 - Results – evaluating the three macroeconomic regimes:

Once all of the results have been aggregated, all observations were classified based on their resulting macroeconomic regime to evaluate the performance during each macroeconomic regime, as well as calculating all of the coefficients used in the macro hedge-fund index and event-driven fund index linear regression formulas. This part covers all of the fundamental and mechanisms used for all macroeconomic regimes, which are the expansion, transition, and crisis regimes.

For the analysis of the expansion regime for both macro and event-driven hedge fund indexes, the main idea was to regroup all of the months classified as belonging to the expansion macroeconomic regime into a single worksheet. The same process was used for separate worksheets evaluating the performance of the transition and crisis regimes. Once the process was completed, linear regressions have been used for the two hedge fund indexes by using the "Data Analysis" function in Excel. For the macro hedge fund index, the Y (dependent variable) data was collected from the excess macro returns calculated earlier, while the X (independent variable) data regrouping the data from the ΔVIX , ΔDXY , R_g , $\Delta \text{t-bill}$, and the MKT. The same goes for the event-driven hedge fund index, with the only change being the usage of the excess event-driven return as the Y dependent variable in the Data Analysis function in Excel. By running those linear regression results, not only we have obtained the coefficients for each variable in the linear regression function, but also other statistical indicators indicating relationships among the observed data, as per follows:

- Multiple R: this coefficient is the main indicator for evaluating the variance in the relationship between a dependent variable Y and one or multiple independent variables X, representing the square root of the R² coefficient.
- R² coefficient: this coefficient is one of the indicators illustrating the relationship that the excess macro return for both macro and event-driven indexes have along with the selected macroeconomic variables in the linear regression, noted as the independent variables X in the model.
- Adjusted R²: this indicator reflects better the relationship between the dependent variable and independent variables in the model, as it allows for the inclusion of n observations in the model, and k scenarios (which is not applied for the existing model, given the presence of only one k scenario that corresponds to each macroeconomic regime).
- Standard Error: the standard error coefficient evaluates the margin between the observed data and actual data existing in the linear regression model, and thus evaluating the accuracy of the conducted linear regression.

Based on these results, the final step, consisting of calculating the performance metrics for each hedge fund index in the expansion regime has been conducted.

The first ratio used was the annual Sharpe Ratio, which consists of evaluating the index's risk-adjusted return. For calculating the Sharpe Ratio, the following formulas have been used for both the macro and event-driven hedge fund indexes:

$$\text{Annual Sharpe Ratio Macro} = \frac{\mu_{\text{Excess Macro}}}{\sigma_{\text{Excess Macro}} * \sqrt{12 \text{ months}}}$$

$$\text{Annual Sharpe Ratio Event-Driven} = \frac{\mu_{\text{Excess Event-Driven}}}{\sigma_{\text{Excess Event-Driven}} * \sqrt{12 \text{ months}}}$$

The next ratio to be used was the Sortino Ratio, which evaluates the return of a hedge fund by integrating the potential maximum drawdown that a macroeconomic regime experiences across the observed period. The first factor to be calculated, for both macro and event-driven hedge fund indexes was the Downside Factor, an indicator that evaluates the largest loss a fund can incur during the period. The formulas used for both the Downside Factor and Sortino Ratio for the two hedge fund indexes are:

$$\text{Downside Macro} = \sigma_D^M = \sqrt{\frac{\sum_{i=1}^n \min(R_{mt} - R_f, 0)^2}{n}}$$

$$\text{Downside Event-Driven} = \sigma_D^{ED} = \sqrt{\frac{\sum_{i=1}^n \min(R_{ted} - R_f, 0)^2}{n}}$$

$$\text{Sortino Ratio Macro} = \frac{\mu_{\text{Excess Macro Returns}}}{\text{Downside Factor Macro} * \sqrt{12 \text{ months}}}$$

$$\text{Sortino Ratio Event-Driven} = \frac{\mu_{\text{Excess Event-Driven Returns}}}{\text{Downside Factor Event-Driven} * \sqrt{12 \text{ months}}}$$

The third ratio used for evaluating the performance of the expansion regime for the two hedge fund indexes was the usage of the Omega Ratio. The Omega Ratio is used to evaluate investment return by including non-normal, skewed, or fat-tail distributed observations, which is efficient for both hedge fund strategies given the presence of derivatives in both indexes. For calculating the following ratio, two major factors have to be determined initially. The first one consists of the Gains Above 0, which evaluates the sum of all positive observations that are above 0, with the Excel formula used being the following one for both macro and event-driven indexes:

$$\text{Gains Above 0 Macro} = G^M = \sum_{i=1}^n \max(R_t^M - R_f, 0)$$

$$\text{Gains Above 0 Event-Driven} = G^{ED} = \sum_{i=1}^n \max(R_t^{ED} - R_f, 0)$$

The same logic and formula were applied for the second factor, which is the Losses Below 0, an indicator evaluating the observed losses within the excess macro and event-driven returns, with the formula for each index being:

$$\text{Losses Below 0 Macro} = L^M = \sum_{i=1}^n \max(-(R_t^M - R_f), 0)$$

$$\text{Losses Below 0 Event-Driven} = L^{ED} = \sum_{i=1}^n \max(-(R_t^{ED} - R_f), 0)$$

By calculating these two factors, the Omega Ratio can be obtained by using the following logic:

$$\text{Omega Ratio Macro} = G^M / L^M$$

$$\text{Omega Ratio Event-Driven} = G^{ED} / L^{ED}$$

The last factor calculated was the Maximum Drawdown factor, an indicator that evaluates the largest average loss a fund can incur during the observed period. For obtaining the Maximum Drawdown, different other factors had to be considered, with the detailed process being explained below:

First step was to calculate the Cumulative Return Index, which evaluates all the resulted returns for macro and event-driven excessive returns throughout the period. The next step was to evaluate the running peak, which observes the highest accumulated return for the period. The formulas used were the following:

$$\text{First-Period CRI Macro} = 1 * \left(1 + \frac{\text{Excessive Macro Return } n}{100}\right)$$

$$\text{CRI Macro} = \text{CRI Macro}_{n-1} * \left(1 + \frac{\text{Excessive Macro Return } n}{100}\right)$$

$$\text{First-Period CRI Event-Driven} = 1 * \left(1 + \frac{\text{Excessive Event-Driven Return } n}{100}\right)$$

$$\text{CRI Event-Driven} = \text{CRI Event-Driven}_{n-1} * \left(1 + \frac{\text{Excessive Event-Driven Return } n}{100}\right)$$

$$f(\text{Running Peak Macro}) = \max(\text{CRI Macro})$$

$$f(\text{Running Peak Event-Driven}) = \max(\text{CRI Event-Driven})$$

Once the two sub-variables were calculated, the logic was followed by determining the Drawdown per each observed period, which allows to see the difference between the observed cumulative period at one point, and the running peak for the same period, with the following formulas being observed. Once all of the periods had been calculated, the Maximum Drawdown can be found by finding the smallest values from all of the Drawdown Per Period data:

$$\text{Drawdown Per Period Macro} = \frac{\text{Cumulative Return Macro } n - \text{Running Peak Macro}}{\text{Running Peak Macro} * 100}$$

$$\text{Drawdown Per Period Event-Driven} = \frac{\text{Cumulative Return Event-Driven } n - \text{Running Peak Event-Driven}}{\text{Running Peak Event-Driven} * 100}$$

$$f(\text{Maximum Drawdown Macro}) = \min(\text{Drawdown Per Period Macro}).$$

$$f(\text{Maximum Drawdown Event-Driven}) = \min(\text{Drawdown Per Period Event-Driven}).$$

3.1.4.6 - Cross-regime analysis:

Another step that has been undertaken involves the results found for all the macroeconomic regimes, and to conduct cross-regime analysis to see whether the average performance of each macroeconomic defined regime could see a change in its current macroeconomic conditions. The first layer is the coefficient stability analysis, which includes the observed coefficients, p-values, and t-values for each macroeconomic regime. Down below is the presentation of each chosen metric, and the reasons of their selection for this analysis:

- Coefficient: this represents mostly the average that each indicator obtained while conducting the linear regression, while it has also been used to evaluate the weight or sensitivity that each factor had across the three defined regimes.
- P-value: representing the probability that observed values for each linear regression factor could occur at any moment to define the statistical significance that each variable holds under each macroeconomic regime.
- T-value: also known as the “signal-to-noise” ratio, its purpose how many standard errors will deviate from the total observation’s sample mean, while also assessing the sensitivity that each factor has for the observed excessive returns in the macroeconomic regimes.

Based on the observed values for each linear regression factor for these three statistical indicators across all macroeconomic regimes, a classification function was applied to evaluate the stability of each observed regime to understand whether returns of hedge funds could be impacted by changes or sensitivity in macroeconomic factors, or whether persistent systematic relationship among factors occurs regardless of the regime in which the fund will persist.

Let β_j^k denote the coefficient of factor j in regime k , and let $1[|t_j^k| > 1.645]$ denote an indicator equal to 1 if the t -value of factor j in regime k exceeds the 10% significance threshold, 0 otherwise. Define the function across regime as such:

$$S_j = 1[|t_j^E| > 1.645] + 1[|t_j^T| > 1.645] + 1[|t_j^C| > 1.645]$$

Therefore, the stability classification T_j for each factor j is defined as the following in the function:

$$T_j = \{ \text{Reversal if } \text{sign}(\beta_j^E) \neq \text{sign}(\beta_j^T) \text{ or } \text{sign}(\beta_j^T) \neq \text{sign}(\beta_j^C) \}$$

$$T_j = \{ \text{Stable if } \text{sign}(\beta_j^E) = \text{sign}(\beta_j^T) = \text{sign}(\beta_j^C) \text{ and } S_j \geq 2. \}$$

$$T_j = \{ \text{Directional if } \text{sign}(\beta_j^E) = \text{sign}(\beta_j^T) = \text{sign}(\beta_j^C) \text{ and } S_j = 1. \}$$

$$T_j = \{ \text{Insignificant if } \text{sign}(\beta_j^E) = \text{sign}(\beta_j^T) = \text{sign}(\beta_j^C) \text{ and } S_j = 0. \}$$

For the Excel function, a reversal signal means that there is a probability that the magnitude of the sensitivity and variations between the observed data and the returns of hedge funds could indicate that the macroeconomic regime is most likely changing, while the directional indicates that a macroeconomic regime will follow a specific trend across the period. Lastly, an insignificant signal could mean that the variations in the observed macroeconomic returns and their distance to the average observed excessive returns won't impact and change the macroeconomic regime in which the hedge fund indexes will operate.

While the cross-regime analysis won't provide further statistical interpretation to the results found, it allows the reader to understand how actual fluctuations or margins in the macroeconomic regimes impacting excess return for both macro and event-driven hedge fund indexes might impact the way macroeconomic conditions could change, and thus giving a signal for an investor on the potential changes that a macroeconomic regime the hedge fund investment might face.

3.1.4.7 - Markov chain build-up:

The penultimate major step in the conduction of my statistical testing includes the application of the Markov chain principle. A Markov Chain is a stochastic process that describes a succession of sequence related to possible events across an observed period of time to provide further evidence on the possible changes that could occur on the observations of a dataset, with the fundamentals being proven from the works of ³⁷Norris JR (1997) and its implications for regime transition and financial meaning for hedge funds thanks to the article of ³⁸Hamilton and J.D (1989). A transition matrix between multiple variables is built to observe all of the combinations and probabilities of switches from a state into another, noted as the transition probabilities.

³⁷ Norris & J.R (1997)

³⁸ Hamilton & J.D (1989)

Across the thesis's build-up, the first step was to construct the transition matrix between the three defined macroeconomic regimes. To do so, a 3-by-3 matrix was build to evaluate the existing probabilities of switch between the regimes across all of the observations in the study. The table can be explained below:

Regime 1/Regime 2	Expansion (x)	Transition (y)	Crisis (z)	Row total
Expansion (x)	x_x	x_y	x_z	$x_x + x_y + x_z$
Transitional (y)	y_x	y_y	y_z	$y_x + y_y + y_z$
Crisis (z)	z_x	z_y	z_z	$z_x + z_y + z_z$

With:

- x_x = counting factor value between expansion regimes observations in macro and event-driven hedge funds.
- x_y = counting factor value between expansion and transitional regimes in macro and event-driven hedge funds.
- x_z = counting factor value between expansion and crisis regimes in macro and event-driven hedge funds.
- y_x = counting factor between transitional and expansion regimes in macro and event-driven hedge funds.
- y_y = counting factor between transitional regimes in macro and event-driven hedge funds.
- y_z = counting factor between transitional and crisis regimes in macro and event-driven hedge funds.
- z_x = counting factor between crisis and expansion regimes in macro and event-driven hedge funds.
- z_y = counting factor between crisis and transitional regimes in macro and event-driven hedge funds.
- z_z = counting factor between crisis regimes in macro and event-driven hedge funds.

To count all of the observations between each specific regime across the previous worksheets, the COUNTIFS, given the set of conditions related to the macroeconomic regime to be searched being crucial for identifying each variable:

$$x_x = \sum_{i=1}^N fn(Observe Period n; Expansion n); fn(Observe Period n; Expansion n)$$

$$x_y = \sum_{i=1}^N fn(Observe Period n; Expansion n); fn(Observe Period n; Transition n)$$

$$x_z = \sum_{i=1}^N fn(Observe Period n; Expansion n); fn(Observe Period n; Crisis n)$$

$$y_x = \sum_{i=1}^N fn(Observe Period n; Transitional n); fn(Observe Period n; Expansion n)$$

$$y_y = \sum_{i=1}^N fn(Observe Period n; Transitional n); fn(Observe Period n; Transitional n)$$

$$y_z = \sum_{i=1}^N fn(Observed\ Period\ n; Transitional\ n); fn(Observed\ Period\ n; Crisis\ n)$$

$$z_x = \sum_{i=1}^N fn(Observed\ Period\ n; Crisis\ n); fn(Observed\ Period\ n; Expansio\ n)$$

$$z_y = \sum_{i=1}^N fn(Observed\ Period\ n; Crisis\ n); fn(Observed\ Period\ n; Transitional\ n)$$

$$z_z = \sum_{i=1}^N fn(Observed\ Period\ n; Crisis\ n); fn(Observed\ Period\ n; Crisis\ n)$$

Once all of the observed probabilities between the macroeconomic factors, as well as the row total for the matrix has been established, transition probabilities successions had to be established the actual statistical chance that a switch in the regime could occur or not.

The first probability matrix table, known as the first-order Markov property, will be the first step to validate the probability of regime switches that could be observed across the selected hedge funds. The matrix takes the form, where each entry $P(I;j)$ representing the conditional probability to transition from regime i to regime j in the following month, with the following formulas being used:

Regime	Expansion	Transitional	Crisis	Total
Expansion	$x_x / x_x + x_y + x_z$	$x_y / x_x + x_y + x_z$	$x_z / x_x + x_y + x_z$	Sum of the rows
Transitional	$y_x / y_x + y_y + y_z$	$y_y / y_x + y_y + y_z$	$y_z / y_x + y_y + y_z$	Sum of the rows
Crisis	$z_x / z_x + z_y + z_z$	$z_y / z_x + z_y + z_z$	$z_z / z_x + z_y + z_z$	Sum of the rows

Once the first-order Matrix table has been completed, successive probabilities matrix tables have been constructed by conducting exclusive matrix multiplication that explains the derivation of multi-period transition regimes, which use the MMULT Excel function that will return the product of two arrays with the same number of columns and rows. In the evaluation of the probability matrix tables, the MMULT was used all the time by taking all of the obtained probabilities in the table during each derivation conducted, which can be expressed as the following.

This can be expressed statistically as the two-step transition probability, given i regime one and j regime j for given intermediate k macroeconomic regime. Noted as P^2 , which will yield the two-month-ahead transition probabilities between all of the regimes, the formula is written as it follows:

$$P^2(i;j) = \sum_k P(i;k) * P(k;j)$$

Once all of the derivations for each month-period succession of transition probabilities between the defined macroeconomic regimes covers all of the observed data from the initial database, the final probability matrix table will include the weights that each macroeconomic regime will serve in the calculation of the performance that a portfolio combining both macroeconomic and event-driven indexes will yield from January 2018 to December 2025.

3.1.4.8 - Portfolio construction:

The last step of the conducted research's methodology includes the construction of an actual portfolio that will evaluate the portfolio's performance based on the representative weights of each macroeconomic regime, as well the weight that each hedge fund strategy will have on the portfolio. The first step for constructing the portfolio was to calculate the mean excess, standard deviations for both macro and event-driven hedge fund indexes across all three macroeconomic regimes, as well as the ³⁹correlation coefficients between the excess returns for the two hedge fund strategies to evaluate whether the performance yielded by the two strategies can be explained by changes in one of the strategies.

First, the average of each observed regime's excess return for both macro and event-driven hedge fund indexes was calculated by using the AVERAGE function on Excel, which can be expressed as follows:

$$\text{Mean Excess Macro-Expansion} = \mu_{me} = \frac{\sum \text{Excess Macro Returns Expansion}}{\sum \text{Total } n \text{ Excess Macro Returns Expansion}}$$

$$\text{Mean Excess Macro-Transitional} = \mu_{mt} = \frac{\sum \text{Excess Macro Returns Transitional}}{\sum \text{Total } n \text{ Excess Macro Returns Transitional}}$$

$$\text{Mean Excess Macro-Expansion} = \mu_{mc} = \frac{\sum \text{Excess Macro Returns Crisis}}{\sum \text{Total } n \text{ Excess Macro Returns Crisis}}$$

$$\text{Mean Excess Event-Driven Expansion} = \mu_{ee} = \frac{\sum \text{Excess Event-Driven Returns Expansion}}{\sum \text{Total } n \text{ Excess Event-Driven Returns Expansion}}$$

$$\text{Mean Excess Event-Driven Transitional} = \mu_{em} = \frac{\sum \text{Excess Event-Driven Returns Transitional}}{\sum \text{Total } n \text{ Excess Event-Driven Returns Transitional}}$$

$$\text{Mean Excess Event-Driven Expansion} = \mu_{ec} = \frac{\sum \text{Excess Event-Driven Returns Crisis}}{\sum \text{Total } n \text{ Excess Event-Driven Returns Crisis}}$$

The next step follows the same logic, with the focus turning instead on calculating on the standard deviations for each macroeconomic regime related to its hedge fund index by using the STDEV Excel function to garner standardized results. The functions can be expressed as follows:

$$\text{Standard Deviation Macro-Expansion} = \sigma^{\text{ME}} = \sqrt{\sum \frac{\sum_{i=1}^n (Rt^m - Rf - \mu_{me})^2}{n-1}}$$

$$\text{Standard Deviation Macro-Transitional} = \sigma^{\text{MT}} = \sqrt{\sum \frac{\sum_{i=1}^n (Rt^m - Rf - \mu_{mt})^2}{n-1}}$$

$$\text{Standard Deviation Macro-Crisis} = \sigma^{\text{MC}} = \sqrt{\sum \frac{\sum_{i=1}^n (Rt^m - Rf - \mu_{mc})^2}{n-1}}$$

$$\text{Standard Deviation Event-Driven Expansion} = \sigma^{\text{EDE}} = \sqrt{\sum \frac{\sum_{i=1}^n (Rt^{ed} - Rf - \mu_{ede})^2}{n-1}}$$

³⁹ Markowitz (1952)

$$\text{Standard Deviation Event-Driven Transitional} = \sigma^{\text{EDT}} = \sqrt{\sum_{i=1}^n \frac{(Rt^{ed} - Rf - \mu_{edt})^2}{n-1}}$$

$$\text{Standard Deviation Event-Driven Crisis} = \sigma^{\text{EDC}} = \sqrt{\sum_{i=1}^n \frac{(Rt^{ed} - Rf - \mu_{edc})^2}{n-1}}$$

The last step was to calculate the correlation coefficient between the excess macro and excess event-driven returns in each macroeconomic regime, defined as follows for each regime:

$$p_i = \frac{\text{Cov}(R_m - R_f; R_{ed} - R_f)}{\sigma_{mi}; \sigma_{edi}}$$

With the inclusion of the coefficients weights for each macroeconomic regime that has been obtained thanks to the probability matrix table in the Markov chain process, the third step was to construct the allocated weight for each hedge fund index, so that it could be eventually applied in the construction of the portfolio's metrics, including its mean, standard deviation, and Sharpe ratio.

The build-up process for weight allocation of the two hedge fund indexes traces from the fundamentals of the mean-variance optimisation framework formulated by Markowitz (1952) integrating matrix probabilities obtained from the Markov chain. The reasoning is that the tested model won't assume a single macroeconomic environment, but rather a multitude amount of regimes defined earlier that have allocated weights to portfolio return and risk, and thus taking into account the direct impact of macroeconomic variables across the hedge fund portfolio. This process will eventually be explained later, once the Sharpe ratio's process is defined, which is correlated directly with the optimized weight allocation for each hedge fund strategy.

The next step was to build and optimize the portfolio construction, with the first idea being around building the mean of the portfolio by taking into account the probability matrix coefficients obtained from the Markov chain process, the observed means for each hedge fund index per macroeconomic environment, as well as the optimized weight allocation for each hedge fund index, with the weights being distributed equally at 50% before optimizing the weights once calculating the Sharpe Ratio. To calculate the portfolio mean, noted as μ_p , the following formula was used:

$$\mu^p = (p_e * (w_m * \mu_{me} + w_{ed} * \mu_{ee})) + (p_t * (w_m * \mu_{mt} + w_{ed} * \mu_{et})) + (p_c * (w_m * \mu_{mc} + w_{ed} * \mu_{ec})))$$

With:

- μ^p = the weighted portfolio mean.
- p_e = the coefficient of the expansion regime.
- w_m = the weight allocation for macro-based hedge fund index strategy.
- μ_{me} = the excess mean of macro-based hedge fund index during the expansion regime.
- w_{ed} = the weight allocation for event-driven hedge fund strategy.
- μ_{ee} = the excess mean of the event-driven hedge fund index during the expansion regime.

- μ_{mt} = the excess mean of the macro-based hedge fund index in the transitional regime.
- μ_{et} = the excess mean of the event-driven hedge fund index in the transitional regime.
- μ_{mc} = the excess mean of the macro-based hedge fund index in the crisis regime.
- μ_{ec} = the excess mean of the event-driven hedge fund index in the crisis regime.

The next step was to determine the standard deviation of the portfolio. Yet, in order to calculate the standard deviation, we had first to find the variance of the portfolio. Different variables, with Var_r (variance within regimes), Var_{br} (variance between regimes), and Var_t (total variance) being defined as following:

$$\text{Var}_r = \sum_i \pi_i [w^2 * \sigma^2_{M,i} + (1 - w)^2 * \sigma^2_{ED,i} + 2w(1 - w)\rho\sigma_{M,i}\sigma_{ED,i}]$$

$$\text{Var}_{br} = \sum_i \pi_i (\mu_{p,i} - \mu_p)^2$$

$$\text{Var}_t = \text{Var}_r + \text{Var}_{br}$$

By taking the total variance of the portfolio based on the intra and inter-regime variances for the macroeconomic regimes, we can thus deduct the standard deviation of the portfolio by applying the following formula:

$$\sigma = \sqrt{\text{Var}_t}$$

Lastly, the Sharpe ratio of the combined hedge fund strategies was obtained by applying the following formula, based on the initial weight allocation per each hedge fund index:

$$\text{Sharpe Ratio} = \frac{\mu_p}{\sigma \sqrt{12}}$$

Nonetheless, the first raw results in the Mean, Standard Deviation, and Sharpe Ratio of the portfolio are not sufficiently enough to determine which hedge fund index is the most ideal to allocate in a designed portfolio. Within this scenario, the Solver function was used, a function that helps finding the most optimal weight value for macro hedge funds (given that macro hedge fund strategy was the main factor of the research thesis) to calculate the final Sharpe ratio. By finding the optimal weights for each hedge fund strategy, the Mean and Standard Deviation of the portfolio would also be optimized further. The idea around the Solver function was to select Sharpe Ratio as the target cell for finding the most optimal and performant value confirming the dynamics of the portfolio's exposure to the underlying macroeconomic factors by changing the weight value proportion of the total allocation for the macro hedge fund index.

The last, and final step, of the statistical approach of the thesis consists of creating indexes evaluating the profitability that each hedge fund index is yielding across all of the observed periods, as well as a general portfolio index. To calculate the indexes, the first step was to calculate the Monthly Portfolio Return, noted as R_p which can be expressed as follows:

$$R_p = w_{mn} * ER_{mn} + w_{evn} * ER_{evn}$$

By doing so, the index coefficients for the portfolio, macro, and event-driven hedge funds can be built. Noted as I_p , I_m , and I_{ev} , the start point was 100 for the first observed period (January 2018), as no previous data was used in the research. Given the parameters used, each index can be calculated as per the following formulas:

$$I_{p\ n} = I_{p\ n-1} * \left(1 + \frac{Rp\ n}{100}\right)$$

$$I_{m\ n} = I_{m\ n-1} * \left(1 + \frac{Excess\ Macro\ Return\ n}{100}\right)$$

$$I_{ev\ n} = I_{ev\ n-1} * \left(1 + \frac{Excess\ Event-Driven\ Return\ n}{100}\right)$$

Once all of the index coefficients have been calculated, a graphic was created to evaluate the dynamics of both the portfolio and the two hedge fund indexes in order to analyze and find patterns on the behavior of each variable, while also trying to find whether the underlying macroeconomic variables could explain results and performance of the different hedge fund strategies tested.

IV – Findings and analysis:

4.1 – Brief review of research question and introduction:

The research question, which is focusing on the way that macro and event-driven hedge funds exhibit risk-mitigation mechanisms over the presence of macroeconomic indicators under different macroeconomic regimes, relies on distributing for each observation a macroeconomic regime, as they were defined during the Data & Methodology section of the paper research.

For the period of observation, which lies from January 2018 until December 2025, grouping a total of 96 observations saw 26 expansion months, 47 transitional, and 22 crisis periods. With the large amount of transitional observed periods, the research paper reflects the cumulation of momentum and uncertainty that lies around macroeconomic environments during periods of crisis, regardless of their nature. With the recent periods of economic stress, such as the trade war between the US and China (2018-2019), the COVID 19 Pandemic period (2020-2022), as well as inflation hikes and tight interest rates (from March 2022 until roughly November 2023) produced mixed signals that regime classification relying on macroeconomic variables and their dynamics showing that hedge fund returns could be sensitive to any potential changes or shocks in any crucial macroeconomic factor.

4.2 – Descriptive statistics per macroeconomic regime:

The first analysis of the Excel model would be around the general tendencies and observations we've got for all of the model's macroeconomic regimes. In this sense, the model comprises 96 observations stretching from January 2018 to December 2025, with a total of 26 months spent in

the Expansion regime, 48 months in the Transitional regime, and 22 months in the Crisis regime. The large presence of transitional regimes comes to no surprise given the large amount of economic uncertainty triggered by the different events, including the trade war between the United States of America and China in 2018, the Covid-19 Pandemic between 2020 and 2021, as well as the inflationary shocks occurring post-Covid and that have been accentuated by the War in Ukraine between 2022 and 2023. Throughout this sub-section, all of the regime's key metrics, including the performance metrics, coefficients for each of the regression's variables, as well as statistical metrics including the p-value, t-value, or R^2 coefficients have been observed and analyzed.

4.2.1 - Expansion regime results:

Regression Statistics		Performance metrics	
Multiple R	0,493516	Annual Sharpe Ratio	0,6672
R Square	0,243558	Sortino Ratio	0,2149
Adjusted R Square	0,054448	Omega Ratio	1,8686
Standard Error	1,600998	Max drawdown	-0,12%
Observations	26	Average excess u	0,3171
		Skewness	0,5363
		Kurtosis	-0,2668

Figure: the regression statistics and performance metrics for the HFRI Macro Index.

In this first small section, we will analyze the results found for both macro and event-driven hedge fund indexes during the expansion macroeconomic regime. For the HFRI Macro Index, we can see that the fund has yielded roughly 0,31% on a monthly basis, with a standard error being roughly around 1,6, while the annualized Sharpe Ratio has resulted in 0,66. Given the values of Sortino Ratio at 0.21 and Omega Ratio at 1.86, downside indicators may have been present, especially given the decrease in the changes of VIX and gold returns from the OLS model, with the former yielding an average at -0,2283, while the latter had seen a coefficient of -1,186. This shows not only that volatility was not significantly impacting the performance and the stability of macro hedge funds in the expansion regime, but also that the decreases in the VIX might relate to a potential similar move in gold returns, all while enabling the Dollar Index to remain more consistent, with the changes in the DXY yielding a coefficient of 0,05 for the environment. The maximum drawdown was contained at -0,12%, which is supported by the strong and stable nature of cumulative return indexes calculated compared to the maximum running return that yielded at 1,09. The observed and structured R^2 coefficient at 0,24, as well as 0,05 for the Adjusted R^2 indicates the idea of importance for certain macroeconomic weights in the results of hedge funds, as the low course of volatility explains also a weaker impact of the other factor drivers. This stability is lastly confirmed by the Skewness of 0,53 and Kurtosis of -

0,2, indicating that no major deviations occurred for macro-based hedge funds during the calm period.

<i>Regression Statistics</i>		Performance metrics	
Multiple R	0,886505	<i>Annual Sharpe Ratio</i>	0,5586
R Square	0,785892	<i>Sortino Ratio</i>	0,0619
Adjusted R Square	0,732365	<i>Omega Ratio</i>	0,5577
Standard Error	1,674732	<i>Max drawdown</i>	-0,24%
Observations	26	<i>Average excess u</i>	0,5220
		<i>Skewness</i>	-2,3194
		<i>Kurtosis</i>	10,500

Figure: the regression statistics and performance metrics for the HFRI Event-Driven index.

Regarding the HFRI Event-Driven index, who also saw stable and positive results in its performance, albeit with some differences compared to the behavior of macro hedge funds. While yielding a higher excess return on average at 0,52, the HFRI Event-Driven index saw a lower annualized Sharpe ratio at 0,55, as well as lower values for the Sortino Ratio (0,06) and Omega Ratio (0,55) given the standard error defined at 1,67. With higher R^2 and Adjusted R^2 coefficients than the HFRI Macro Index at 0,88, respectively 0,78, indicate that stability of the general market is much more favorable, as per the 0,74 coefficient for the 5-year yield change and market factor at 0,74. This is surprisingly noticeable though, given the negative Skewness of -2,319, respectively high 10,5 of Kurtosis, showing that event-driven funds can fail if IPOs underperform or if M&A deals might yield into lower financial metrics than expected, even during normal and stable conditions. Even so, a weaker power in currency and slight positive swings in the VIX might be preferable for the HFRI Event-Driven index given the strategy's favoring of economic announcements or announced corporate events, as per the third hypothesis of this research.

4.2.2 - Transitional regime results:

<i>Regression Statistics</i>		Performance metrics	
Multiple R	0,591118	<i>Annual Sharpe Ratio</i>	0,4887
R Square	0,349421	<i>Sortino Ratio</i>	1,2343
Adjusted R Square	0,271971	<i>Omega Ratio</i>	0,7827
Standard Error	1,307107	<i>Max drawdown</i>	-0,01%
Observations	48	<i>Average excess u</i>	0,2161
		<i>Skewness</i>	-0,4927
		<i>Kurtosis</i>	0,8910

Figure: the regression statistics and performance metrics for the HFRI Macro Index.

The next regime to analyze will be the transitional regime, which has the highest number of observations, with 48 months of the 2018-2025 period classified as such. For the HFRI Macro Index produced a total excess return of 0,2161, all while yielding an annualized Sharpe ratio at 0,4887, all while having a standard error slightly higher at roughly 1,30. Thanks to the Sortino ratio of 1,23 and Omega Ratio of 0,78, the downside effects of volatility have been controlled by the macro hedge funds, although the macroeconomic uncertainty is higher than in the expansion regime. This statement can be reported further by the maximum drawdown of -0,01% yielded for the regime in a context where the running cumulative returns had a better-than-expected effect. The Skewness coefficient of -0,49 and the Kurtosis coefficient at 0,89 depict that, although the left tail distribution was much fatter, no excess returns have deviated from the average excess return that the HFRI Macro Index has shown, indicating a better stability than expected. The R^2 of roughly 0,35 as well as the adjusted R^2 of 0,27 indicate that stability of the main sources of returns for macro hedge funds might be more impacted, especially given the higher coefficient for the changes in volatility index, reported at 0,11 per the OLS model.

Regression Statistics		Performance metrics	
Multiple R	0,851977	Annual Sharpe Ratio	0,3603
R Square	0,725865	Sortino Ratio	1,2289
Adjusted R Square	0,69323	Omega Ratio	1,1776
Standard Error	0,973439	Max drawdown	-0,15%
Observations	48	Average excess u	0,1828
		Skewness	0,0416
		Kurtosis	-0,0986

Figure: regression statistics and performance metrics for the HFRI Event-Driven Index.

On the other hand, the HFRI Event-Driven Index saw an excess average return of 0,18, while its Sharpe ratio was at 0,36. The interesting observation is the 1,22 Sortino ratio, which is almost similar to the 1,23 Sortino ratio of HFRI Macro Index, showing that downside movements are mostly similar on average between the two hedge fund strategies. With strong R^2 and adjusted R^2 coefficients at 0,72, respectively 0,69, event-driven hedge funds behave normally during the transitional period, showing that slow climbs in the VIX or 5-year treasury bill might not affect at first glance the performance of event-driven hedge funds. Although the maximum drawdown is higher, reported at -0.15%, the higher cumulative returns, as well as the lack of deviations in the distribution of excess-returns, as per shown by the Skewness coefficient (0,04) and Kurtosis (-0,09) indicate the relative stability of event-driven hedge funds amid a higher uncertainty over the macroeconomic environment.

4.2.3 - Crisis regime results:

Regression Statistics	Performance metrics
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Multiple R	0,561627
R Square	0,315425
Adjusted R Square	0,101495
Standard Error	1,60068
Observations	22

<i>Annual Sharpe Ratio</i>	-0,2156
<i>Sortino Ratio</i>	-0,5970
<i>Omega Ratio</i>	1,8594
<i>Max drawdown</i>	-0,08%
<i>Average excess u</i>	-0,1051
<i>Skewness</i>	1,3081
<i>Kurtosis</i>	4,1989

Figure: the regression statistics and performance metrics for HFRI Macro Index.

<i>Regression Statistics</i>	
Multiple R	0,834296
R Square	0,696049
Adjusted R Square	0,601064
Standard Error	0,913267
Observations	22

Performance metrics	
<i>Annual Sharpe Ratio</i>	1,0909
<i>Sortino Ratio</i>	3,0964
<i>Omega Ratio</i>	1,1479
<i>Max drawdown</i>	-0,05%
<i>Average excess u</i>	0,4554
<i>Skewness</i>	-0,1860
<i>Kurtosis</i>	-0,7873

Figure: the regression statistics and performance metrics for HFRI Event-Driven Index.

The last macroeconomic regime to analyze is the crisis regime, which might be the most skewed and sensitive regime, given the total of 22 observations as per the macroeconomic classification guideline. For the HFRI Macro Index, the OLS model yielded an average excess return of -0,1, while the annualized Sharpe ratio was at -0,21, confirming the sensitivity of exposure that macro hedge funds behave related to the changes in the macroeconomic environment. This is further confirmed by the Sortino ratio at -0,597. Surprisingly, the main paradox is the 1,5894 Omega ratio of the model, showing a parallel with the research done by Fung and Hsieh (2004) showing that the convexity of the macro returns can help these hedge funds stabilizing their source of income amid lower performances. With a Skewness of 1,30 and Kurtosis of 4,19, this further confirms that while the negative period affected the average performance of macro hedge funds, the highest peaks in returns, which are more frequent than in any other regime have been more impactful for the final performance of the index. Even more, the fund is still affected overall by the stability of the regime, given the R^2 of 0,31, as well as the adjusted- R^2 of 0,1. Furthermore, by analyzing the coefficients in the changes of VIX (-0,31) and return of gold prices (3,62), macro hedge funds can be sensible to the higher volatility that is occurring on the market, which is at a higher frequency than during normal periods, all while being able to adjust their hedging by taking positions in commodities.

For the HFRI Event-Driven Index, the situation is completely different, and much different than the performance garnered by the HFRI Macro Index. In effect, with an average excess return of 0,45, an annualized Sharpe ratio at 1,09, as well as Sortino and Omega ratio at 3,09, respectively 1,14 show a surprising effect, and thus yielding the strongest stable performance for the two hedge fund strategies for the whole observed macroeconomic regimes. Fewer extreme observations compared to the average are observed, as the Skewness was reported at -0,16, respectively -0,78. The strong 0,69 and 0,6 results for the R^2 , respectively adjusted R^2 , show the robustness of the strategy amid stress-level conditions. From the analysis of the coefficients, and especially the changes in VIX (0,03), return in gold prices (-3,4), event-driven hedge funds might take advantage of the crisis for finding any potential investment opportunities, especially in distressed debt securities or fixed-income positions. While the results are surprisingly noticeable, the low number of observations might put the yielded results under question, especially given the evaluated period of 2018 to 2025.

4.3 – Cross-regime analysis and Markov chain:

In this section, we will analyze two of the most important parts of the research. The first analysis will be around the cross-regime analysis which will investigate further the impact of the coefficients, p-values and t-values for each factor of the linear regression under each macroeconomic regime, and for each hedge fund strategy, all while understanding the impact that these factors have on the potential switch between regimes. As a brief reminder from the previous sections, any p-value that is below or equal to 0.05 would show statistical significance for the results that have been calculated, while any value above the 0.05 benchmark would garner statistical insignificance, albeit it could leave for further financial and economic interpretation of the results. At the same time, the t-value will also show how much deviations in each variable might influence the results yielded for each factor of the linear regression. Lastly, the probabilities of switch or stability for each regime will be tested further by explaining the Markov chain probability scheme. Before starting the analysis of each coefficient along with its p-values and t-values, the summarized results can be found in this graphic:

HFRI Macro Index Analysis											
Parameter	Expansion Coef	Expansion P Value	Expansion t-value	Transition Coef	Transition P Value	Transition t-value	Crisis Coef	Crisis P Value	Crisis t-value	Stability - regime swi	
α	0,280	0,454	0,763	0,031	0,878	0,154	-0,338	0,457	-0,763	Reversal	
Changes Volatility Im	-0,228	0,082	-1,833	0,114	0,068	1,874	-0,313	0,027	-2,430	Reversal	
Changes US Dollar In	0,052	0,864	0,173	0,127	0,236	1,201	0,215	0,316	1,036	Insignificant	
Return Gold Prices	-1,119	0,906	-0,119	4,904	0,352	0,942	3,623	0,663	0,444	Reversal	
5-year yield change	0,729	0,563	0,589	2,692	0,016	2,523	0,126	0,915	0,109	Insignificant	
Market Factor (MKT)	0,063	0,485	0,711	0,146	0,003	3,216	-0,198	0,173	-1,427	Reversal	
HFRI Event-Driven Index Analysis											
Parameter	Expansion Coef	Expansion P Value	Expansion t-value	Transition Coef	Transition P Value	Transition t-value	Crisis Coef	Crisis P Value	Crisis t-value	Stability - regime swi	
α	-0,048	0,895	-0,133	0,078	0,607	0,519	0,196	0,450	0,775	Reversal	
Changes Volatility Im	0,041	0,757	0,314	0,052	0,252	1,161	0,032	0,671	0,433	Insignificant	
Changes US Dollar In	-0,905	0,009	-2,901	-0,028	0,720	-0,361	0,042	0,725	0,358	Reversal	
Return Gold Prices	-27,006	0,012	-2,746	-2,111	0,589	-0,544	-3,409	0,474	-0,733	Stable	
5-year yield change	0,748	0,570	0,577	-1,023	0,205	-1,288	-0,717	0,295	-1,084	Reversal	
Market Factor (MKT)	0,474	0,000	5,144	0,289	0,000	8,574	0,297	0,002	3,743	Insignificant	
Summary Statistics											
	Expansion - macro	Expansion - Event Driv	Transition - macro	Transition - Event Driv	Crisis - Macro	Crisis - Event Driven					
R Square	0,244	0,786	0,349	0,726	0,315	0,696					
Adjusted R Square	0,054	0,732	0,272	0,683	0,101	0,601					
Sharpe Ratio	0,667	0,559	0,489	0,360	-0,216	1,091					
Sortino Ratio	0,215	0,062	1,234	1,229	-0,597	3,096					
Omega Ratio	1,869	0,558	0,783	1,178	1,859	1,148					
Maximum Drawdown	-0,001	-0,002	0,000	-0,002	-0,001	0,000					

4.3.1 - Cross-regime analysis:

4.3.1.1 - General overview of the alpha's evolution:

The first observation would lie around the main values that have been generated in the cross-regime analysis by analyzing each coefficient, p-value, and t-value for all of the factors constituting the linear regression model, including the alpha (α), Δ in the VIX, Δ in the DXY, return of gold prices, Δ 5-year treasury yield, as well as the market factor for all macroeconomic regimes, as well as for both the HFRI Macro Index and HFRI Event-Driven Index. By looking briefly at the alpha coefficients obtained in the HFRI Macro Index, we can notice the similar patterns and economic trends observed earlier in the interpretation of data from all three regimes. From a stable coefficient at 0,26 during expansion regime to -0,338 in a crisis period, the behavior espoused by macro hedge-funds might relate to the assumptions of our hypotheses, in a way that macro hedge funds are more sensible to the context and regime in which they do operate. Nonetheless, the low values observed show that the model doesn't show any statistical significance, all while the hedge funds might not always be able to beat directly their benchmark (in the research's context the S&P 500's monthly returns) during unstable macroeconomic conditions.

On the other hand, the HFRI Event-Driven Index's alpha coefficients show a different pattern, which confirms the earlier observations around the average excess returns and Sharpe ratios calculated during all three macroeconomic regimes. While the initial coefficient is significantly low during the expansion regime at -0,048, the coefficient grows later, from 0,078 during transition up to 0,196 while being in the crisis regime. This later confirms one of the findings during the interpretation of the crisis regime results interpretation, during which the robustness of event-driven hedge funds was confirmed by the latter's strong R^2 and adjusted- R^2 coefficients.

As a conclusion, the yielded alpha coefficients for the two hedge fund strategies in the three selected macroeconomic environments may confirm the findings that have been found earlier during the general analysis of the descriptive statistics for each regime. However, given the low coefficient scores, as well as the high p-values which are for all macroeconomic regimes above 0.05 (averaging 0.59 for the HFRI Macro Index and 0,65 for the HFRI Event-Driven Index) and the high deviations in the observed variables as it can be seen in the t-values (averaging 0,051 for HFRI Macro Index, respectively 0,651 for the HFRI Event-Driven Index), no statistical evidence can be yielded from the OLS model, and thus concluding that we cannot confirm whether alpha opportunities can actually be dependent on the macroeconomic regime in which the hedge funds operate.

4.3.1.2 - VIX analysis:

The next variable from our model is the interpretation of the VIX's changes, with the HFRI Macro Index having the most significant statistical results. In fact, with a coefficient of 0,228 in the expansion regime and a specific statistical inference (p-value equal to 0,082 and t-value at -1,83), a coefficient of 0,114 in the transitional environment (p-value equal to 0,068 and t-value at

1,87), as well as a coefficient of -0,313 during a crisis environment (with p-value equalling 0,027 and t-value at -2,43) indicate that volatility changes during macroeconomic regimes correlate with the movements that macro hedge funds are espousing, and thus confirming the initial idea of Ang & Timmermann (2012), all while demonstrating briefly that VIX might have a significant power in the behavior of hedge funds, as it was formulated during the research's hypotheses. These statistical data also confirm that VIX can hurt macro hedge funds during calm periods, all while enabling those hedge funds to garner potential returns during transitional phases, as hedge funds will capture volatility-based positions that could benefit from the volatility swings that appear more probable during the transitional and crisis periods.

At the same time, the presence of negative returns in both the coefficient and t-values observed show that exploiting volatility portions are negatively impacting the performance of the macro-driven hedge funds, given the higher exposure these hedge funds represent. It joins further the observations noted by Ang and Timmermann in which volatility mechanisms can translate into higher drawdowns for hedge funds. The presence of p-value coefficients also show that VIX are driving the movement and logic of macro hedge funds, confirming further the point that VIX is one of the main factors influencing the movement of macro hedge funds, regardless of the macroeconomic regime.

Nonetheless, the observed results for the HFRI Event-Driven Index cannot be fully interpreted, given the statistical non-significance of the results (p-value averaging at 0,56 for the three regimes, and t-value averaging at 0,71), concluding that no direct relationship could be determined between the behavior of event-driven hedge funds and the changes in volatility, reflecting the primary nature of event-driven hedge funds in which corporate deals or public economic announcements will serve as the main drivers for event-driven strategies.

4.3.1.3 - DXY and treasury-yield analysis – a monetary approach:

In this section, we will announce the way that DXY changes, as well as changes in the 5-year treasury yield bill could affect the hedge fund studied, given the general impact that monetary policies may have on the macroeconomic environment as a whole, all while influencing the hedging positions that both macro and event-driven hedge funds might undertake, especially around fixed-income securities.

Regarding the changes in the 5-year treasury yield in the HFRI Macro Index, the 0,72 coefficient in the expansion regime (although insignificant given its 0,56 p-value), the 2,69 coefficient (statistically significant at 0,016 for the p-value) in the transitional regime, as well as 0,12 coefficient (statistically insignificant, with the p-value = 0,915) are signaling that the treasury yield announcements and changes might not have a specific significance for the behavior of macro hedge-funds. The significance around the transitional regime is mostly explained by the uncertainty macroeconomic environment between 2022 and 2023, during which the monthly movements saw the biggest swings, averaging a total average movement of 0,12 during this period, as well as the hawkish stances that the FED took against the gripping inflation during the

same period. In this sense, the 5-year treasury yield changes act more as a second driver for macro hedge funds, working under scenarios where the monetary policy becomes tighter. This is also explained by the observations in the changes of the Dollar Index, as none of the coefficients obtained for the three macroeconomic regimes are statistically significant, given the high p-values observed (averaging around 0,47 for the three regimes), making the factor insignificant for a potential macroeconomic regime switch.

In the context of the HFRI Event-Driven Index, the interpretation of the two factors is also eventually insignificant from a statistical point of view. Although potential reverse could be driven partially by the changes in the DXY and 5-year treasury yield, the statistical significance is cancelling out any potential weight that these two factors may have on the event-driven hedge funds. Regarding DXY, alpha coefficients of -0.9 in the expansion regime (p-value = 0,09, which would most likely be explained by the favorable macroeconomic environment where borrowing costs are significantly low), -0,028 in the transitional environment (p-value at 0,72) and 0,042 for the crisis period (p-value at 0,725) do not drive the behavior of event-driven hedge funds, who are more likely to face tighter constraints regarding the announced economic or corporate deals. For the 5-year treasury yield changes, the yield coefficients of 0,748 in expansion, -1,023 in transitional and -0,717 during crisis cannot be explained due to the very high p-values obtained. One pattern for event-driven hedge funds could be that rising yields may help funds at stable levels but could impact significantly if the variations are too severe, which would constrain borrowing costs or incurred financing opportunities for corporate deals, and thus indirectly impacting the performance of event-driven hedge funds.

4.3.1.4 - Evaluation of returns in gold:

For the HFRI Macro Index, the alpha coefficients at -1,11 for expansion, 4,9 for transitional and 3,62 in the crisis environment show no statistical significance given the very high nature of the yielded p-values (averaging 0,64 for all three regimes). Nonetheless, the implications are not observed from a statistical point of view, but rather in the economic perspective, as the movements in gold prices returns could signal a potential implicit change of the environment on the middle run. Gold sensitivity is significantly high during transitional and crisis periods in which gold futures positions' prices tend to soar, and thus potentially triggering long positions for macro-driven hedge funds during those periods. Macro hedge funds holding long gold positions might benefit in a context where the VIX changes significantly and when the monetary policy becomes tighter, given the opposite movements that return gold prices take around the two factors, with the correlation between the VIX changes and return in gold prices being at -0,028, while the correlation with the 5-year yield change being at 0,551, and thus showing net benefits for specific macro hedge funds opting for long positions in gold contracts.

Interestingly, in the HFRI Event-Driven Index, return in gold prices are the only factor from all the research showing stability for the regime's behavior, all while showing the highest sensitivity values, with the alpha coefficient of -27 in expansion (statistically significant at p-value = 0,012),

-2,11 in transitional (statistically insignificant at $p\text{-value} = 0,589$) and -3,4 for crisis ($p\text{-value}$ at 0,47, and thus being insignificant from a statistical perspective). Although most of the observations are significant, it can explain the “stable” stability of any regime: in fact, event-driven hedge funds might move in an opposite direction from the generated returns in gold prices, as gold prices soar during periods in which M&A deals or IPOs might decrease or crash significantly, a condition to which event-driven depend crucially on for their alpha opportunities. This is further supported by the low level of gold prices that happened during the early phases of the research’s period, in which corporate events that a higher frequency than changes in gold prices.

4.3.1.5 - Evaluation of the Market Factor (MKT):

The last evaluated variable of the linear regression model includes the Market Factor (MKT), which will evaluate the influence that the benchmark of this research (S&P 500’s monthly returns) may have or not on the way both hedge fund strategies will move.

Regarding the HFRI Macro Index, the coefficient of 0,063 in expansion (statistically insignificant given the $p\text{-value} = 0,48$), 0,14 in transitional (significant at $p\text{-value} = 0,003$) and -0,19 during crisis (insignificant with $p\text{-value}$ at 0,17) might indicate some underlying patterns about macro hedge funds in relationship with the general behavior of stock markets that hasn’t been demonstrated in both the research of Fung & Hsieh (2004), as well as the article of Ang & Timmermann (2012).

In effect, during expansion regime, macro hedge funds will have a negligible movement regarding their equity market exposure, while the transition environment signals that the development of a 0,146 market beta, and thus showing that macro hedge fund managers may allocate further weight around equity-correlated beta positions ahead of potential significant movements in the environment, and thus proving the idea of dynamically and weight-adjusted idea shown by Patton & Ramadorai (2013). Even so, with the coefficient being at -0,19 during crisis, macro hedge funds will try to reduce their exposure and weight allocation around equity-based movements. Given that only the transitional regime shows a significant statistical inference, this would mean that macro hedge funds are the most optimized from the two strategies for their direct relationship to equity-market positions rather than during both optimal and crisis regimes.

For the HFRI Event-Driven index, all three regimes show significant and reliable coefficients, with 0,47 for expansion ($p\text{-value}$ below 0,0001), 0,28 for transitional ($p\text{-value}$ still below 0,0001) and 0,297 in crisis ($p\text{-value}$ at 0,002). This is the clearest and most interpretable finding from the cross-regime analysis, also indicating further movements, during which the equity-beta hold for event-driven funds remains stable, regardless of the macroeconomic regime. This also implies that the risk-adjustment mechanisms are different for event-driven hedge funds, which might not allocate huge amounts of weight or change frequently their exposure to market beta, confirming

once again that their main driver would rely more around the frequency of positive corporate deals rather than market opportunities.

4.3.2 - Analysis of the Markov chain's probabilities:

4.3.2.1 - First-order transition matrix:

	Expansion	Transitional	Crisis	Row Total	
Expansion	14	11	0	25	
Transitional	12	23	13	48	
Crisis	0	13	9	22	

Based on the first process, which evaluates the number of times during which regimes switch between them or stay in the same regime, the process might not be the most optimized for hedge fund managers that allocate financial leverage in their positions. In fact, regardless of the hedge fund strategy evaluated, we can see that none of the regimes are directly switching from one extreme to another (Expansion to Crisis and Crisis to Expansion having 0 observations), while most of the behavior shows that regimes would stay within their respective regime for a certain period (Expansion-Expansion factor being at 14 observations, Transitional-Transitional being at 23 observations while Crisis-Crisis being at 9 observations), as well as showing that all of regimes are passing by a transitional phase before completing their reversal. This could imply that managers would wait for the environment to show clear signals for at least one month: yet, this could be risky for managers that use the Markov chain as a market predictive model, given the high frequency of changes that could occur within their market, which might also impact their leverage costs for their positions.

4.3.2.2 - Interpretation of regime-switch coefficients – initial phase:

Probabilities				
56,00%	44,00%	0,00%	1	2,272727273
25,00%	47,92%	27,08%	1	1,92
0,00%	59,09%	40,91%	1	1,692307692

The expansion regime sees a steady 56% probability to remain the same for the upcoming month, a 47,95% probability that the transitional regime will remain the same for the next month and 40,91% that crisis will remain the same for the next month.

The fact that expansion regime sees the most stable robustness and stability from all three regimes comes to no surprise, given the low volatility environment which tends to favor strong and stable performances for funds. Crisis is also the least stable, which is also followed by a 59% probability to switch for a transitional regime, showing that shocks might be severe but that can be offset quickly, especially in the context of the 2022-2023 which saw partial but severe shocks, such as the War in Ukraine, inflation cycle, and hawkish interest rates announced by the FED to

tackle inflation, with the latter eventually reducing the risk of a deep crisis on the long term. The transitional regime's relative stability around 47% shows that transitional signs could move in either direction, with a 25% probability to switch for an expansion regime, as well as a 27% probability to reverse into a crisis regime, showing that transitional factors might still include relative uncertainty around the actual movement that markets take, showing that managers must take both long and short positions to tackle any abrupt movement within the market, and thus confirming the dynamical structure of the regimes.

4.3.2.3 - Two-month switch dynamics:

Probability 2		
42,36%	45,72%	11,92%
25,98%	49,96%	24,06%
14,77%	52,49%	32,74%

For the two-month period transition, several dynamics can be noticed, with the most interesting ones being the decrease of the stability of expansion and crisis regimes, with the expansion-expansion probability shifting at 42,36% while the crisis-crisis probability being at 32,74%. At the same time, the transitional-transitional probability increased slightly at 49,96%. At the same time, the convergence from expansion to crisis seems more defined at 11,92% of probability to switch from expansion to crisis, as well as the opposite movement, from crisis to expansion reaching a probability of 14,77%, illustrating the capacity for hedge fund managers to better capture and understand further the factors that might influence the macroeconomic context.

4.3.2.4 - Three-month switch dynamics:

Probability 3		
31,58%	48,47%	19,95%
27,54%	49,47%	22,99%
24,73%	50,16%	25,11%

By three months ahead, the rows and dynamics between the macroeconomic regimes are converging further. The probabilities of stability for both expansion and crisis regimes see further reductions, with 31,58% of probability for expansion-expansion, as well as 25,11% for crisis-crisis probability, while the transitional probabilities dominate almost all the columns at roughly 50% for each row. All information about the initial regimes in which funds are reduced, since hedge fund managers are already acknowledging the changing conditions.

4.3.2.5 - Four and five-month switch dynamics:

Probability 4		
28,26%	49,29%	22,45%
28,01%	49,35%	22,64%
27,83%	49,40%	22,77%

Probability 5

28,04%	49,35%	22,62%
28,04%	49,35%	22,62%
28,04%	49,35%	22,62%

By the four and fifth months of observations, values are almost identical, confirming the dominance that transitional switch periods represent, as well as the gradual but slow reduction in the stability of expansion and crisis regimes. Acknowledging the current regime provides no actual information about the future movements, reflecting the stability and knowledge that hedge fund managers dispose of.

4.3.2.6 - Six-month switch dynamics – evaluation of the steady state:

Probability 6

28,04%	49,35%	22,62%
28,04%	49,35%	22,62%
28,04%	49,35%	22,62%

By the last table, which evaluate the six-month dynamics, shows the full convergence of the regimes, showing 28,04% in expansion, 49,35% for transitional, and 22,62% for crisis, which closely matches the distribution of the total observations that had been garnered in the early phase of the data research, showing that the 2018-2025 period captures all potential macroeconomic regimes, as well as full information that could be optimal for hedge funds and their strategy adoption. At the same time, Transitional probabilities count for roughly half of the probabilities for all dynamics, meaning that a Markov-based portfolio analyzing the stability of regimes will position its trades around futures or securities that are dynamically reacting to transitional periods, even during either expansion or crisis regimes. Lastly, this system would only work for managers that project their positions on the middle run, and this specifically for a 6-month period as per the calculated Markov chain process, showing that any hedge fund manager should limit their weight exposure for each regimes to the probabilities found, meaning an allocation of 28.04% for expansion-favorable positions, 49,35% for transitional-favourable positions and 22,62% for crisis-favorable positions.

4.4 – Analysis of the weighted portfolio:

The last step of the research concludes all of the findings that have been collected earlier by constructing a weighted portfolio that compromises efficient allocation of weights for both macro and event-driven hedge funds. For this to be feasible, the first values to be found are descriptive statistics of the excess means, standard deviations and correlation coefficients for

both macro and event-driven hedge funds during the three constructed macroeconomic regimes. Furthermore, given the resulting coefficients for the three regimes thanks to the construction of the Markov chain, the mean, standard deviation and Sharpe ratio were calculated, all while optimizing the allocated weight thanks to the usage of Solver which relies on Matrix and GRG Non-Linear equations based on constraints that will be detailed within the following sub-sections. Lastly, for each month observed between 2018 and 2025, a specific portfolio index, macro and event-driven indexes would be calculated to observe the dynamics of all feasible strategies and to determine whether a separate hedge fund strategy or a combined hedge fund strategy would garner more returns for managers.

4.4.1 - Descriptive statistics – a general overview:

The general brief results from the conducted calculations of descriptive statistics per each hedge fund strategy and each macroeconomic regime are represented in the following table:

	Expansion	Transitional	Crisis
Mean Excess - Macro	0,317	0,216	-0,105
Mean Excess - Event Driven	0,522	0,183	0,455
Std - Macro	1,646	1,532	1,689
Std - Event Driven	3,237	1,758	1,446
Correlation	0,381	0,159	0,226

4.4.1.1 – Expansion regime results:

For the HFRI Macro Index, we can notice a 0,317-mean excess for the expansion regime and a standard deviation of 1,64 over the period, while the HFRI Event-Driven Index has a mean excess of 0,522 and 3,23 of standard deviation, being more than double of the volatility espoused by the macro hedge fund, which shows further explanation over the high skewness and kurtosis observed previously for the event-driven hedge funds in the expansion regime results. The correlation of 0,381 is the closest relationship between the two hedge fund strategies, suggesting a moderate positive relationship between the two strategies during calm periods, while returns differ significantly during the transitional and crisis periods.

4.4.1.2 – Transitional regime results:

For both the HFRI Macro Index and HFRI Event-Driven Index, the general excess averages are decreasing significantly at 0,21 for the macro hedge fund, respectively 0,18 for the event-driven hedge fund strategy. At the same time, the standard deviation for the HFRI Event-Driven Index decreases significantly, reaching 1,758, while the standard deviation of the HFRI Macro Index decreases slightly at 1,532. Nonetheless, the correlation between the two funds is the lowest in the transitional regime, reaching a total of 0,16 approximately, showing that the mechanisms of buffer against potential risks have a different nature. This could push a hedge fund manager to

combine long or short positions per each hedge fund strategy given the nearly 50% of allocation for transitional positions, as well as given the independent nature of the two returns.

4.4.1.3 – Crisis regime results:

The crisis regime produces the sharpest gap between the two hedge fund strategies regarding their excess mean, with the HFRI Macro Index resulting in a -0,1 average excess return, and thus being the sole observation that yields a negative value, while the HFRI Event-Driven Index generates a strong-stable 0,45 average excess return. The standard deviations remain stable for the period, with 1,69 for macro-based hedge funds and 1,44 for the event-driven strategy: nonetheless, these observations are surprising, since it gives a better stability for the event-driven hedge funds, even amid the tight macroeconomic regime. The correlation of 0,226 in this context is low and insignificant, though it means that the sensitivity that both macro and event-driven returns face is similar, especially regarding the stress signals that might explain drawdown for both regimes.

4.4.2 – Construction of the portfolio:

4.4.2.1 – Structuration of the formulas:

As explained in the Portfolio construction sub-part of the Data & Methodology section, different mathematical principles have been used for determining the mean, standard deviation and Sharpe ratio of the combined portfolio, all while finding the optimal weight allocation for both the macro , respectively event-driven hedge fund strategies. Based on the regime-switch probabilities found in the six-month switch dynamics, the 0,28 coefficient score for the expansion regime, 0,49 for transitional regime and 0,22 for the crisis regime were served as the main observations.

Once these weights have been integrated in the file, the different formulas of the portfolio have been built, with the first one calculating the mean of the portfolio, which follows the weighted average method.

The next process is finding the standard deviation of the portfolio, which is decomposed in two major parts. The first compoennt is the within-regime variance of the three economic regimes that combines both the allocated weights for each hedge fund strategy, the coefficients found for each economic regime, the standard deviations of each hedge fund strategy per macroeconomic regime and the respective correlations found, which combines the simple two-stock variance formula, while the second variance is between-regime variance which will evaluate the volatility between each regime. By totalling the two variance, we will obtain the total standard deviation of the portfolio. Lastly, the Sharpe ratio of the portfolio was calculated, which evaluates the relationship between the mean of the portfolio and the square root of the standard deviation within the portfolio. However, in order to find the Sharpe Ratio, the Solver function in Excel has been used in order to find the maximized Sharpe ratio, with the weight in the macro hedge fund being selected as the main constraint, given the higher exposure that macro hedge funds can bring to the Sharpe ratio of the whole portfolio. Furthermore, no constraints or limits in the

(0,522, 0,183, 0,455) are higher than the excess returns resulting from the macro-driven hedge fund strategy, which is also affected by the negative performance in the crisis regime (-0,105) and the low crisis regime coefficient allocation at 22,61%. Though, the lower weight allocation for macro-driven hedge funds isn't minimal to the effects it has on the whole portfolio's performance, given the lower sensibility to potential volatility shocks as per the average standard deviation across the three regimes (average standard deviation of the macro-based hedge fund being at 1,62, compared to 2,15 for event-driven hedge fund), as well as the superior annualized Sharpe ratio during the expansion regime, as per found in the initial observation of the Expansion regime results from the OLS model run.

With the results yielded in the portfolio, some parallels can be found with the fourth hypothesis formulated within the research, in which the Markov-based weight allocation would produce a superior portfolio allocation, while the macro strategy's macroeconomic exposure would limit the actual true performance of a weight-distributed portfolio, especially in a context where another factor of the hypothesis, with that being the likelihood of the regimes staying consistent over 6 months rather than frequently switching to another macroeconomic regime being proven as well during the calculation of the research's Markov chain.

4.4.3 – Construction of the performance indexes – analysis of indexes:

The last step of the research consists of evaluating the monthly performance for all the months covering the 2018-2025 period by integrating the monthly portfolio returns, as well as specific indexes integrating the mixed portfolio, as well as each hedge fund strategy separately. The first step was to determine the monthly portfolio return of each monthly observation by attributing the corresponding excess macro and excess event-driven returns to the weight allocations found through the usage of Solver. Once all monthly portfolio returns had been calculated, the indexes for all three hypothetical scenarios were determined, with the first one being the portfolio index. Starting on a 100-point basis for the first month (January 2018), the results for the other monthly observations were determined by using a cumulative return logic in which the previous period's portfolio index was used along with the addition of the corresponding monthly portfolio return. The same starting point logic was used for the two other indexes, although the macro index is using the monthly period's excess macro return, while the event-driven index uses the excess event-driven return corresponding to the monthly observation. Once all the monthly portfolio returns and indexes for all hypotheses have been found, a graphic depicting the evolution of the indexes per each hedge fund strategy along with the portfolio index returns to evaluate directly which strategy might yield the highest returns on average.

4.4.3.1 – General results & observations:

By December 2025, the event-driven index was resulting at 133,54, being the strongest performer of all three indexes, while the macro index was at 113,24, being the lowest performer of all three strategies, which might come as a surprise given the highest Sharpe ratio observed during the expansion regime. Meanwhile, the combined portfolio index reached a total of 125,21,

reflecting the negative constraints that macro funds are posing amid uncertain macroeconomic conditions.

For all three indexes, the nadir point was reached in March 2020 where all markets were heavily impacted by the Covid-19 pandemic, especially in a period where lockdown measures have been introduced by many governments worldwide. In this period, the most impacted index was the event-driven index which was at 85,10, followed by the portfolio index at 89,75, while the macro index was surprisingly the highest at 95,67, which could be explained by the lower volatility amplitude that has been observed in February prior to the period. The nadir point of the event-driven index is mainly explained by the collapse in corporate deals and lack of liquidity in equity markets during the pandemic.

The strongest monthly performer was observed in November 2020 for the portfolio monthly return, when it reached a 4,42% increase from the previous month. Driven mostly by an increase in the event-driven index (reaching 104,56), this strong performance is mostly explained by the announcement of the first Covid-19 vaccines, as well as strong stabilized equity performance for the third quarter for most companies.

Regarding the macro index, the total peak was reached in 121.24 in October 2022, with the main driver being the hawkish FED interest rates around 3,25%, an environment that mostly pushed for efficient long positions in commodities and short positions in equity-related market positions. Though, given the final value reached in December 2025 at 113,24 shows a gradual normalization of volatility on the long-term, as well as a more relaxed environment characterized by an ease in the FED interest rate decisions, as well as better performance of gold price returns.

On the other hand, the event-driven index sees increased and gradual increases in its observed indexes from March 2020 onwards with only very minor disruptions, which is shown by the higher amount of M&A deals and IPOs concluded during the post-Covid period, as well as the performance of the hedge fund strategy during transitional periods.

The graphic below shows further the evolution of the observed indexes:

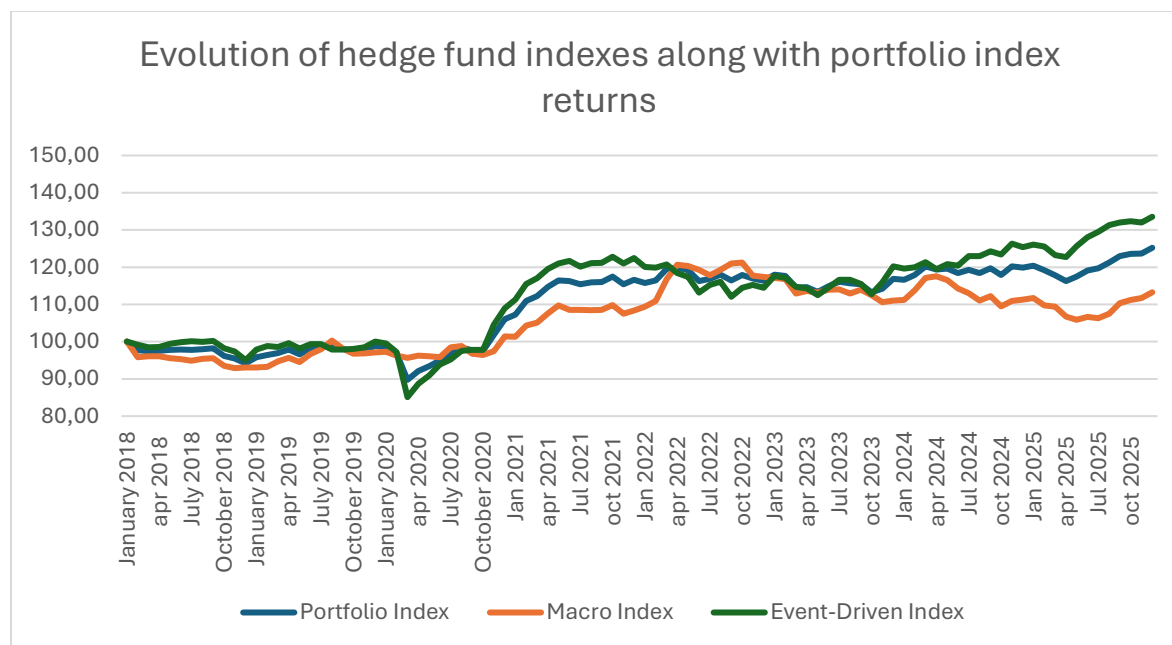


Figure – Graphic depicting the evolution of indexes for each hedge fund industry along the portfolio index returns from January 2018 to December 2025.

As we can see from the graphic, we can see two main dynamics. The first one is the period between January 2018 and February 2020 during which the performances yielded by all three indexes were more or less similarly aligned, albeit the event-driven index was still performing better than the macro index by roughly 4 points on average. From March 2020 until present, there are some periods of inconsistencies, and mainly during the period of Covid 19 in which the macro index was the most severely impacted index by the highly correlated risks of the index with direct macroeconomic exposure. Although the strong performance during October 2022 was observed, the following underperformance of the macro index is confirmed, with the gap increasing significantly between the macro and event-driven index. This related further with the fourth hypothesis of the research, which indicates that direct macro-economic exposure for macro-based hedge funds is not optimal during high-volatility windows or when macroeconomic conditions are normalized. Lastly, the Markov chain's actual allocation around the transitional regime observations shows the diversification benefit between the two strategies in which diversification or increased intake of different hedging positions could help strategies at stabilizing their returns even amid higher uncertainty around the macroeconomic environment.

4.4.4 – Final implications and limitations of the conducted research.

4.4.4.1 – Final interpretation and evaluation of the formulated hypotheses:

As an initial final interpretation of the research's findings, each hypothesis that has been formulated in the Research issue and hypotheses sub-section of the Literature Review will be interpreted to see whether the actual findings could claim partially a certain segment or not of each hypothesis.

H1 - Regimes might have a factor exposure to the correlation between variables:

The general verdict is that the findings of the research are partially supporting the initial claim. In effect, the directional evidence described during the cross-regime analysis and Markov chain can explain certain movements or mechanisms espoused by the two hedge funds during the defined macroeconomic regimes. Nonetheless, a lack of concrete statistical inference, given the high p-values for the evaluated coefficients, are limiting the actual projection of the research in fully responding to the hypothesis. The changes in VIX's coefficient for the macro index at -0,228 to 0,114 in transition and back to -0,313 during the crisis regime show that regime concentration is the main factor that would explain the performance of hedge funds, rather than the actual strategy or garnered return. With a p-value equal to 0.027, VIX sensitivity depends on the regime rather than the stability of the market, while the coefficients shift in gold returns (-1,12 for expansion, 4,9 in transitional and 3,62 during crisis) show further protection that macro hedge funds may need when conditions are shifting quickly. However, with a very high p-value for both transitional and crisis (p-value = 0,352; p-value = 3,623), we cannot conclude the actual statistical inference around this potential hypothesis.

For the assumption of correlated movement between the VIX, DXY and gold return changes, partial support can be concluded. From an initial observation across all monthly observations, the movements seem to be scarce, as the correlation between VIX and DXY is merely at 0.267, while the VIX moves differently on average from gold return changes at -0.028, showing potential low power that could explain their simultaneous movements during the different macroeconomic regime. While taking a closer look at each coefficient obtained, the relationship seems to be closer during the expansion phase, given the macro hedge fund's exposure to the macroeconomic environment during which the factors are serving as a signal for the movements of the hedge fund. Though, the DXY doesn't hold any significant power in explaining the potential movement of a macroeconomic regime, all while remaining static during stressed regimes during which VIX and gold prices move together while the DXY holds different movements compared to the VIX. This could also result from the different sub-periods of the research's historical framework: in fact, the 2022 inflationary context seems to contradict the base of the hypothesis, as the VIX saw major swings in its score whereas the DXY had better stability overall. Therefore, directional evidence could be concluded for certain periods but not conditional for all regimes, as well as from a statistical perspective as there's been only 22 observations in the crisis period, showing some potential statistical weakness.

H2 – The strength and weight of certain macroeconomic variables as drivers in the performance of both strategies.

For macro-based hedge funds, the hypothesis can be supported. The changes in VIX are the only ones that can be backed by statistical inference, as all coefficients for the changes in the VIX were linked to p-values well below the 0.05 threshold required, with no other factor achieving this result for all three macroeconomic regime, while the classification has showed clearly that changes in the VIX could be a sign of reversal of the macroeconomic regime. With the further

collapse of HFRI Macro Index's adjusted R^2 from transition to crisis (0.272 to 0.1) illustrates that the stability and sensitivity of macro hedge fund's performance deteriorated during stress regimes, with further non-linear deviations being observed, showing the strength of the VIX.

Nonetheless, the claim can only be partially supported for HFRI Event-Driven Index. The Market Factor (MKT) is the main factor explaining the stability or reversal of the regime for this specific strategy given the statistical inference from the p-values obtained for each coefficient of the MKT during all the defined macroeconomic regimes. VIX is driving small changes in the regime for event-driven hedge funds, with coefficients of 0,04 in expansion, 0,05 in transition and 0,03 during the crisis period, albeit not statistically supported given the high p-values obtained, which directly links to the core strategy of event-driven hedge funds, which is the focus on the number of corporate deals or announcements rather than equity-market drivers.

As a conclusion, the hypothesis clearly supports macro hedge funds as the VIX dominance is clearly visible but requires further investigation for event-driven hedge funds where the actual market context and factor are the main catalysts.

H3 - Asymmetric performance of macro and event-driven hedge funds:

There is some partial support for this claim, with different subparts explaining further findings. During expansion, HFRI Macro Index had a Sharpe ratio of 0.67 compared to 0.55 for HFRI Event-Driven Index: yet, the generated average excess return is in favor of HFRI Event-Driven Index at 0.522 compared to 0.317, mainly explained by the high skewness and kurtosis observed for the event-driven hedge fund, as well as the presence of a 0.588 Omega ratio for the event-driven hedge fund, indicating that the fund experiences both very high and abnormal returns in their performance and sharp decreases in their alpha opportunities. The macro's reliable and stable skewness and kurtosis confirm further that macro funds can adjust their risk-performance mechanism to a higher degree given the calm macroeconomic context.

Some partial claim for the hypothesis can also be concluded in the transitional phase. In fact, the annualized Sharpe ratio remains higher for the macro hedge fund at 0,48 compared to 0,36 for the event-driven hedge fund, as well as a better adjustment in its average excess returns at 0,21 compared to 0,18 for the event-driven hedge fund. The high 2,69 5-year treasury yield coefficient obtained during the transitional period (confirmed by a p-value of 0,016) captures the macro's ability at capturing adjustments and potential gaps existing during the tightening of monetary policies, especially during the 2022-2023 hawkish monetary cycle.

Nonetheless, the opposite observations and phenomena have been noticed during the crisis period. Contradiction of the hypothesis can be clearly seen for both the annual Sharpe ratio and average excess returns, as the Event-Driven severely outperforms its Macro counterpart, with a Sharpe ratio at 1,09 versus -0,21 for the macro-based hedge fund (with it being the only negative Sharpe ratio during the period), as well as an average excess return of 0,45 compared to -0,1 for the macro hedge fund (which is, once again, the only negative excess return yielded in the whole research). These results are contradictory to the consensus of the hypothesis, as well as the main

existing literature review and research of Fung and Hsieh (2004) where adjustment to the economic risks can drive funds upward. The crisis months between 2018 and 2025 are all characterized by higher FED interest rates and abrupt changes in the 5-year treasury yield, as well as more stressful geopolitical risks, which were not present during the 2008 crisis, and thus allowing event-driven hedge funds to exploit positions around subordinated debts to spur return, whereas macro hedge funds are trapped in their default mechanisms, which is based on the leverage opportunities (which are becoming tighter due to hawkish monetary rates) and market noise (which becomes distressed). The high Omega ratio of the HFRI Macro Index at 1.859 (the highest Omega ratio of the research observed) shows that high returns exist during the period, albeit representing a very small portion of the total period given the negative performance overall. Lastly, the lack of significance, as per the 22 observations classified as belonging in the crisis period might interpret of lack of concrete observation, which would be contrasting to the findings of Ang & Timmermann (2012) where the usage of observations during the 2008 period supported their assumptions around the macroeconomic factors directly impacting the behavior of hedge funds.

As a conclusion, the hypothesis is partially supported only during expansion and transitional regimes but empirically reverted during crisis periods where event-driven hedge funds gain the uphold on macro hedge funds.

H4 - Consistency of regimes and signal around regime-switching periods:

The hypothesis around the consistency of regimes regarding their stability and frequency of switch between the regimes is supported. In effect, the Markov chain process in the first month forward persistence shows probabilities of 0.556 of maintenance in expansion, 0,479 in transition and 0,409 in crisis, showing that the most likely outcome in the mid-term is the maintenance of regimes under their own. The mid-term framework is further supported by the derivations across the 6 months for all three regimes, with 2.27 months for expansion, 1.92 months in transition and 1.69 in crisis, confirming that the economic dynamics rely more on capturing mid-term momentum than facing month-to-month swings.

With a high number of monthly observations from 2022 to 2023 classified in the transitional regime (a total of 14) provides further support for the claim. With the last derivation of the probabilities convergence, we can notice that all probabilities are at 28,04% for the expansion regime, 49,35% for transitional and 22,62% for the crisis, confirming the close alignment between the Markov chain's structure and the proportion of observations for each regime during the conducted research, as per demonstrated in the descriptive statistics for each regime. Because of the different economic and geopolitical events occurring from 2018 to 2025, the core of the Markov chain's basis, which is long run in the probabilities, is demonstrated in all the regimes of this research.

Lastly, regarding the signals concerning the switch probability between regimes can be partially confirmed. Although there are some probabilities to switch from expansion to crisis, or from

transitional to crisis, most of the switches are coming from the expansion and crisis regimes switching to a transitional environment rather than the opposite, suggesting potential explanation, albeit not fully confirmed.

H5 - Optimizing weight allocation of the two strategies:

The portfolio optimization through the usage of Solver function on Excel confirms a better Sharpe ratio and performance of the portfolio based on an optimal weight allocation for both hedge fund strategies, with the optimal allocation of 43% in macro hedge fund and roughly 57% in the event-driven hedge fund proves a better Sharpe ratio than an equally weighted portfolio.

The fundamental reason for the lower weight allocation for the macro strategy is because of the actual nature of HFRI Macro Index's performances during each macroeconomic regime. With a negative average return of -0.105% during the crisis regime and its annualized Sharpe ratio of -0.21, as well as a 22.6% steady-state weight, the macro hedge fund represents a potential risk of allocation for the portfolio, even during optimal macroeconomic conditions such as the expansion or transitional phases in which the HFRI Macro Index outperforms the HFRI Event-Driven Index. The latter, on the other hand, generates positive returns during all three regimes, as well as a superior 1.09 annualized Sharpe ratio during the crisis regime, giving a better stability framework for the event-driven strategy.

By looking at the performance of the cumulative indexes, we can notice some further patterns that link to the allocated weights. On one side, the Macro Index peaks at 121.24 in October 2022 during the hawkish-rate period of the FED, with the performance declining substantially to reach a total of 113.24 in December 2025, with this underperformance reflecting the mechanism of this hypothesis during which macro hedge funds could yield strong performances during uncertain periods but the alpha opportunities and returns are flat during normal conditions. The event-driven index sees a steady and sustained growth from the Covid-19 period towards December 2025, with only a few declines being noticed around the 2022-2023 period in which the high inflation environment prevented the emergence of new corporate deals. Nonetheless, the real change started in July 2023 when the Event-Driven Return Index started to outperform the Macro Index consistently, eventually ending at 133.54 in December 2025.

The hypothesis's main prediction was that event-driven would see more weight allocation than the macro hedge fund given the latter's higher degree of exposure to direct macroeconomic risks. The research partially concludes this point, with the most thrilling revealing being during the crisis period where macro hedge funds generate negative returns, while having stronger performance during the expansion period, which results in an idea that the hypothesis didn't precisely mention that crisis would be the actual beneficiary of the strategies implied. As a conclusion, macro's main contribution comes from both the transitional and expansion phases, with positive returns and significant market factors (MKT), while event-driven strategy would provide stability during the crisis but also expansion periods. The annualized portfolio's Sharpe ratio at 0.57 captures the diversification core of the portfolio that incorporates both strategies.

4.4.4.2 – Limitations of the current research:

While the research has provided insightful findings that could confirm most of the hypothesis which are at the core of the central research issue of the thesis, there have been some limitations around the way this research has been conducted methodologically, both regarding the sample structure, the lack of further macroeconomic variables, or issues around the framing of existing factors applied for each specific hedge fund strategy.

1 – Sample size constraints around regimes:

Given the classification methodology that has been used for distributing each monthly observation of each macroeconomic variable included in the research for a specific macroeconomic regime, there has been some issues around the structuration of the sample size, and notably for the crisis regime, who had only a total of 22 observations. Because of this, the OLS running model only have a total degrees of freedom of 16 given the 5 other variables included in the OLS structure as per the defined linear regression model developed in the statistical methodology of the research. This explains notably the crisis-regime coefficients around the VIX and gold price returns show economically meaningful insight on the direction that both hedge fund strategies are taking during these periods but fail to achieve a statistical significance sphere given their low p-values. Because of this, data limitation is the most likely reason over the reason why most of the macroeconomic underlying factors might not explain the full picture of hedge fund's behavior during stress regimes. A better improvement in the foreseeable future would be to expend the range of observation for the periods starting from 2008, which would incorporate further the 2008 financial crisis and the 2011 Eurozone's Sovereign Debt crisis, which would garner further monthly observations that may be classified as belonging to the crisis period, as the research's observed period from 2018 to 2025 reflects mostly the market outlook methodologies used by the asset management departments of large banks, including Goldman Sachs or Blackrock. This research would be conducted in a further post-thesis research conducted independently by the author of the research.

2 – OLS linearity assumption and presence of non-normal return distributions:

The regression framework is based on Excel's OLS (Ordinary Non-Linear Least Squares) method which assumes a direct linear relationship between the dependent excess returns observed and the integrated macroeconomic variables in the linear regression model. Both assumptions are challenged empirically by the actual data garnered, especially around the highly observed skewness and kurtosis in the event-driven hedge fund during expansion and macro hedge fund for the crisis period. These observations are severe deviations from the original relationship, which also explains further the non-statistical significance of the p-values in the different coefficients per each regime. However, these observations might reflect the real return nature of hedge funds, and especially the macro hedge funds during extreme events in which unusual distributions or non-linear payoffs are observed, as per the conclusions of Fung and Hsieh (2004). The OLS framework, though, cannot capture those non-linear relationships, and thus resulting into higher deviations from the original stable averages observed. A further idea

would be to explore further quantitative methodologies that could drive hedge fund's performance, including the structuration of a Markov-based regression system given the more direct relationship between the regimes and their own regression system including specific parameters, as well as an easier way to translate it into the Markov chain probabilities system.

3 – The actual index data and the absence of position-level transparency.

As it has been stated in the introduction of this research, one of the limits around the current literature review, which is also specific to the actual nature of hedge fund's framework is the absence of actual return and alpha data around the most performant hedge funds for each investment strategy chosen. First, the real nature of index data and the actual smoothed values can also reflect on the actual generated Sharpe ratios, as well as Alpha and Beta, which can also reflect on the actual level of sensibility of the hedge funds around specific macroeconomic factors, including the sensitivity of performing and underperforming periods related to the changes in the VIX. At the same time, the survivorship and backfill biases are inflating the actual index data of returns for all the period since survivorship bias will remove the defunct or very underperforming funds from the aggregate, while backfill bias will only allocate significant weight on the early returns or returns during stable conditions, meaning that the actual annualized Sharpe ratio, Sortino and Omega ratios, as well as the actual structured portfolio might be biased due to the biases existing within the reporting of the returns for both hedge fund strategies from HFRI.

4 – Lack of specificities for the regime classification methodology:

The composite system scoring method employed used in the research are based on monthly observations around macroeconomic factors which are also at the core of the independent factors of the OLS model. As an example, the changes in the VIX and DXY have been used in the linear regression while the original basis level, which is the VIX and DXY scores have been attributed as classifiers for attributing to each monthly observation an actual macroeconomic regime as per the classification used. This overlap of the factors can create an endogeneity concern in which the values used for classifying the regime are actually used also as factors to explain the returns for each hedge fund strategy, which could reflect some unusual exogenous relationships in the models of the research. At the same time, some of the classification might not always work for the different other hedge fund strategies existing, including the HFRI Diversification Index that incorporates fixed-income securities, or even the HFRI Event-Driven Index. A solution would be to use non-financial metrics that are more related to the general macroeconomic outlook, including the GDP growth, unemployment rate would eliminate partially this issue, albeit not as deep as the current research's framework that is more accurate for month-to-month observations. This is mainly explained by the usage of VIX level for the classification of regimes whereas the changes in VIX (ΔVIX) has been used along with other changes from other macroeconomic factors in the OLS models built, which have a concrete relationship without the two being identical variables. Through my future horizons around this research, one idea could be the usage of other factors that are more explicit to the current horizon, such as GDP growth, unemployment

rate changes, while volatility or dollar index would be solely used as regression inputs without including their month-to-month fluctuations to limit the power or high weight that these two factors may represent.

5 – Memoryless within the Markov chain framework:

The assumption around the first-order Markov is that the probability of transitioning to any future regime depends only on the current regime in which a hedge fund is positioned. Nonetheless, macroeconomic regimes path dependence over the underlying factors, something that the Markov chain ignores: as an example, the transition regime's stability as per observed in the 2022-2023 period might imply further increased probability of transitioning into a crisis framework given the tighter effects that hawkish FED interest rates and liquidity mismatches build over the period. A higher-order Markov model incorporating further lags from extended period could capture better the actual dependence path that macroeconomic regimes, but it would link further to the first limitation of the research, which is the extension of the observed period from the actual 2018-2025 period used by the research.

6 – Geopolitical factors:

The inclusion of different political and geopolitical events, such as the USA-China trade war, the Covid-19 Pandemic, the War in Ukraine, the Israel-Gaza conflict or the US presidential elections cycle are events partially captured by the macroeconomic factors, but not characterized as discrete events, given the lack of a structured stress level coefficient that is explained by geopolitical shock, since the actual classification method is only structured around macroeconomic shock drivers. Geopolitical risks are still hard to quantify, given the lack of direct primary or secondary data which would support the actual: even the inclusion of returns in gold prices cannot provide a direct framework, although gold prices are moving usually given the fluctuation related to geopolitical events. As a future sub-thesis research from this topic, the author of the research would look further at the research of ⁴⁰Caldara and Iacoviello (2022) that measures geopolitical risk through the Geopolitical Risk Index that would be used eventually as a model for classifying geopolitical shocks and their impact on hedge fund returns.

7 – Static weight allocation, optimisation and structuration biases:

The last limit of the research relates is about the actual allocation of weights, the assumptions from the hypotheses' idea of allocating specific weights based on the performance of each hedge fund, as well as the general structuration of the research's period. The portfolio construction framework relies on the state-probabilities found in the Markov chain to throughout the entire portfolio and indexes optimization. On the contrary, a regime that is aware of the sentiment that investors share and the current update frequency of regimes would work more on a longer horizon. The static use of steady-state weights in this research illustrates that the portfolio actually doesn't use the predictive information within the regime in which the portfolio may

⁴⁰ Caldara, D. & Iacoviello (2022). Measuring geopolitical risk. *American Economic Review*, 112(4), 1194-1225

operate eventually. A dynamic balancing approach that would update frequently weights for each macroeconomic regime would be more consistent with new observations being directly available could solve both the issues around the actual data available for users, as well as a more maximized annualized Sharpe ratio, given the very low change in the Sharpe ratio calculated due to the Solver's findings. The latter point also addresses the idea that Sharpe ratio would be optimized without short-selling constraints or other factors that haven't been counted directly in the research, including management and performance fees.

Finally, the findings of this research are mostly specific for the 2018-2025 period, a short period that reflects both historically low interest rates, periods of uncertainty related to the Covid-19 Pandemic and the different conflicts that had occurred worldwide. With the distribution of 27.1% in expansion, 48.9% in transitional and 22.9% in crisis is only capturing a short-term essence, without any potential long-term implications that can be deducted directly from the research. At the same time, the dominance of the transitional phase given the uncertainty around the 2022-2023 period or the presence of the trade war context from 2018 to 2019 between the USA and China might imply that transitional regime phases may not have the same weight structure on longer runs, which is why a further sub-research with a longer period framework could actually lead to results that might attribute further weight to expansion and crisis regimes. In conclusion, the findings around the underperformance of macro-driven hedge funds compared to event-driven hedge funds might apply for this research, but its actual implication on the long run might only be determined (or not) by extending the period of research.

V – Conclusion and future regards.

To conclude, this research study offers a comprehensive and methodological research around the core question that had always fascinated people interested regarding hedge funds on how hedge fund strategies, and especially macro and event-driven hedge funds can adjust their mechanisms to mitigate risks during specific macroeconomic conditions. By combining different articles already existing, the monthly return data for the period January 2018 to December 2025, as well as applying different mathematical concepts around OLS, linear regression, computation of performance metrics, Markov chain and fundamentals of statistics, the research moves beyond the unexplained underlying macroeconomic variables that may influence the behavior of hedge funds, with the main findings supporting the idea that macroeconomic regimes are significantly shaping the way risk-adjusted performance and sensitivity of both macro and event-driven hedge funds.

The primary contribution of the research consists around the first point indicating the risk-adjustment observed for macro and event-driven hedge funds. The results are contradictory to the main narrative of the existing literature from Fung and Hsieh (2004), as well as Ang and Timmermann (2012). The HFRI Macro-Index indicates normal and stable returns during expansionary and transitional regimes, outperforming consistently compared to the HFRI Event-

Driven index, concluding the idea that stability can help macro hedge funds at finding consistency in their performance. At the same time, further confirmation around the importance that certain factors have on the performance of the macro hedge fund, including changes in the volatility index prove the sensitivity and direct exposure to which macro-based hedge funds are exposed to. On the other hand, the surprising results that are contradictory to the literature research are quite important to observe: with a negative return and lower annualized Sharpe ratio for the HFRI Macro Index during the crisis period compared to its event-driven counterpart, which also resulted in the best risk-adjusted performance from all the research. This inversion is not just purely observed from these metrics, but the robust Sortino ratio, Omega ratio and maximum drawdown obtained are also noticeably strong, with the most explainable reason being the coverage period: in fact, the period 2018 to 2025, which is a short-term timeframe, captures a diverse amount of reactive market moments, including the Covid-19 Pandemic, the geopolitical uncertainty around the War in Ukraine and conflict in Gaza, as well as the diverse dynamics of the different economic variables, including the FED interest rate which saw both flat periods and hawkish instances. In this sense, macro hedge funds will have a higher exposure since tighter monetary policies could impact the access to capital markets, while also signaling higher inflation, conditions that could limit the leverage that macro hedge funds can have. At the same time, distressed debts or lowly-performing debts could be an option for event-driven hedge funds, given the underperformance of unstable debt or bonds on the fixed-income markets, which might push event-driven funds to opt for long buying positions on their distressed securities while short selling on positions prone to direct exposure on the market. The final research in the portfolio construction that led to the development of the indexes of the portfolio, as well as each hedge fund's strategy confirm further this point, as the event-driven index constantly outperformed the macro index since the summer of 2023 in the context where conditions were normalizing, which resulted in an increased number of corporate deals that are beneficial for event-driven hedge funds.

This research could provide some further insights for hedge fund managers, risk managers or any person that is passionate to understand further about the mechanisms of hedge funds. From the idea that regime-aware dynamic allocation between the two strategies produces superior risk-adjusted outcome, all while having steady-state convergences demonstrated in the Markov chain can define effective rebalancing stability for hedge funds that seek to hold their positions or stability of the regime that they are placed into, while also using the different factors related to the market factor or the treasury yield to guide allocation of positions in either one of the two strategies amid regime changes. The consistent reversal of VIX can also illustrate the way budgeting and positions are allocated for managers, all while handling arbitrage trading opportunities. At the same time, the lack of concrete statistical inference may also mean that macro funds have their most reliable and stable factors around factors that are impacting heavily during tight monetary frames. Lastly, the absence of any direct expansion-to-crisis transitions, or the preference of regimes to stay within their framework in the initial period due to lack of information has an operational implication for risk managers, as the movements suggest that

managers should allocate a single month for preparing to any potential adverse events by adjusting leverage costs, putting further positions in their discretionary long-short directions, which would allow for a better preparation all while incurring less maximum drawdown during a crisis regime.

With that being said, it is important to note that the research also holds its several limitations. In fact, as we have demonstrated across the section that the lack of concrete observations for the crisis regime might not explain deeply the real underlying causes that could explain the significant outperformance of event-driven hedge funds compared to their macro peers. The limits of the OLS' linearity assumption by the non-normal distributions and highly observed skewness and kurtosis, the lack of concrete explanations around the optimal static weight allocation can create limits on actually interpreting the accuracy of the results. The memoryless assumption around the Markov chain ignores the path dependence of the macroeconomic regimes, with the static nature of the weight allocation not capturing the actual real-time dynamics on how the macro and event-driven hedge fund strategies capture their positions at a frequent time. Finally, the window of the period chosen for the research might not capture periods that would explain further ideas around the reaction of the funds during acute crises, including the 2008 Financial crisis or the 2011 EU Sovereign Debt crisis.

For the future research, three principal extensions can be pursued for future purposes. The first one would be the extension of the research environment to include a higher number of years, with the prospect being to observe the period 2008-2025 in a new research so that further macroeconomic or crises events may be incorporated, such as the 2008 financial crisis or the 2011 EU Sovereign Debt crisis as highlighted in the limitations section. By extending the period of observation, different biases such as the low amount of periods classified as "Crisis" or issues around observed statistical data may be solved, which would improve the cross-regime analysis and the statistical significance of the model, especially around the resulting p-values and alphas for each macroeconomic regime. At the same time, an extended framework would also require further reviewing around the classification threshold used for each macroeconomic indicator so that pre-2015 monetary policies are also accurately reflected in this perspective.

Another extension that is possible to this research would be to observe and analyze other hedge fund strategies, including hedge funds around fixed income, or traditional equity long/short positions which are among the most common hedge fund strategies present within the industry. By having a multi-strategy framework, further underlying causes can be determined around the causes of differences in the performance of each hedge fund, while also attributing further evidence for specific macroeconomic factors which may demonstrate some of the results of this research, but that cannot be proved entirely due to the low statistical significance and reliability that the current model proposes. Moreover, the enriched framework around other strategies can also prove further value to the methodology used, and especially around the usage of OLS or Markov Chain, while also presenting further evidence for new statistical and mathematical

calculations such as the integration of eigenvalues or improved simplex method to find the right parameters for the performance of multi-strategy portfolios.

Lastly, the third extension would respond to the main limitations of the current research which lays around the statistical insignificance of specific variables or macroeconomic factors for certain regimes. By including further articles or research done previously, as per the examples of Brunnermeier and Pedersen's (2009) research around leverage and its effects in the performance of hedge funds, as well as the creation of a factor model signaling specific signal or noise for each hedge fund strategy could enhance the value added to this research around the potential correlation between regimes and specific macroeconomic regimes. Furthermore, the inclusion of the Caldara and Iacoviello's (2022) Geopolitical Risk Index can also represent further quantitative emphasis around the impact that geopolitical issues or events may have on specific hedge funds and that would help further at classifying periods under crisis conditions.

As for the personal and professional implications of this research thesis, this result has been garnered through the hard work I have completed across four years in two different educational institutions, two different professional perspectives, as well as my extra-curriculum research which have all contributed to the analytical framework of this research.

The main foundation lies around my first two years of education received at emlyon business school which allowed me to develop my basic financial knowledge and background, from financial planning, understanding investment choices, getting a grip around finance for managers or learning about financial accounting information, as well as mathematics and statistics applied to organisations all helped me at develop and garner further my quantitative and analytical framework. All the courses received at emlyon, including "Finance for managers", "Business economics and markets" or "Understanding accounting information and data" introduced me to the fundamentals of the relationship between financial data and its implications and usage for organizations, while other courses including "Financial Planning" or "Investment choices" allowed me to visualise further the quantitative implications of different financial fundamentals across companies, which allowed me to develop my quantitative appetite. The chance of following the course of Applied Research in Social Sciences, as well as my completed mission for Field studies helped me understand briefly how to structure and conduct market research for a centralised question topic, all while developing my competencies around thriving and collecting both primary and secondary data on the behalf of a question topic. A two-month internship in a publicly certified accounting firm in France (Sogeco Prorior) provided the first encounter to real-world finance data, with the chance of completing financial statements and auditing reports with the firm's CEO and other directors enabling me to understand that theoretical framework isn't sufficient at understanding the real implication of finance for companies, but the right approach in real environments being the most crucial factor.

However, the real turning point came during my second year's experience at SES Satellites and the ability to work in the firm's fixed accounting and finance departments, in which I was able

not only to report simple financial acquisitions the firm was doing across their departments, but also to gain further exposure around corporate finance, including continuous development around the M&A department DCF valuation, LBO modelling, as well as a real exposure around macroeconomic conditions and their implications for the firm's operational ambitions. Working on the market research and actual financial structuration allowed me to understand that garnering and understanding financial data is not just a simple process, but rather as a way indicating me about how macroeconomic factors could determine the way a company chooses its operations. By observing these factors are linked to the major implication of this research thesis, which is built around the macroeconomic underlying factors that do influence the performance of hedge funds in real time, especially around structured strategies.

Furthermore, the intellectual stimuli came mostly while stepping foot in Sungkyunkwan's academic environment. In effect, the chance of taking specialized courses around finance such as Intermediate Investments, Current Topics of Finance, Derivatives and Risk Management, Banking Finance and Intermediates or Behavioural Finance and Fintech allowed me to understand concepts related to market finance which were beyond my initial baggage of knowledge, with notions around VIX implications, FED's interest rates, futures pricing, knowledge around puts/calls driving my appetite at understanding the further implications that macro fluctuations can have around financial decisions. At the same time, the ability to get knowledge around equity and hedge fund strategies, specific theoretical concepts as per the example of the CAPM or Fama French Model or understanding crucial valuation ratios as per the example of the Sharpe and Sortino ratios have provided a crucial factor in the understanding of portfolio diversification and building, which have been proven as the main factors for the structure of this research thesis. Thanks to these courses, I have managed to expose my knowledge towards advanced quantitative and mathematical concepts, as proven by my courses of Business Statistics or Quantitative Analysis for Business have linked myself to the understanding of algebra, statistics or matrix, with different concepts including Markov chain, OLS or matrix operations as per the case of the linear programming concepts or Gaussian elimination providing the mathematical framework of this research. By being curious to explore the vibrant financial ecosystem of Seoul in a post-Covid climate helped me at understanding further the importance of macroeconomic factors and regimes of economies and their impact on portfolio performance.

With my post-undergraduate studies commencing in UCL's MSc Finance program in London, I will enhance my journey further, with the curriculum's emphasis around valuation and further implications of strategies around private equity and hedge funds will help me at not only enhancing further my professional perspective, but also to respond to the limitations and future outlooks of this existing research topic. Thanks to the observation of new mechanisms or other direct skills related to market finance, as per the example of Bloomberg terminal sessions or institutional portfolio construction could help me bridging further my gap between the academic methodology I have built and the actual macroeconomic implications that are existing and affecting specific hedge funds.

References

- Ackermann, C., McEnally, R., & Ravenscraft, D. (1999). The performance of hedge funds: Risk, return, and incentives. *Journal of Finance*, 54(3), 833–874.
- Agarwal, V., & Naik, N. Y. (2004). Risks and portfolio decisions involving hedge funds. *Review of Financial Studies*, 17(1), 63–98.
- Agarwal, V., Daniel, N. D., & Naik, N. Y. (2009). Role of managerial incentives and discretion in hedge fund performance. *Journal of Finance*, 64(5), 2221–2256.
- Amenc, N., Géhin, W., Martellini, L., & Meyfredi, J. C. (2008). Passive hedge fund replication: A critical assessment of existing techniques. *Journal of Alternative Investments*, 11(2), 69–83.
- Ang, A., & Timmermann, A. (2012). Regime changes and financial markets. *Annual Review of Financial Economics*, 4, 313–337.
- Aragon, G. O., Ergun, A. T., Getmansky, M., & Girardi, G. (2017). Hedge fund liquidity management. *Working Paper, Securities and Exchange Commission*.
- Asness, C. S., Krail, R., & Liew, J. M. (2001). Do hedge funds hedge? *Journal of Portfolio Management*, 28(1), 6–19.
- Bali, T. G., & Gokcan, S. (2003). Alternative approaches to estimating VaR for hedge fund portfolios. *Journal of Alternative Investments*, 5(4), 7–21.
- Billio, M., Getmansky, M., Lo, A. W., & Pelizzon, L. (2012). Econometric measures of connectedness and systemic risk in the finance and insurance sectors. *Journal of Financial Economics*, 104(3), 535–559.
- Bloomenthal, A. (2025, January). World's top 10 hedge fund firms. *Investopedia*. <https://www.investopedia.com/articles/personal-finance/011515/worlds-top-10-hedge-fund-firms.asp>
- Boyson, N. M., Stahel, C. W., & Stulz, R. M. (2010). Hedge fund contagion and liquidity shocks. *Journal of Finance*, 65(5), 1789–1816.
- Brunnermeier, M. K., & Pedersen, L. H. (2009). Market liquidity and funding liquidity. *Review of Financial Studies*, 22(6), 2201–2238.
- Chen, J. (2025, February). Understanding performance fees in hedge funds: Examples and insights. *Investopedia*. <https://www.investopedia.com/terms/p/performance-fee.asp>
- Depresio, G. (2026, January). How hedge funds use leverage to amplify returns. *Investopedia*. <https://www.investopedia.com/ask/answers/102015/how-do-hedge-funds-use-leverage.asp>
- Fung, W., & Hsieh, D. A. (2001). The risk in hedge fund strategies: Theory and evidence from trend followers. *Review of Financial Studies*, 14(2), 313–341.
- Fung, W., & Hsieh, D. A. (2004). Hedge fund benchmarks: A risk-based approach. *Financial Analysts Journal*, 60(5), 65–80.
- Fung, W., & Hsieh, D. A. (2009). Measurement biases in hedge fund performance data. *Financial Analysts Journal*, 65(3), 36–38.
- Fung, W., Hsieh, D. A., Naik, N. Y., & Ramadorai, T. (2008). Hedge funds: Performance, risk, and capital formation. *Journal of Finance*, 63(4), 1777–1803.

- Greene, W. H. (2003). *Econometric analysis* (5th ed.). Prentice Hall.
- Hamilton, J. D. (1989). A new approach to the economic analysis of nonstationary time series and the business cycle. *Econometrica*, 57(2), 357–384.
- Jensen, M. C. (1968). The performance of mutual funds in the period 1945–1964. *Journal of Finance*, 23(2), 389–416.
- Keating, C., & Shadwick, W. F. (2002). A universal performance measure. *Journal of Performance Measurement*, 6(3), 59–84.
- Kenton, W. (2025, March). Understanding global macro hedge funds: Strategies and examples. *Investopedia*. <https://www.investopedia.com/terms/g/global-macro-hedge-fund.asp>
- Lo, A. W. (2001). Risk management for hedge funds: Introduction and overview. *Financial Analysts Journal*, 57(6), 16–33.
- Magdon-Ismail, M., & Atiya, A. F. (2004). Maximum drawdown. *Risk Magazine*, 17(10), 99–102.
- Markowitz, H. (1952). Portfolio selection. *Journal of Finance*, 7(1), 77–91.
- Moreira, A., & Muir, T. (2017). Volatility-managed portfolios. *Journal of Finance*, 72(4), 1611–1644.
- Norris, J. R. (1997). *Markov chains*. Cambridge University Press.
- Patton, A. J., & Ramadorai, T. (2013). On the high-frequency dynamics of hedge fund risk exposures. *Journal of Finance*, 68(2), 597–635.
- Patron, H. (2004). Are "market neutral" hedge funds really market neutral? *Journal of Alternative Investments*, 7(2), 49–57.
- Preqin. (2023). *History of the hedge fund industry*. <https://www.preqin.com/academy/lesson-3-hedge-funds/history-of-the-hedge-fund-industry>
- Sharpe, W. F. (1964). Capital asset prices: A theory of market equilibrium under conditions of risk. *Journal of Finance*, 19(3), 425–442.
- Sharpe, W. F. (1966). Mutual fund performance. *Journal of Business*, 39(1), 119–138.
- Sortino, F. A., & van der Meer, R. (1991). Downside risk. *Journal of Portfolio Management*, 17(4), 27–31.
- Standard Chartered. (n.d.). *Hedge funds explained: How they work to diversify your portfolio*. <https://www.sc.com/ae/wealth-insights/hedge-funds-explained/>
- Teo, M. (2011). The liquidity risk of liquid hedge funds. *Journal of Financial Economics*, 100(1), 24–44.
- Ubide, A. (2006, June). *Demystifying hedge funds*. International Monetary Fund — Finance and Development. <https://www.imf.org/external/pubs/ft/fandd/2006/06/basics.htm>
- Van de Wetering, S. (2025). Why do hedge funds fail? *Empaxis*. <https://www.empaxis.com/blog/reasons-hedge-funds-fail>
- Fama, E.F., & French, K.R. (1993) – Common risk factors in the returns of stocks and bonds. *Journal of Financial Economics*, 33(1), 3–56.

Boyson M. (2018). Hedge Fund Performance Persistence: A New Approach. Taylor & Francis Online, 64(6), 27-44.

1 Viral V.Acharya, Lasse Heje Pedersen (2005). Asset pricing with liquidity risk. Science Direct, 77(2), 375-410.

1 Citadel. (2025). Citadel – Who we are. <https://www.citadel.com/who-we-are/>

1 Bridgewater Associates. Bridgewater Associates. <https://www.bridgewater.com/>

1 AQR Capital Management. (2025). Our Approach. <https://www.aqr.com/What-We-Do/Our-Approach>

1 H. Jacobs, S. Muller (2020). Anomalies across the globe: Once public, no longer existent. Journal of Financial Economics. 135, 213-230.

1 Brunnermeier, M.K. & Nagel, S. (2004). Hedge funds and the technology bubble. Journal of Finance, 59(5), 2013-2040.

Y. Naik. T. Ramadorai (2007). Capacity Constraints and Hedge Fund Strategy Returns. European Financial Management, 13(2), 239-256.

1 J. Patton. Ramadorai (2013). On the High-Frequency Dynamics of Hedge Fund Risk Exposures, The Journal of Finance. 68 (2), 597-635.

HFR (Hedge Fund Research). HFRI Event-Driven (Total) Index. <https://www.hfr.com/indices/hfri-event-driven-total-index/>

HFR (Hedge Fund Research). HFRI 400 (US) Macro Index (M). <https://www.hfr.com/indices/hfri-400-us-macro-index-m/>

J.P Morgan Asset Management (2026). Guide to the Markets. J.P Morgan Chase & Co. <https://am.jpmorgan.com/us/en/asset-management/adv/insights/market-insights/guide-to-the-markets/>

Goldman Sachs Asset Management (2026). Investment Outlook 2026 – Seeking Catalysts Amid Complexity. Goldman Sachs.

Blackrock. Tom Becker. (2026). A global macro-outlook for 2026: patience. Blackrock. <https://www.blackrock.com/institutions/en-us/insights/2026-macro-outlook>

Lioui, A.,& Maio, P. (2014). Interest rate risk and the cross-section of stock returns. Journal of Financial and Quantitative Analysis, 49(2), 483-511.

Investing.com (2025). Historical data: CBOE Volatility Index, FED Interest Rate, Gold Spot Price, US 5-year treasury yield, Federal Funds Rate, S&P 500. Investing.com. <https://www.investing.com/>

Caldara, D. & Iacoviello (2022). Measuring geopolitical risk. *American Economic Review*, 112(4), 1194-1225.

